



Sai-PavanBata / CMOS_CIRCUIT_DESIGN



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 Sai-PavanBata [Revise threshold voltage equation and lecture header](#)  e28ee12 · now 

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Revise threshold voltage equation ...

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README



CMOS-Circuit-Design-Spice-Simulation-using-Sky130nm-technology

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VirtualBox Setup

Follow the steps below to load the provided **CMOS VDI file** in **Oracle VirtualBox**.

Step 1: Install 1

- Download **Oracle VirtualBox** from: <https://www.virtualbox.org/wiki/Downloads>
- Complete the installation using the default options.

Step 2: Create a New Virtual Machine

1. Launch **VirtualBox**
2. Click **New**
3. Enter the following details:

Option	Selection
Type	Linux
Version	Ubuntu 18.04 (Bionic Beaver) – 64-bit

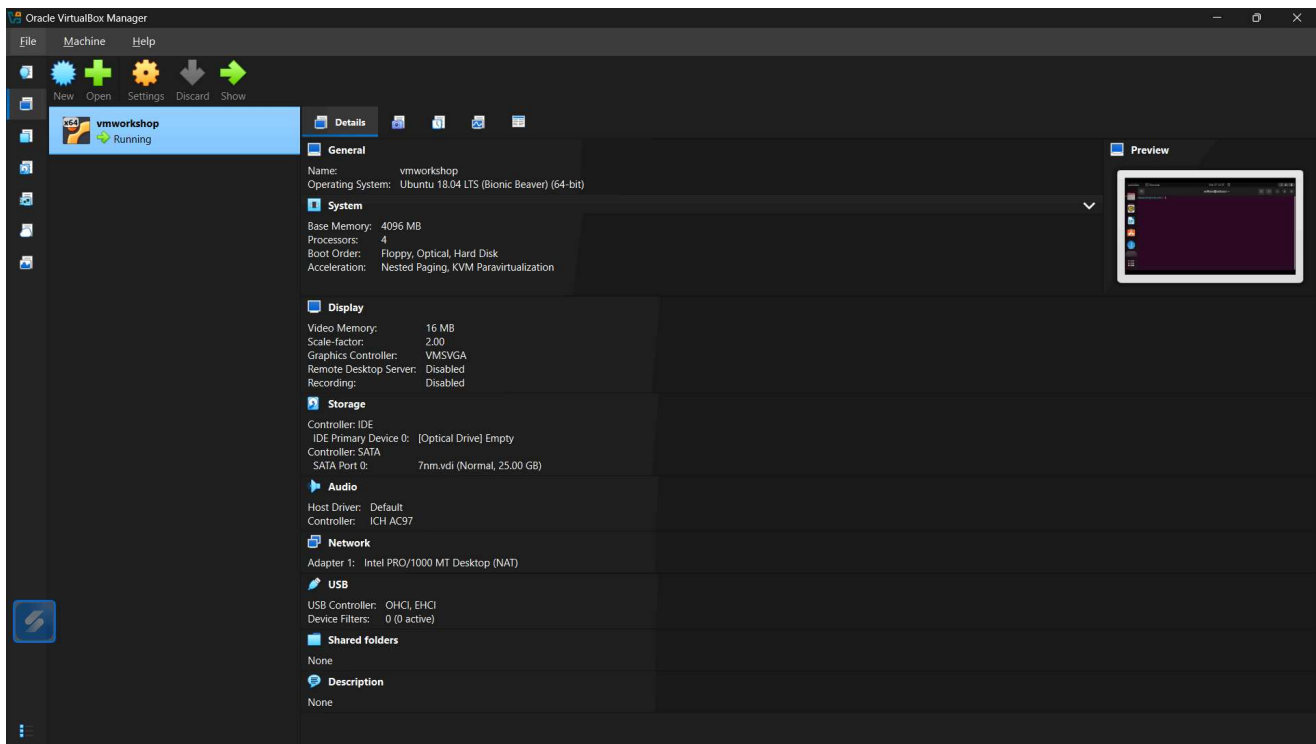
4. Click **Next**

Step 3: Memory Allocation

- Set the RAM size as required
- **Recommended:** 4096 MB
- Click **Next**

Step 4: Load the CMOS VDI File

1. Choose **Use an existing virtual hard disk file**
2. Click the **folder icon** to browse
3. Locate and select the extracted **CMOS VDI file**
4. Click **Open**
5. Click **Next**, then **Finish**



Step 5: Launch the Virtual Machine

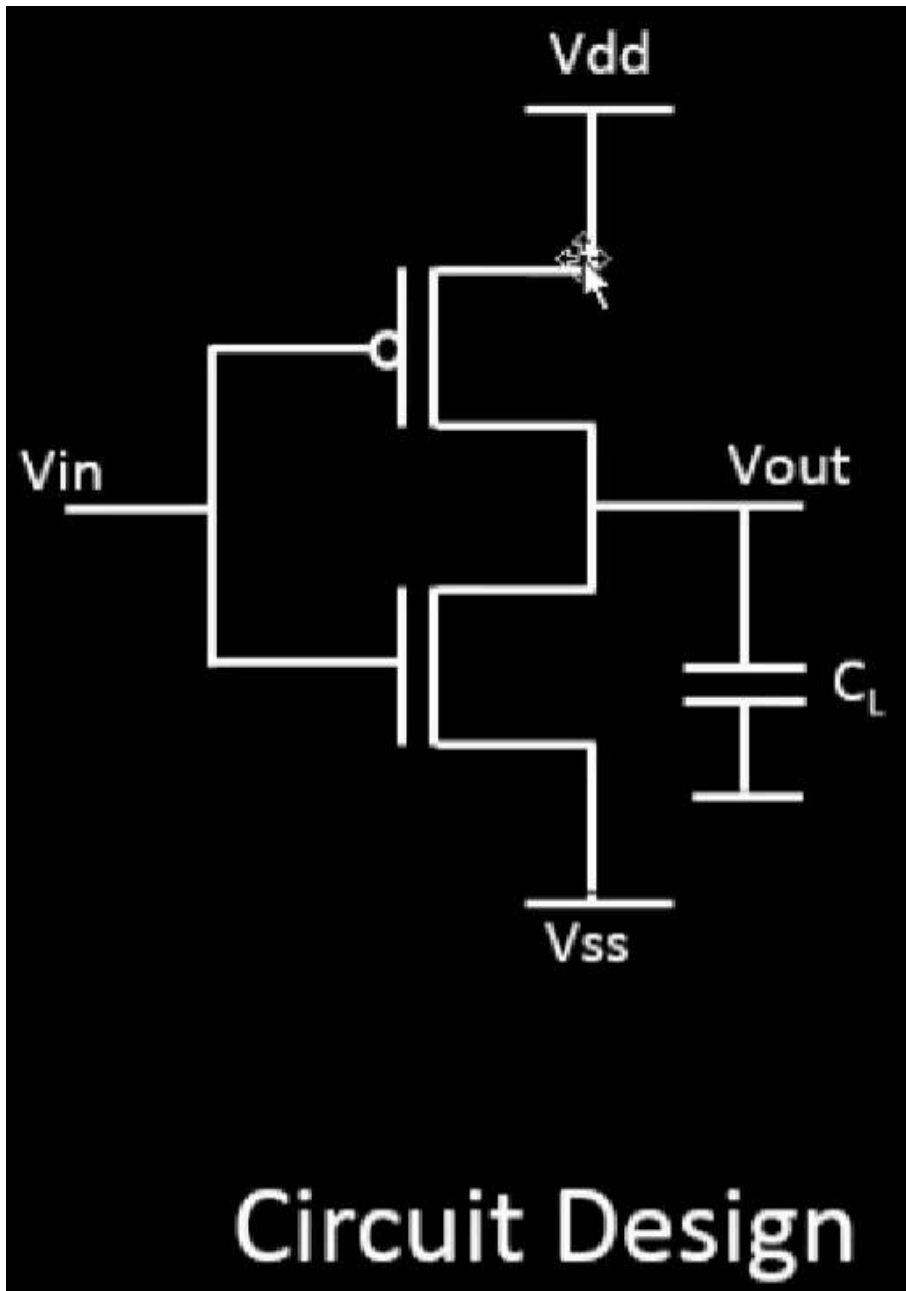
1. Highlight the newly created virtual machine
2. Click **Start** to boot the system

Ngspice SKY130 – Day 1: Basics of NMOS Drain Current (I_d) vs Drain-to-Source Voltage (V_{ds})

Chapter 1: Introduction to Circuit Design and SPICE Simulations

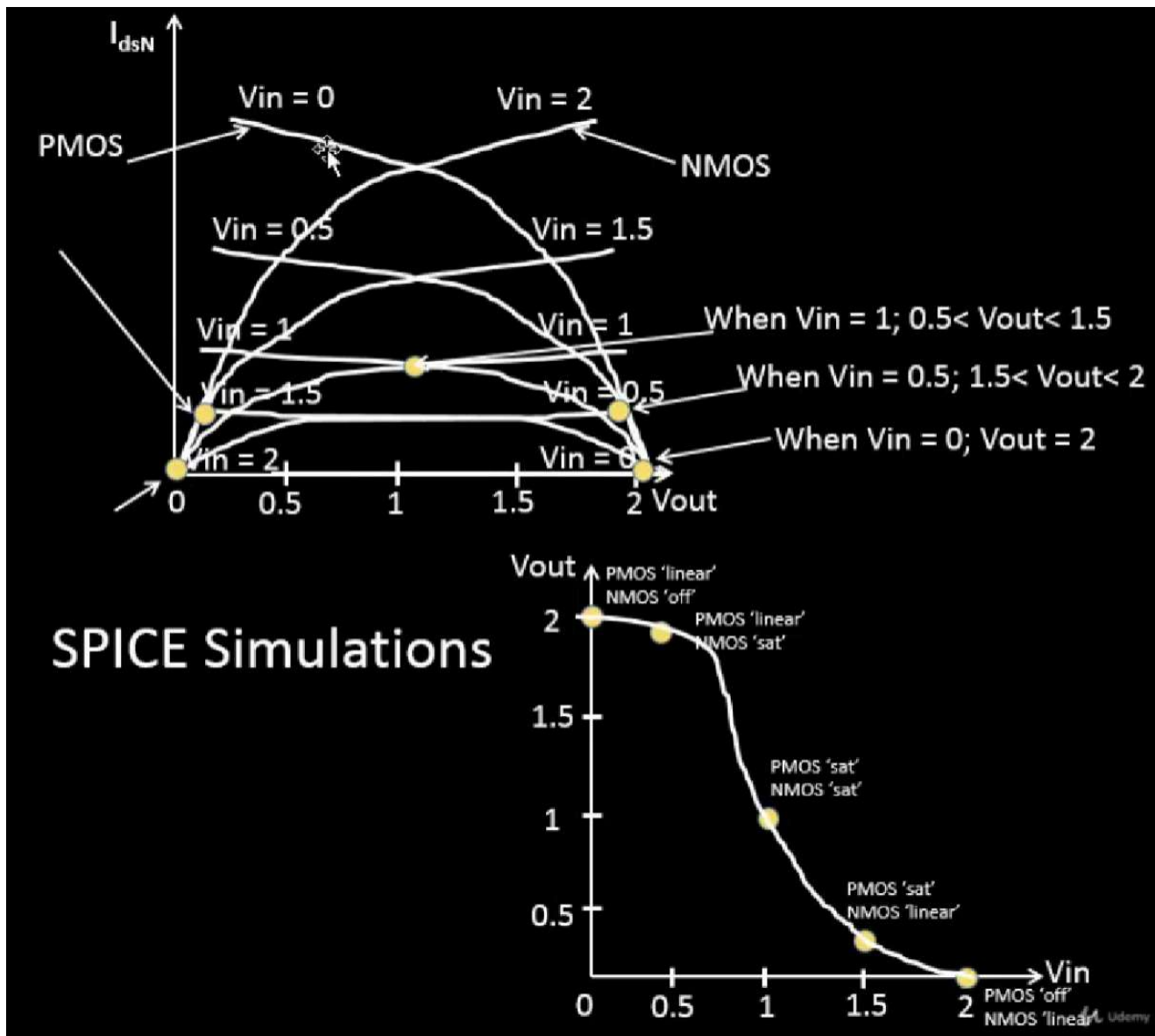
Lecture 1: Why Do We Need SPICE Simulations?

- Digital circuits are constructed using **PMOS** and **NMOS** transistors connected together to form logic gates such as:
 - NAND
 - NOR
 - AND
 - OR
 - Inverter



The above inverter will have the following characteristics. We will run SPICE simulations to determine the delay and thus obtain the W/L ratio of the particular transistor.

- Each logic gate exhibits **electrical characteristics** such as:
 - Propagation delay
 - Rise time
 - Fall time
- These characteristics **cannot be accurately estimated analytically** and therefore require **SPICE simulations**.



- SPICE simulations help in:
 - Calculating delay
 - Determining appropriate W/L ratios of transistors
 - Validating circuit performance before fabrication

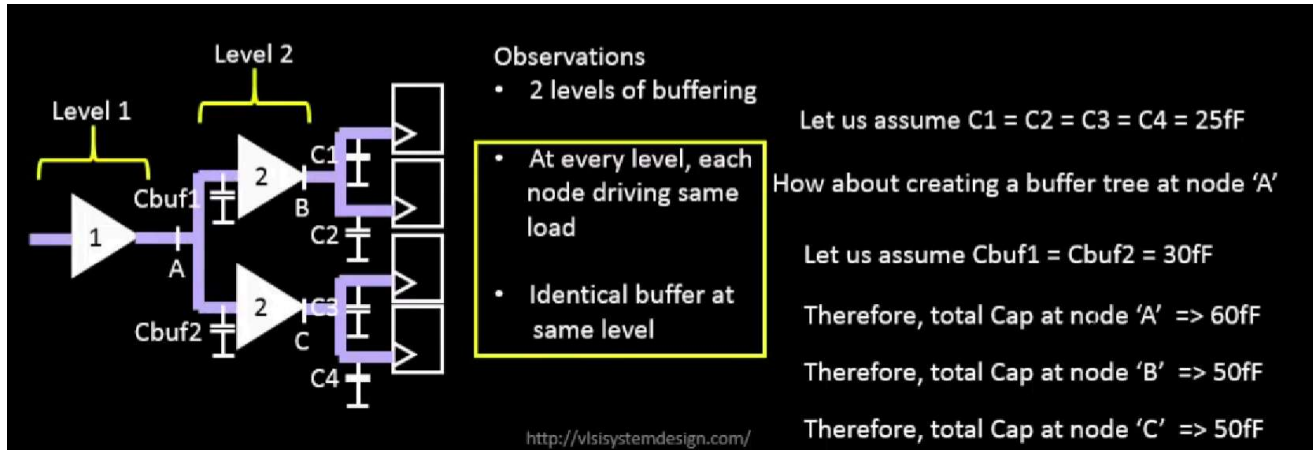
Role of SPICE in VLSI Physical Design

- SPICE (Simulation Program with Integrated Circuit Emphasis) is the backbone of:
 - Timing analysis
 - Clock Tree Synthesis (CTS)
 - Crosstalk and signal integrity analysis
- Without SPICE-based delay estimation:
 - Timing closure is not possible
 - CTS optimisation becomes meaningless

- Crosstalk analysis has no practical reference

Example: Delay Extraction Using SPICE

- Consider a circuit where **Clock Tree Synthesis** has been performed.
- Buffers are inserted to drive **different capacitive loads** at the output.



- After running SPICE simulations:
 - A **Delay Table** is generated
 - The table consists of:
 - Input slew values
 - Output load capacitance values
- The **delay** is obtained from the **intersection of input slew and output load**.

Delay Table for CBUF '1'								Delay Table for CBUF '2'							
		Output Load								Output Load					
		10fF	30fF	50fF	70fF	90fF	110fF			10fF	30fF	50fF	70fF	90fF	110fF
Input slew	20ps	x1	x2	x3	x4	x5	x6	Input slew	20ps	y1	y2	y3	y4	y5	y6
	40ps	x7	x8	x9	x10	x11	x12		40ps	y7	y8	y9	y10	y11	y12
	60ps	x13	x14	x15	x16	x17	x18		60ps	y13	y14	y15	y16	y17	y18
	80ps	x19	x20	x21	x22	x23	x24		80ps	y19	y20	y21	y22	y23	y24

- Separate delay tables are generated for:
 - Level 1 buffers
 - Level 2 buffers
- These delay values are used as **standard timing inputs** in digital and physical design flows.

Steps to find buffer delay

1. **Select the buffer cell** Use the correct cell (e.g., CBUF1 or CBUF2).

2. **Get input slew** Take the input transition time at the buffer input.
3. **Compute output load** Sum the wire capacitance and the fanout input capacitances.
4. **Find bounding table entries**
 - Input slew bounds (e.g., 40 ps and 60 ps)
 - Output load bounds (e.g., 50 fF and 70 fF)
5. **Read four corner delays** Delays at the 4 combinations of (Slew, load).
6. **Interpolate**
 - First in load direction
 - Then in the slew direction (bilinear interpolation)
7. **Result** The interpolated value is the buffer propagation delay.

Question 1

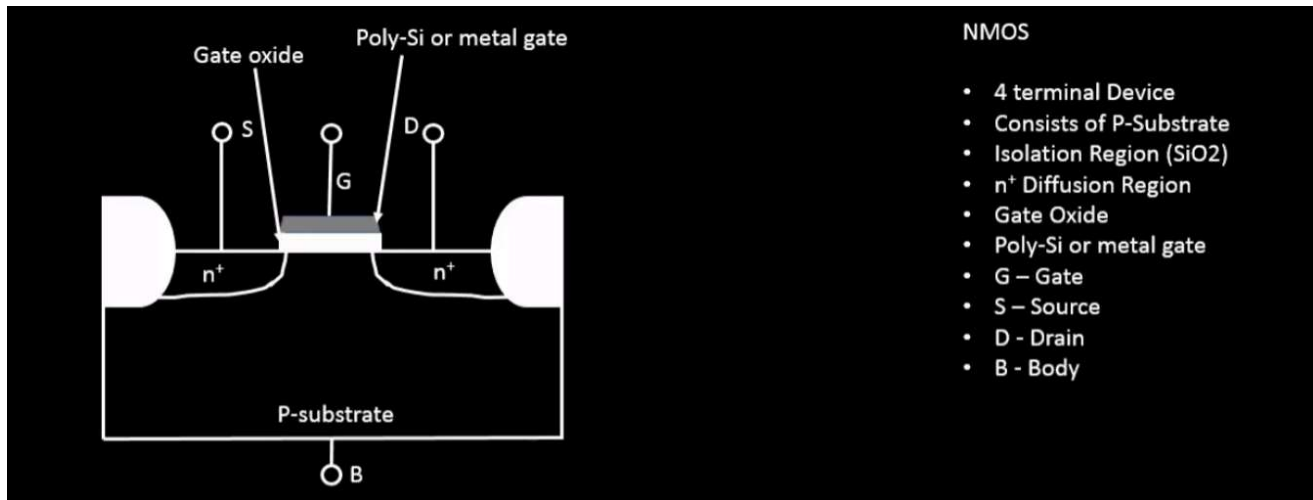
From where do these delay values originate? What is their source? Are these delay models accurate enough to be used in a SPICE model?

Answer

- The delay values are **derived from SPICE simulations** performed at the **circuit level**.
- During characterisation:
 - A specific **input slew** is applied
 - A defined **output load capacitance** is connected
 - The circuit is simulated using **SPICE**
- The resulting **propagation delay** measured from the simulation is recorded as the delay value.
- To verify the accuracy of the delay model:
 - The **same input slew and output load** are applied again
 - A fresh **SPICE simulation** is run
 - The obtained delay is **compared with the tabulated delay value**
- If both values closely match, the delay model is considered **accurate and valid** for use in SPICE-based timing analysis.

Lecture 2: Introduction to Fundamental Elements in Circuit Design – NMOS

Structure of an NMOS Transistor



The figure shows the cross-sectional structure of an NMOS transistor.

1) NMOS as a 4-Terminal Device

NMOS consists of four terminals:

- G — Gate
- S — Source
- D — Drain
- B — Body (Substrate)

2) Physical Regions

- **P-Substrate**
 - Base semiconductor material
 - Forms the **Body (B)** terminal
- **n⁺ Diffusion Regions**
 - Heavily doped n-type regions
 - Form **Source (S)** and **Drain (D)**
- **Gate Oxide (SiO₂)**
 - Thin insulating layer
 - Electrically isolates the gate from the substrate
- **Poly-Silicon or Metal Gate**
 - Controls channel formation

- **Isolation Oxide**
 - Electrically separates adjacent devices

Working Principle

- When $V_{GS} = 0$:
 - No inversion channel exists
 - No conduction between Source and Drain
- When $V_{GS} > V_{th}$:
 - An **inversion layer (n-channel)** forms
 - Current flows when V_{DS} is applied
- The resulting current is called **Drain Current**, denoted as I_D
-

Body Effect and Threshold Voltage Tuning

Usually, the body is connected to the ground. However, if a voltage difference exists between Source and Body:

$$V_{SB} \neq 0$$

The threshold voltage changes.

Body Effect

- Increasing V_{SB} (reverse-biasing source-body junction):
 - Increases depletion width
 - Requires a higher gate voltage for inversion
 - Therefore increases V_{th}

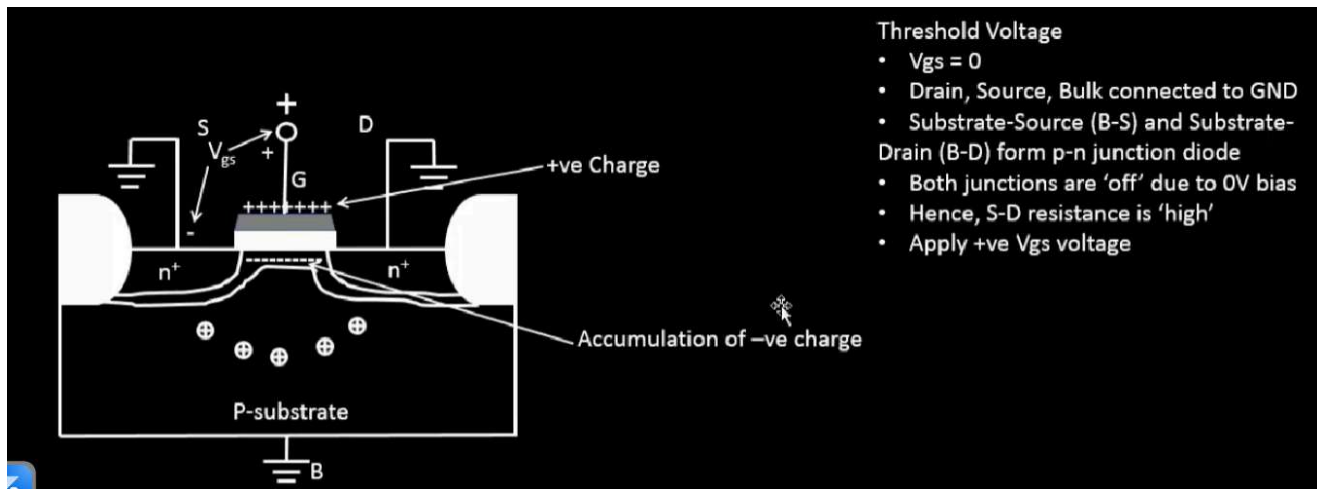
Threshold Voltage with Body Effect

$$V_{th} = V_{th0} + \gamma \left(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|-2\phi_F|} \right)$$

Where:

- V_{th0} = Threshold voltage when $V_{SB} = 0$
- γ = Body-effect coefficient
- ϕ_F = Fermi potential
- V_{SB} = Source-to-body voltage
-

Threshold Voltage



NMOS Threshold Voltage – Physical Understanding

1. Importance of Threshold Voltage

- Threshold Voltage (V_T) is the most critical parameter in NMOS modeling.
- It determines:
 - When the NMOS turns ON
 - Channel formation
 - Accuracy of SPICE simulations
- In SPICE, V_T is not a constant but a **model parameter**, expressed as a function:

$$V_T = f(x, y, z)$$

- Advanced SPICE models use complex equations, but analysis starts with the **basic threshold voltage model**.

2. Initial Condition: $V_{GS} = 0$

- Gate, Source, Drain, and Body are all connected to GND.
- NMOS structure consists of:
 - n^+ Source
 - n^+ Drain
 - p-type Substrate (Body)

Junction Behaviour

- Source–Body (S–B) and Drain–Body (D–B) form PN junction diodes.
- At 0 V bias:

- Both junctions are **OFF**
- No conduction path exists

Electrical State

- Source–Drain resistance is **very high**
- NMOS behaves as an **open switch**
- Drain current:

$$I_D \approx 0$$

3. Applying a Small Positive Gate Voltage ($V_{GS} > 0$)

- Gate, oxide, and substrate form a **MOS capacitor**.
- Gate is treated as a **metal plate**.
- Oxide (SiO_2) acts as a **dielectric**.

Charge Redistribution

- Positive voltage on the gate causes:
 - Accumulation of **positive charge on gate**
 - Repulsion of **holes** (majority carriers) from substrate surface
- Holes move deeper into the substrate.

4. Depletion Region Formation

- Removal of holes near the surface leaves behind:
 - **Fixed negatively charged acceptor ions**
- This creates a **depletion region** below the gate oxide.

Key Concept

- Depletion region contains:
 - No mobile carriers
 - Only an immobile negative charge
- Still **no conduction channel** exists.

5. Toward Threshold Voltage

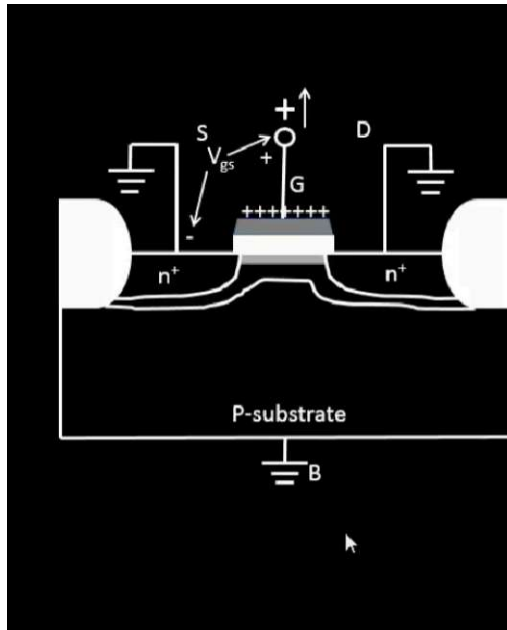
- Increasing V_{GS} further:
 - Expands depletion region
 - At a critical voltage, the surface **inverts** to n-type

- This voltage is called **Threshold Voltage (V_T)**.

$$V_{GS} = V_T \Rightarrow \text{Onset of inversion}$$

- Inversion layer forms a **conductive channel** between Source and Drain.

Lecture 3: Strong Inversion and Threshold Voltage



Threshold Voltage

- $V_{gs} = 0$
- Drain, Source, Bulk connected to GND
- Substrate-Source (B-S) and Substrate-Drain (B-D) form p-n junction diode
- Both junctions are 'off' due to 0V bias
- Hence, S-D resistance is 'high'
- Apply +ve V_{gs} voltage
- Increase gate voltage ' V_{gs} '
- This phenomenon is called 'strong inversion'
- The ' V_{gs} ' voltage at which 'strong inversion' occurs is called threshold vge (V_t)
- Further, increase ' V_{gs} '
- No Change in depletion layer width
- Electrons from heavily doped 'n+' source region are drawn in region under gate 'G'
- Continuous n-channel formation from S-D, whose conductivity is modulated by ' V_{gs} '

1. Recap: Small Positive V_{GS}

- Applying small positive V_{GS} :
 - Repels holes (majority carriers in p-substrate)
 - Leaves behind fixed negative acceptor ions
- A **depletion region** forms under the gate.
- Behavior similar to a **reverse-biased PN junction diode**.

2. Increasing Gate Voltage

- As V_{GS} increases:
 - More holes are repelled
 - Depletion width increases
- Eventually, the surface concentration of electrons increases.

3. Strong Inversion

At a critical gate voltage:

- Surface of p-substrate becomes **n-type**
- A thin **inversion layer** forms.
- This is called **Strong Inversion**.

Definition

$$V_{GS} = V_T$$

- The gate voltage at which strong inversion occurs is called **Threshold Voltage** (V_T).

4. After Threshold ($V_{GS} > V_T$)

- Depletion region width remains approximately constant.
- Additional gate voltage:
 - Does NOT significantly increase depletion width.
 - Increases **channel charge density**.
- Electrons are attracted from a heavily doped **n^+ source**.
- Channel width increases.

Result

- A **continuous n-channel** forms between Source and Drain.
- Now current can flow if V_{DS} is applied.

5. Region of Operation

- Before inversion → **Cutoff Region**
- After inversion → Channel exists
- Current flows only when:

$$V_{DS} > 0$$

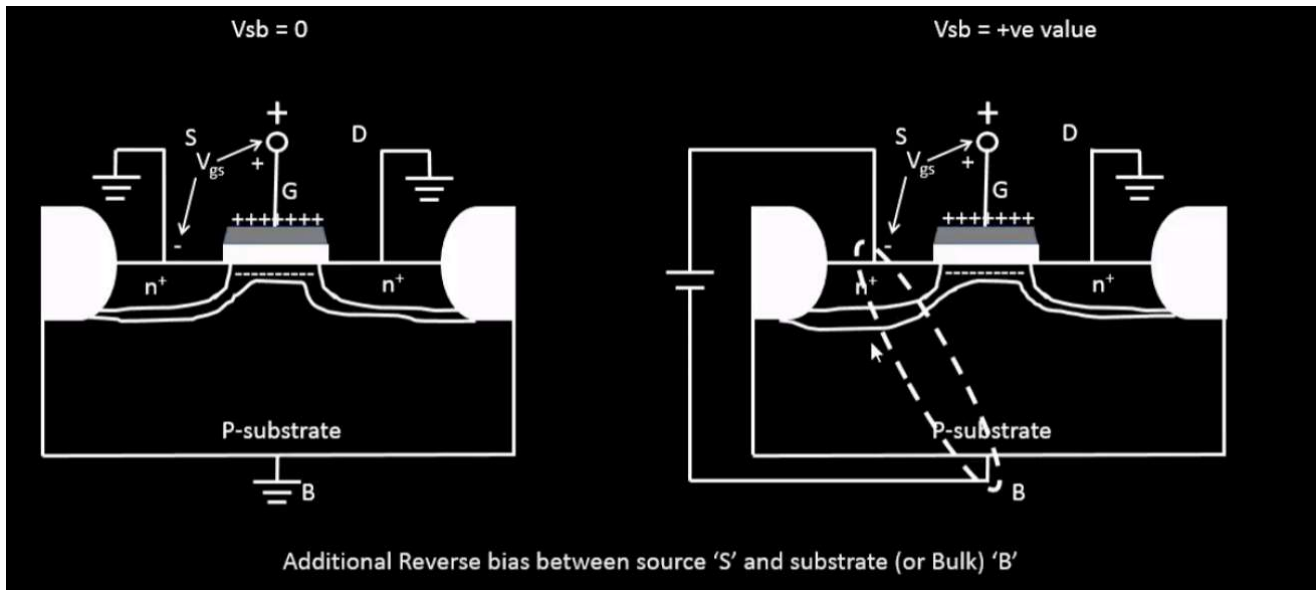
(Subthreshold current ignored for now.)

6. Effect of Body Terminal (Body Effect Introduction)

Consider two cases:

Case 1: $V_{SB} = 0$

- No additional bias between source and body.
- Standard depletion region width.
- Threshold voltage = normal V_T .



Case 2: $V_{SB} > 0$ (Source positive w.r.t Body)

- Source-body PN junction becomes **more reverse biased**.
- Depletion width near the source increases.
- Strong inversion requires a higher gate voltage.

Key Observation

- For same V_{GS} :
 - Inversion happens earlier when $V_{SB} = 0$
 - Inversion is delayed when $V_{SB} > 0$

7. Body Effect Concept

Increasing V_{SB} :

- Increases depletion width
- Increases required inversion charge
- Increases Threshold Voltage

$$V_T \uparrow \quad \text{as} \quad V_{SB} \uparrow$$

This is called the **Body Effect**.

- No standalone parentheses
- Proper headings only
- Image insertion clearly indicated
- All equations GitHub-compatible
- Zero formatting ambiguity

Lecture 4 Threshold Voltage with Positive Substrate Potential (Body Effect)

1. Objective of This Lecture

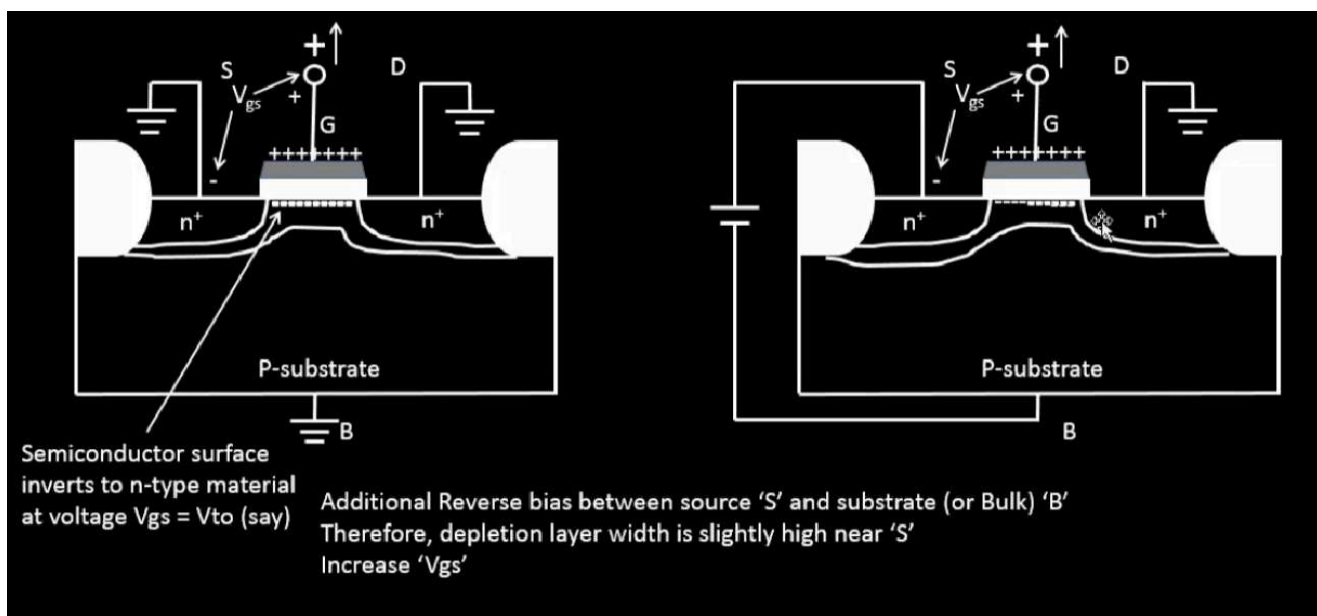
- Study the effect of **positive source-to-body voltage** ($V_{SB} > 0$) on:
 - Depletion region
 - Channel formation
 - Threshold voltage
- Understand the **body effect**
- Relate physical behavior to **SPICE model parameters**

2. Case A: Zero Substrate Bias ($V_{SB} = 0$)

- Source and Body are at the same potential.
- No additional reverse bias across the Source–Body PN junction.
- As V_{GS} increases:
 - Holes are repelled from the surface
 - Depletion region grows under the gate
 - Surface eventually inverts to n-type

$$V_{GS} = V_{T0} \Rightarrow \text{Strong inversion begins}$$

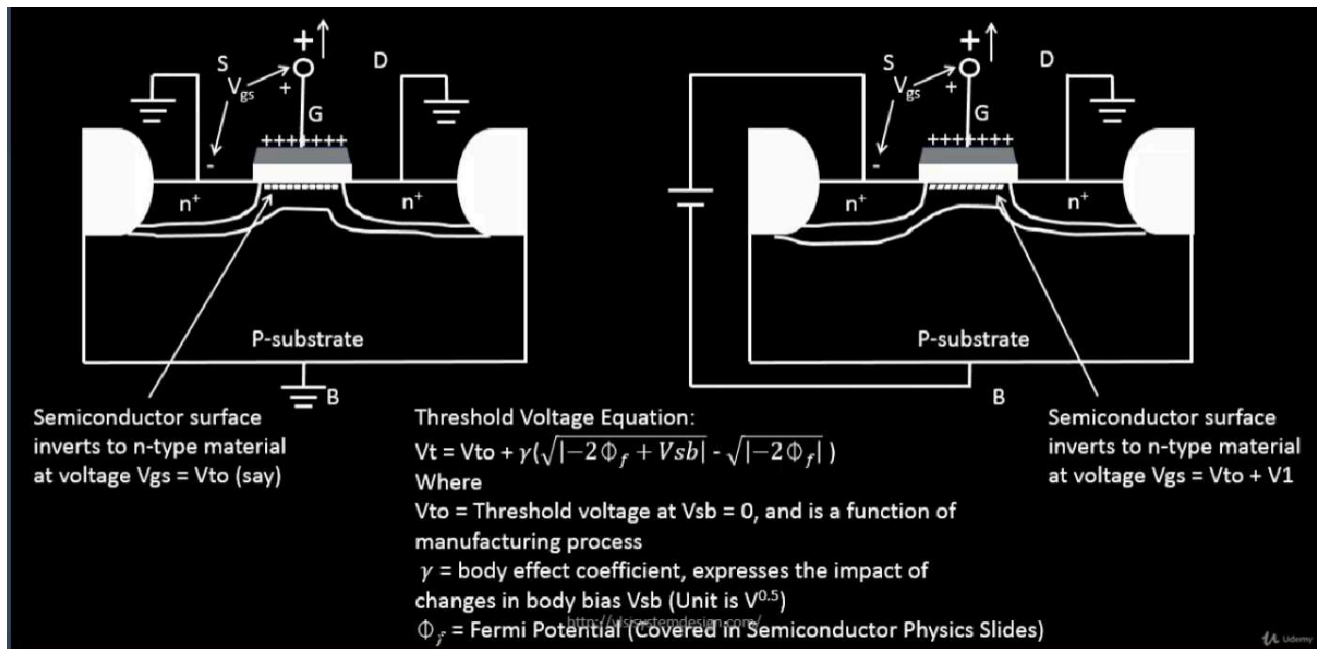
- V_{T0} is defined as:
 - Threshold voltage when $V_{SB} = 0$
 - A function of device and technology parameters



3. Case B: Positive Substrate Bias ($V_{SB} > 0$)

- Source is at a higher potential compared to the Body.
- The Source–Body PN junction becomes **more reverse biased**.
- This introduces an **additional depletion region near the source**.

4. Key Physical Observation



- In both cases, increasing V_{GS} :
 - Repels holes from the surface
 - Initially increases depletion width
- Unique behavior when $V_{SB} > 0$:
 - Channel electrons are pulled toward the Source
 - Positive Source potential attracts a negative charge
- This redistribution does not occur when $V_{SB} = 0$

5. Effect on Surface Inversion

- For the same gate voltage:
 - Surface inversion occurs earlier when $V_{SB} = 0$
 - Surface inversion is delayed when $V_{SB} > 0$
- Reason:
 - Increased depletion width requires additional gate voltage
 - Channel formation near the Source is resisted

6. Definition of Threshold Voltages

Threshold Voltage at Zero Body Bias

$$V_T = V_{TO} \quad \text{when } V_{SB} = 0$$

- V_{TO} is the reference threshold voltage.

Threshold Voltage with Body Bias

$$V_T = V_{TO} + V_1 \quad \text{when } V_{SB} > 0$$

- V_1 is the additional voltage required due to body bias.

7. Threshold Voltage Equation Including Body Effect

$$V_T = V_{TO} + \gamma \left(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|-2\phi_F|} \right)$$

Parameter Definitions

- V_T : Threshold voltage with body bias
- V_{TO} : Threshold voltage at zero body bias
- γ : Body effect coefficient
- ϕ_F : Fermi potential
- V_{SB} : Source-to-Body voltage

8. Body Effect Coefficient

$$\gamma = \frac{\sqrt{2qN_A\epsilon_{si}}}{C_{ox}}$$

$$\gamma = \frac{\sqrt{2qN_A\epsilon_{si}}}{C_{ox}}$$

ϵ_{si} = relative permittivity of silicon = 11.7

N_A = doping concentration

q = charge of the electron

C_{ox} = oxide capacitance

All parameters are technology-dependent and provided by the foundry.

9. Fermi Potential

$$\phi_F = -\phi_T \ln \left(\frac{N_A}{n_i} \right)$$

- ϕ_T = Thermal voltage
- n_i = Intrinsic carrier concentration

(Treated as a constant here; covered in device physics.)

10. Key Conclusions

- Increasing V_{SB} :
 - Increases depletion width
 - Delays surface inversion
 - Increases threshold voltage

- Summary:

$$V_{SB} = 0 \Rightarrow V_T = V_{T0}$$

$$V_{SB} > 0 \Rightarrow V_T > V_{T0}$$

- This phenomenon is called the **Body Effect**.

Chapter 2: NMOS Resistive and Saturation Regions of Operation

Lecture 1: NMOS Resistive Region and Saturation Region of Operation

Regions of NMOS Operation

- NMOS operates in three regions:
 - Cutoff region
 - Resistive (Linear) region
 - Saturation region

Cutoff Region

- Condition:

$$V_{GS} < V_T$$

- No inversion channel
- Drain current approximately zero

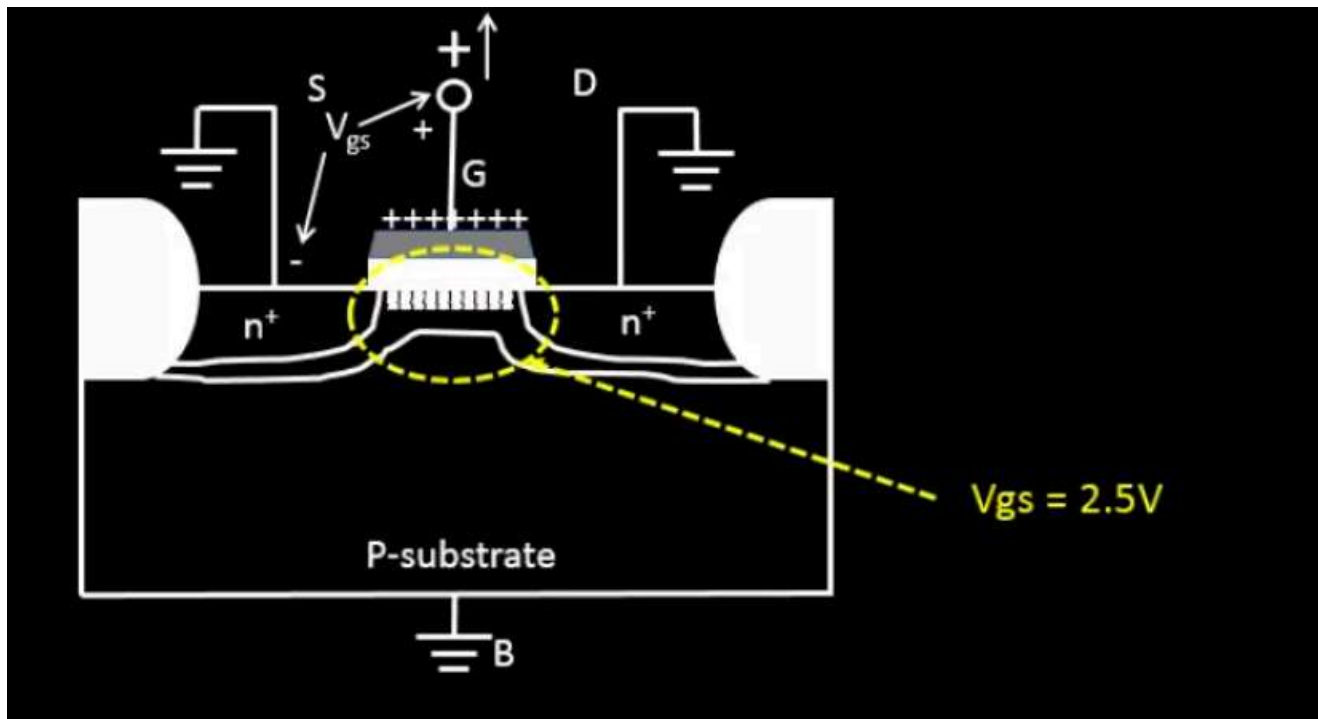
Strong Inversion Condition

- When gate voltage reaches threshold:

$$V_{GS} = V_T$$

- Surface of p-substrate inverts to n-type
- Conducting channel forms between Source and Drain

Resistive (Linear) Region of Operation



(NMOS with uniform channel and small drain-source voltage)

Conditions

$$V_{GS} > V_T$$

$$V_{DS} < (V_{GS} - V_T)$$

Physical Interpretation

- Channel exists along the entire length
- Applying a small V_{DS} :
 - Creates a voltage gradient from Source to Drain
 - Channel remains continuous
- NMOS behaves like a **voltage-controlled resistor**

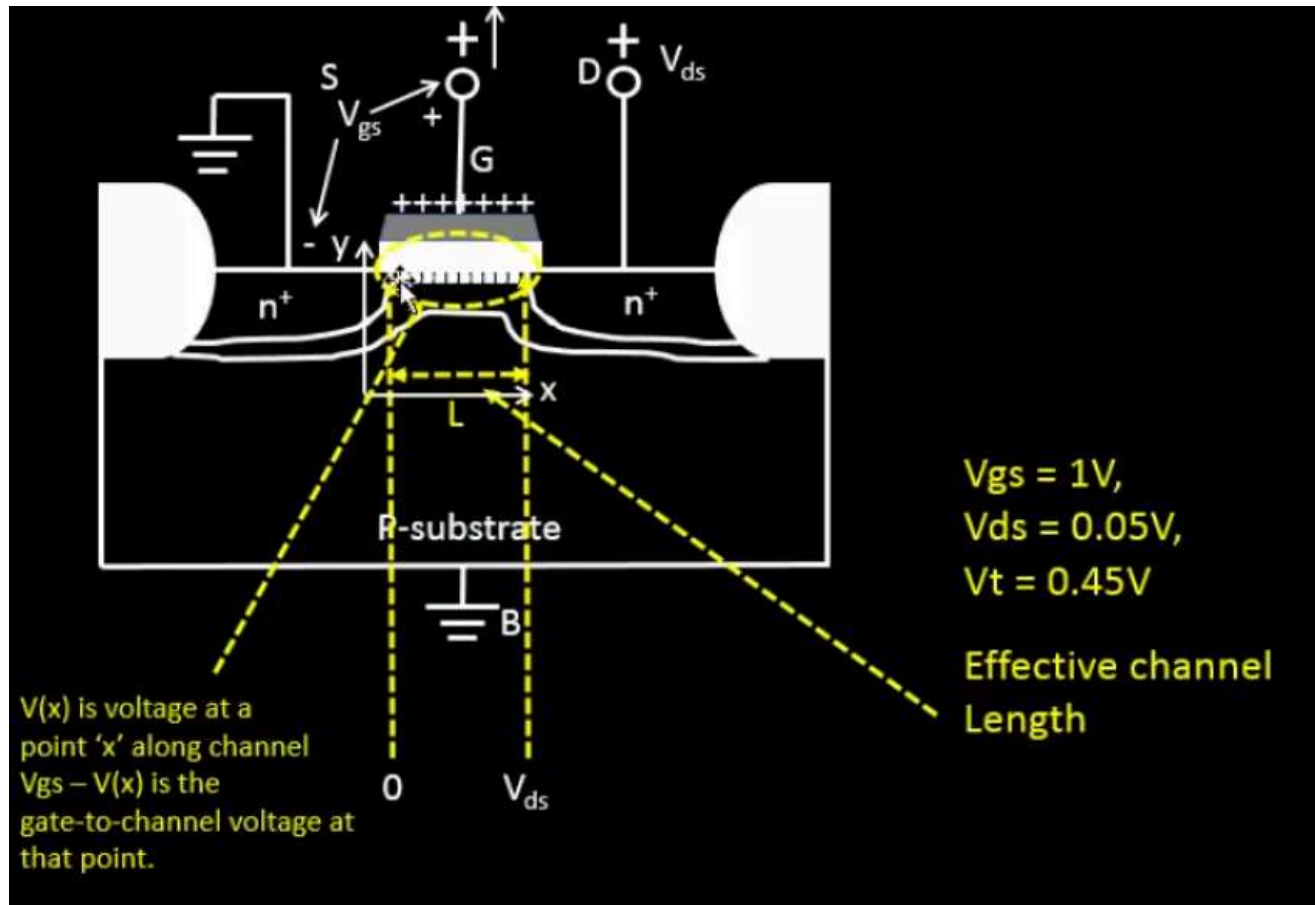
Channel Charge Behaviour

- Channel charge density is proportional to:

$$V_{GS} - V_T$$

- Increasing V_{GS} :
 - Increases electron concentration
 - Increases channel width
 - Reduces channel resistance

Voltage Distribution Along Channel



*Channel potential variation along the channel length

- (Channel potential varies: $0 \rightarrow V_{DS}$).
- Define effective channel length as L
- Let x be the position along the channel

$$V(0) = 0 \quad (\text{Source end})$$

$$V(L) = V_{DS} \quad (\text{Drain end})$$

- Channel voltage varies continuously between 0 and V_{DS}

Effective Gate-to-Channel Voltage

- At any point x :

$$V_{GC}(x) = V_{GS} - V(x)$$

- Channel charge is **not uniform**
- Drain current must be derived using position-dependent quantities

Saturation Region of Operation

Insert Image 3 here (*Channel pinch-off near drain*)

Condition

$$V_{DS} \geq (V_{GS} - V_T)$$

Physical Interpretation

- Channel near the drain gets pinched off
- Increasing V_{DS} :
 - Does not significantly increase channel charge
 - Current becomes weakly dependent on V_{DS}
- Drain current mainly controlled by V_{GS}

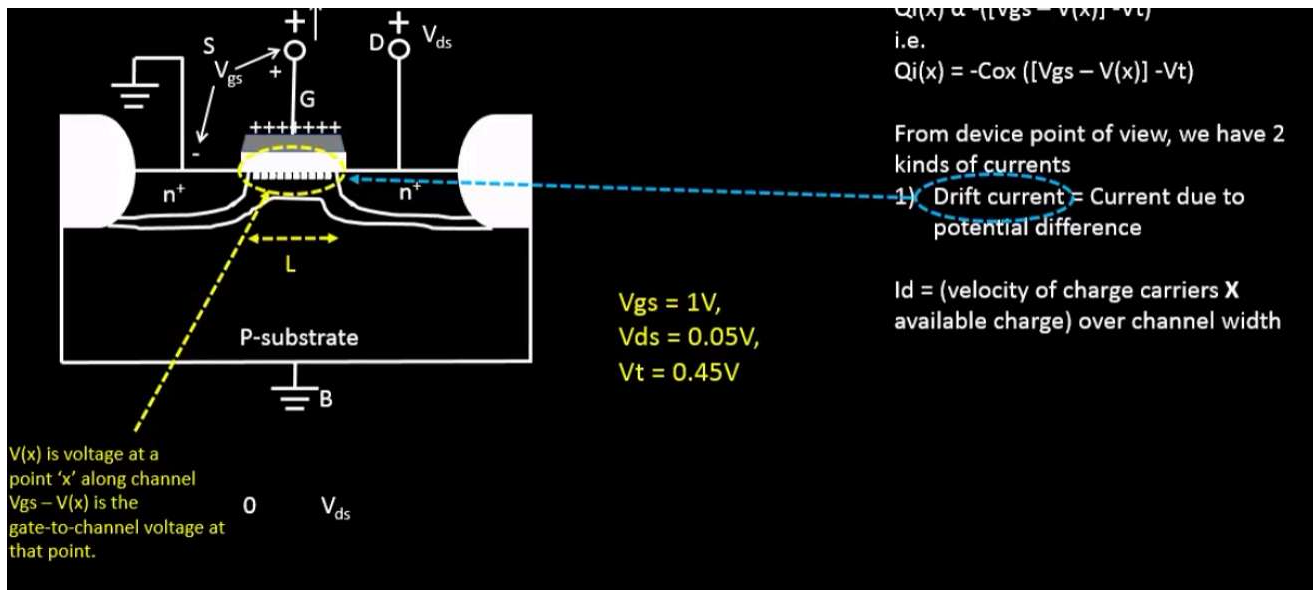
Key Comparison

Parameter	Resistive Region	Saturation Region
Channel	Continuous	Pinched near drain
V_{DS}	Small	Large
I_D control	V_{GS} and V_{DS}	Mostly V_{GS}
Behavior	Acts as resistor	Acts as current source

Lecture 2: Drift Current Theory (Resistive Region with Small V_{DS})

1. Channel Potential Variation with Applied V_{DS}

- Consider the NMOS operating in strong inversion with:



- $V_{GS} = 1.0 \text{ V}$
- $V_T = 0.45 \text{ V}$
- A small drain-source voltage $V_{DS} = 0.05 \text{ V}$
- At the **source end** of the channel:
 - Channel potential $V(x) = 0$
 - Effective gate-to-channel voltage is:

$$V_{GS} - V(x) = 1.0 \text{ V}$$

- At the **drain end** of the channel:
 - Channel potential $V(x) = V_{DS} = 0.05 \text{ V}$
 - Effective gate-to-channel voltage becomes:

$$V_{GS} - V(x) = 0.95 \text{ V}$$

- Therefore, the channel potential is **not constant**.
- A **voltage gradient exists along the channel**, caused purely by the applied V_{DS} .
- This gradient is the fundamental reason current can flow from Source to Drain.

2. Effective Gate Voltage Along the Channel

- When $V_{DS} = 0$, every point along the channel sees the same gate voltage V_{GS} .
- When $V_{DS} > 0$, each point along the channel experiences a different effective gate voltage.
- At any position x along the channel:

$$V_{GC}(x) = V_{GS} - V(x)$$

- Since $V(x)$ varies from 0 at the source to V_{DS} at the drain:
 - The effective gate voltage decreases gradually from source to drain.

- As a result:
 - Channel charge density is **higher near the source**
 - Channel charge density is **lower near the drain**

3. Induced Channel Charge in Presence of V_{DS}

- The inversion charge in the channel depends on:
 - Effective gate voltage
 - Threshold voltage
- At any position x , the induced inversion charge per unit area is:

$$Q_i(x) = C_{ox} (V_{GS} - V(x) - V_T)$$

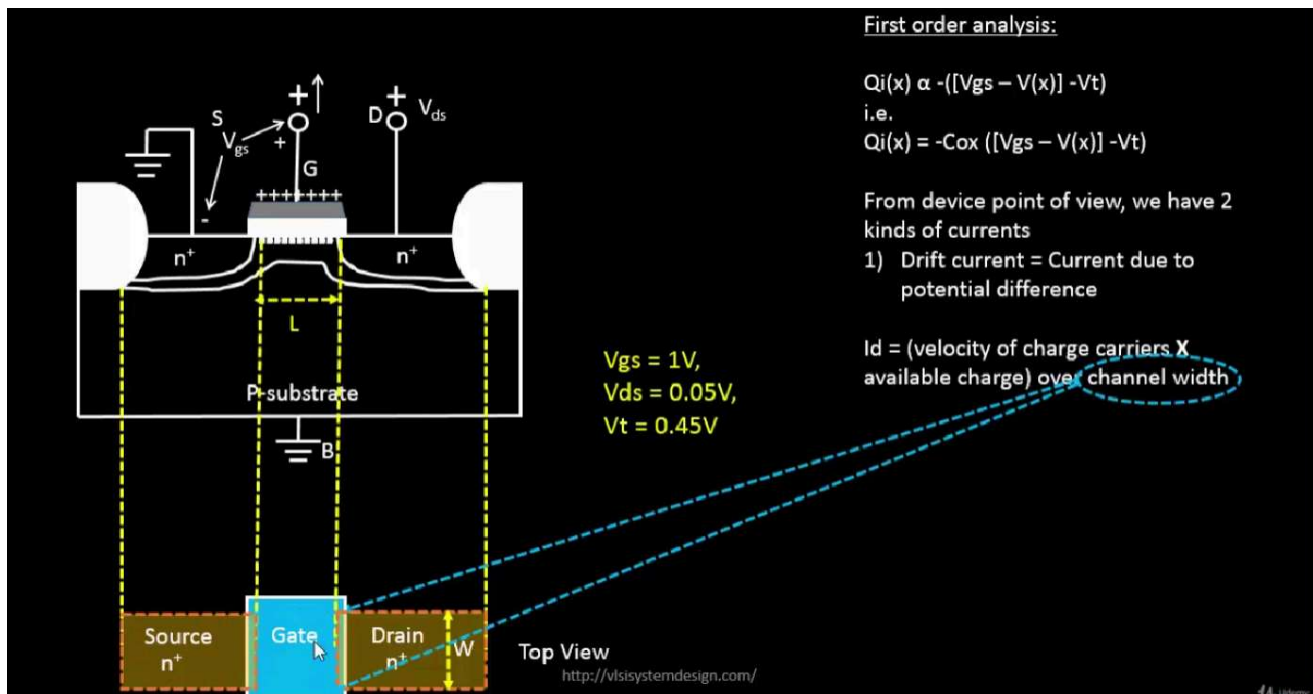
- Threshold voltage V_T represents the minimum voltage required to form inversion.
- Only the voltage **above** V_T contributes to useful channel charge.
- This expression shows clearly that:
 - Channel charge is **position-dependent**
 - Charge decreases gradually as we move from source to drain

4. Gate Oxide Capacitance and Technology Parameters

- Gate oxide capacitance per unit area is given by:

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

- Where:
 - $\epsilon_{ox} = 3.97, \epsilon_0$
 - t_{ox} is the gate oxide thickness
- t_{ox} is:
 - Fixed for a given technology node
 - Provided by the foundry
- Parameters such as C_{ox} , V_T , and t_{ox} :
 - Are **technology constants**
 - Are directly used in SPICE models
- These expressions form the physical foundation before moving to simulations.



5. Drift Current Concept in NMOS

- From device physics, two current mechanisms exist:
 - Drift current
 - Diffusion current
- In this lecture, the focus is on **drift current**.
- Drift current arises due to:
 - An electric field created by a potential difference
- In the NMOS resistive region:
 - Source is at 0 V
 - Drain is at V_{DS}
 - An electric field exists along the channel
- Electrons drift from source to drain under this electric field, producing drain current.

6. Drain Current Dependence on Channel Charge

- Drain current is fundamentally related to:
 - Amount of charge present in the channel
 - Velocity of charge carriers
 - Channel dimensions
- Conceptually:

$$I_D \propto (\text{channel charge}) \times (\text{carrier velocity}) \times (\text{channel width})$$

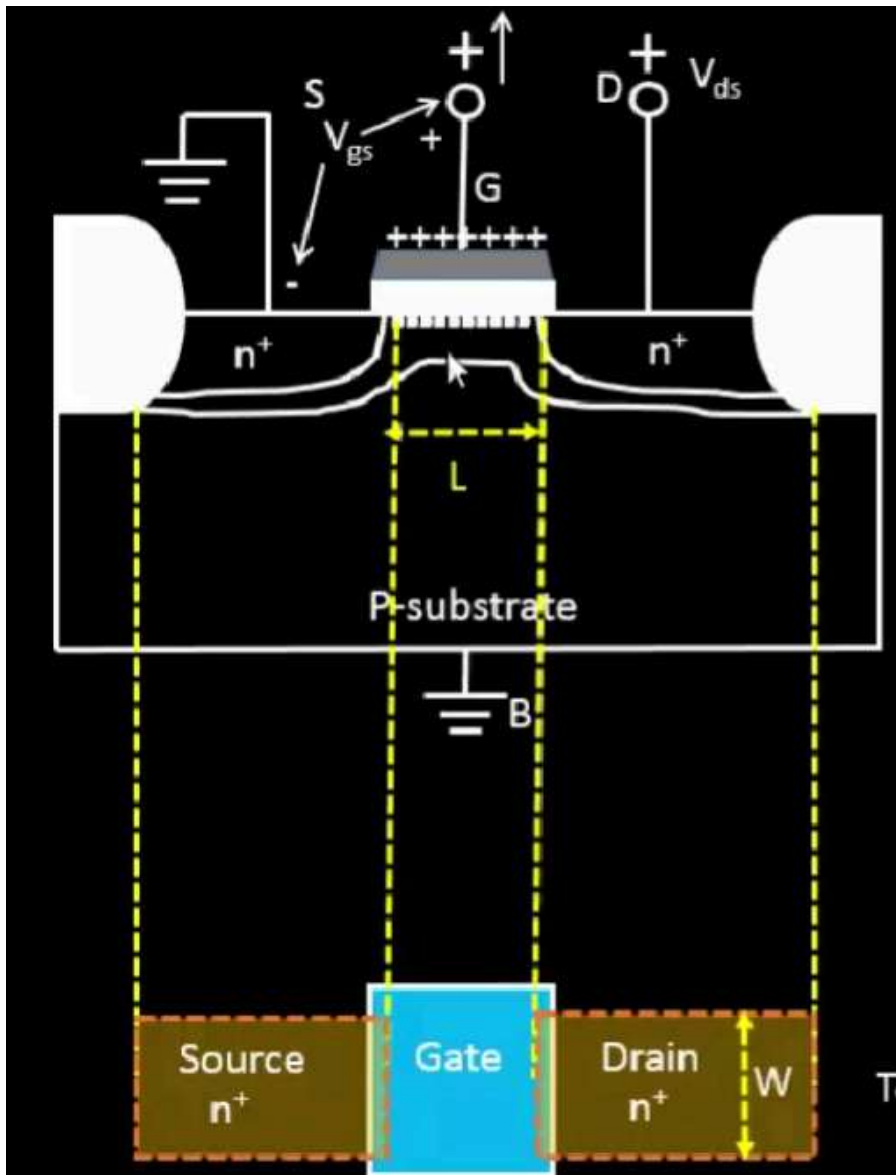
- Channel width W is observed from the **top view** of the MOSFET.
- Channel length L is the distance between source and drain.
- To obtain the final drain current equation:
 - Channel charge expression is combined with carrier velocity
 - Resulting expression is integrated along the channel length
- This derivation leads directly to the **NMOS resistive-region current equation**, which is used in circuit design and delay analysis.

Lecture 3: Drain Current Model for Linear (Resistive) Region of Operation

1. Objective and Physical Basis of Drain Current Model

- Drain current in NMOS is physically understood as the product of:
 - **Velocity of charge carriers**
 - **Available inversion charge in the channel**
- Goal of this lecture:
 - Develop a **simple, first-order drain current model**
 - Suitable for understanding and later use in **SPICE simulations**
- Although SPICE uses more accurate and complex models, this basic model:
 - Explains the underlying physics
 - Forms the foundation for advanced models

2. Channel Geometry and Voltage Distribution



*(NMOS cross-section showing effective channel length L and width W)

- Channel is the gate-overlap region between source and drain.
- Effective channel length is denoted as L .
- Channel width W is visible only in the **top view** of the MOSFET.
- With a small drain-source voltage V_{DS} applied:
 - Source end of channel is at 0 V
 - Drain end of channel is at V_{DS}
- Channel potential varies gradually along the length.

Define:

$$V(x) = \text{Channel voltage at position } x$$

Boundary conditions:

$$V(0) = 0$$

$$V(L) = V_{DS}$$

3. Induced Channel Charge Along the Channel

Inversion charge varying along the channel due to $V(x)$

- Effective gate-to-channel voltage at any point x :

$$V_{GC}(x) = V_{GS} - V(x)$$

- Induced inversion charge per unit area:

$$Q_i(x) = -C_{ox} (V_{GS} - V(x) - V_T)$$

- Negative sign indicates electron charge.
- C_{ox} is gate oxide capacitance per unit area:

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

- t_{ox} and ϵ_{ox} are **technology constants** provided by the foundry.
- Channel charge:
 - Maximum near the source
 - Minimum near the drain

4. Drift Current Expression and Integration

- Drift current arises due to the electric field along the channel.
- Electron velocity is proportional to the electric field:

$$v_n(x) = \mu_n \frac{dV}{dx}$$

- Drain current is given by:

$$I_D = -v_n(x) Q_i(x) W$$

- Substituting velocity and charge expressions:

$$I_D dx = \mu_n C_{ox} W (V_{GS} - V(x) - V_T) dV$$

- Integrate:
 - x from 0 to L
 - V from 0 to V_{DS}

Resulting expression:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

5. Linear Region Approximation and Final Model

- Define process transconductance parameter:

$$k'_n = \mu_n C_{ox}$$

- Drain current equation becomes:

$$I_D = k'_n \frac{W}{L} \left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2} \right]$$

- For **small** V_{DS} , such that:

$$V_{DS} \ll (V_{GS} - V_T)$$

- The quadratic term can be neglected:

$$\frac{V_{DS}^2}{2} \approx 0$$

- Final **linear-region drain current model**:

$$I_D = k_n (V_{GS} - V_T) V_{DS}$$

Where:

$$k_n = k'_n \frac{W}{L}$$

6. Linear Region Condition and Key Observations

- NMOS operates in **resistive (linear) region** when:

$$\begin{aligned} V_{GS} &> V_T \\ V_{DS} &< (V_{GS} - V_T) \end{aligned}$$

- In this region:
 - Drain current is **linear with** V_{DS}
 - NMOS behaves like a **voltage-controlled resistor**
- Model parameters k'_n , V_T , W , and L :
 - Are supplied through **foundry SPICE model files**
- This model is widely used for:
 - Hand analysis
 - Understanding delay
 - Initial circuit design intuition

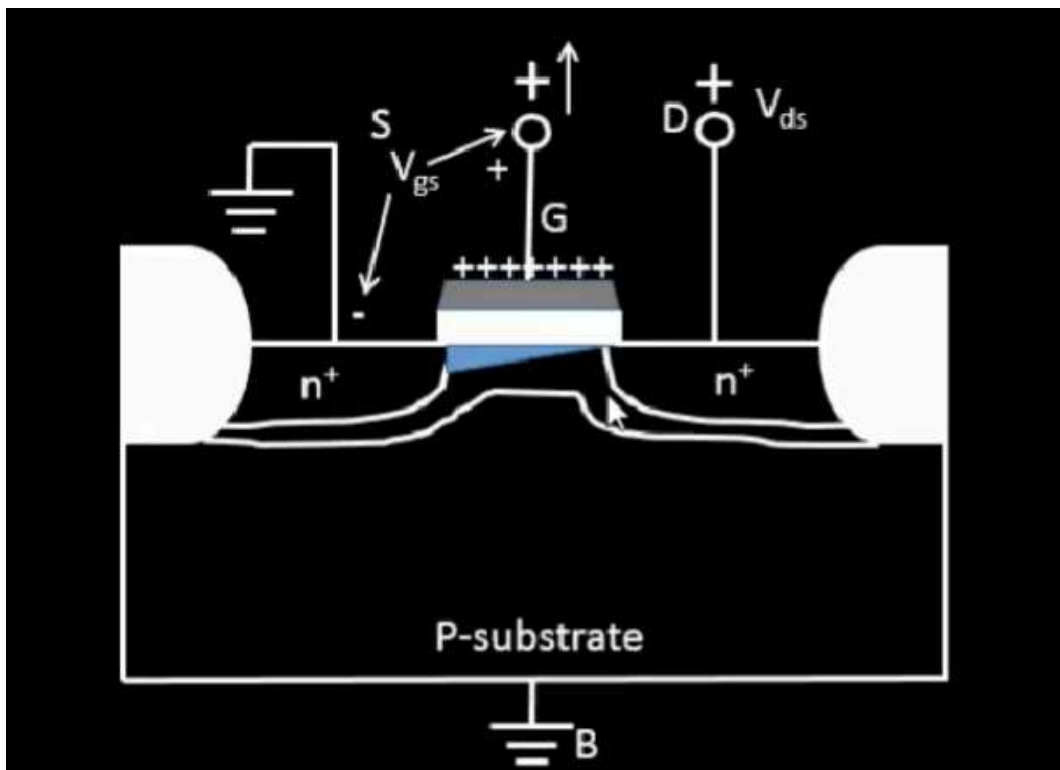
Lecture 4 & Lecture 5

SPICE Conclusion to Resistive Operation and Pinch-Off Condition

1. Motivation for SPICE-Based Analysis

- From earlier lectures, we derived a linear-region drain current equation valid when:
 - $V_{GS} > V_T$
 - $V_{DS} < (V_{GS} - V_T)$
- Manually calculating I_D for:
 - Multiple values of V_{GS}
 - Sweeping V_{DS} from 0 to $(V_{GS} - V_T)$ is impractical and time-consuming.
 - Practical circuit design therefore relies on **SPICE simulators**, which:
 - Use compact models
 - Automatically sweep voltages
 - Compute drain current accurately using foundry-provided parameters
- Before running SPICE, it is essential to **physically understand** what happens as V_{DS} increases.

2. Channel Voltage Concept with Increasing V_{DS}



(NMOS cross-section showing channel and definition of $V(x)$)

- Channel voltage at any point x is denoted as $V(x)$.
- Effective gate-to-channel voltage at that point is:

$$V_{GC}(x) = V_{GS} - V(x)$$

- For a fixed $V_{GS} = 1 \text{ V}$ and $V_T = 0.45 \text{ V}$:
 - When V_{DS} is small, $V_{GC}(x)$ is **greater than** V_T **everywhere**
 - A continuous inversion channel exists from source to drain
- Channel voltage at the drain end is:

$$V_{GC, \text{ drain}} = V_{GS} - V_{DS}$$

- As long as:

$$V_{GS} - V_{DS} > V_T$$

The channel remains intact across the full length.

3. Sweep of V_{DS} and Channel Validity

Vgs	Vds	Vgs - Vds		Vt
1	0.05	0.95	>	0.45
1	0.15	0.85	>	0.45
1	0.25	0.75	>	0.45
1	0.35	0.65	>	0.45
1	0.45	0.55	>	0.45
1	0.55	0.45	=	0.45
1	0.65	0.35	<	0.45
1	0.75	0.25	<	0.45
1	0.85	0.15	<	0.45
1	0.95	0.05	<	0.45

(Table showing V_{GS} , V_{DS} , $V_{GS} - V_{DS}$ compared to V_T)

- Keep $V_{GS} = 1 \text{ V}$ constant.

- Gradually increase V_{DS} from 0.05 V upward.
- Observe:
 - For small V_{DS} , $V_{GS} - V_{DS} > V_T$
 - Channel exists throughout the device

- At the critical point:

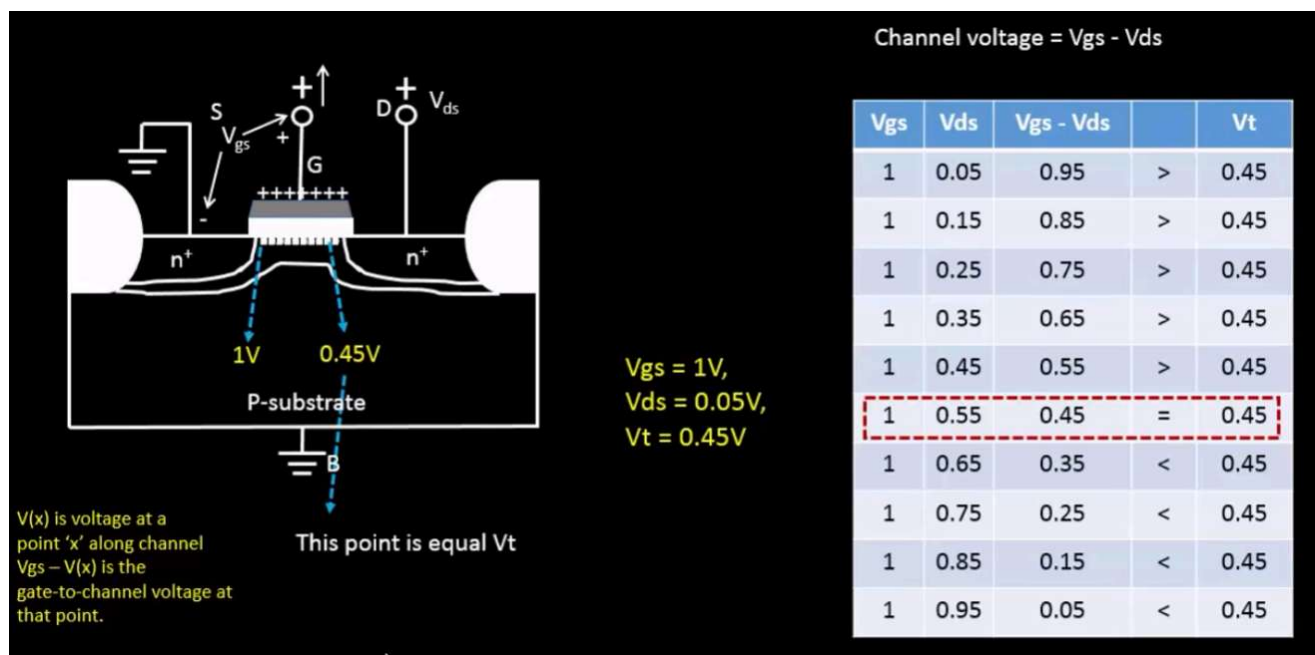
$$V_{DS} = V_{GS} - V_T = 0.55 \text{ V}$$

- Drain-end channel voltage equals threshold voltage:

$$V_{GS} - V_{DS} = V_T$$

- This marks the **onset of channel degradation at the drain side**.

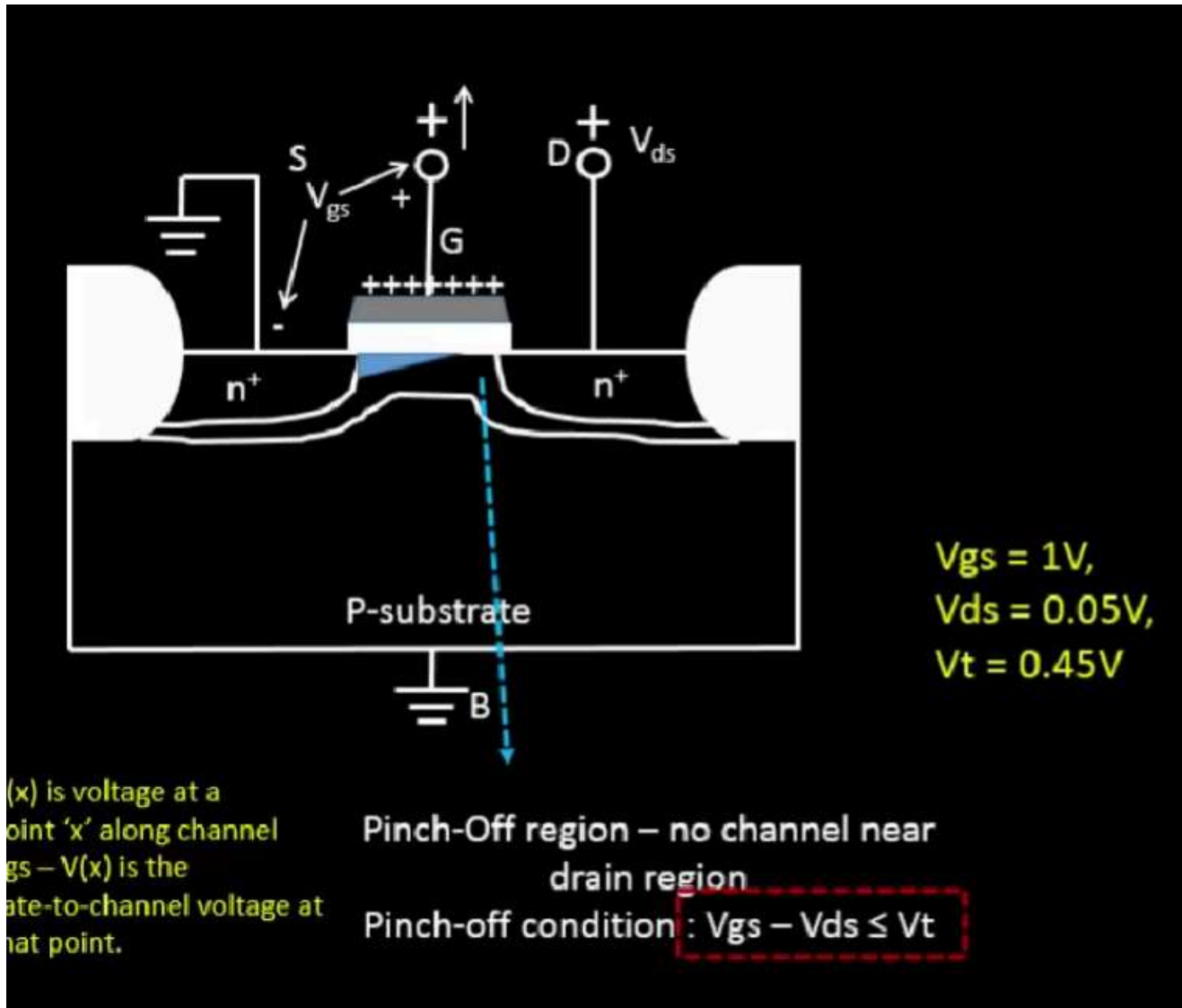
4. Physical Meaning of Pinch-Off Initiation



- Threshold voltage represents the condition for **surface inversion**.
- When $V_{GS} - V_{DS} = V_T$:
 - Drain-end surface is **just at inversion**
 - Any further increase in V_{DS} causes:
 - $V_{GS} - V_{DS} < V_T$ near drain
 - Inversion channel **disappears near the drain**
- This phenomenon is called **pinch-off**.
- Important observation:
 - Channel still exists near the source

- Channel disappears progressively toward the drain
- Pinch-off does **not** mean current stops flowing.

5. Transition from Resistive to Saturation Region



1	0.65	0.35	<	0.45
1	0.75	0.25	<	0.45
1	0.85	0.15	<	0.45
1	0.95	0.05	<	0.45

(Pinch-off region clearly visible near drain)

- Once pinch-off begins:
 - Drain current is no longer linearly dependent on V_{DS}
- Saturation condition is defined as:

$$V_{DS} \geq (V_{GS} - V_T)$$

- Beyond this point:
 - Increasing V_{DS} increases the depletion region near the drain
 - Channel length effectively shortens
 - Drain current becomes **almost constant**
- This operating mode is called the **saturation region**.

6. Key Conclusions and Link to SPICE Models

- Summary of regions:
 - **Resistive (Linear) Region:**

$$V_{GS} > V_T, \quad V_{DS} < (V_{GS} - V_T)$$

- **Pinch-Off / Saturation Region:**

$$V_{DS} \geq (V_{GS} - V_T)$$

- SPICE simulators:
 - Automatically detect region transitions
 - Switch between linear and saturation current models
- Foundry-provided model parameters ensure:
 - Accurate current computation
 - Reliable delay and timing analysis
- Next step:
 - Derive the **saturation-region drain current equation**
 - Validate behaviour through SPICE simulations

Lecture 6: Drain Current Model for Saturation Region of Operation

1. Channel Voltage Behaviour in Saturation

- In the linear region, channel voltage varies as:

$$V_{GC}(x) = V_{GS} - V(x)$$

- In saturation:
 - Channel voltage can no longer increase beyond $(V_{GS} - V_T)$.

- The channel voltage effectively becomes **constant**.
- Therefore, instead of using V_{DS} directly, we replace it with:

$$V_{DS} = V_{GS} - V_T$$
- This substitution reflects the physical fact that:
 - Any additional V_{DS} drops across the **drain depletion region**, not the channel.

2. Drain Current Equation in Saturation (Ideal Case)

- Starting from the linear-region drain current equation:

$$I_D = k_n \left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2} \right]$$

- Substitute:

$$V_{DS} = V_{GS} - V_T$$

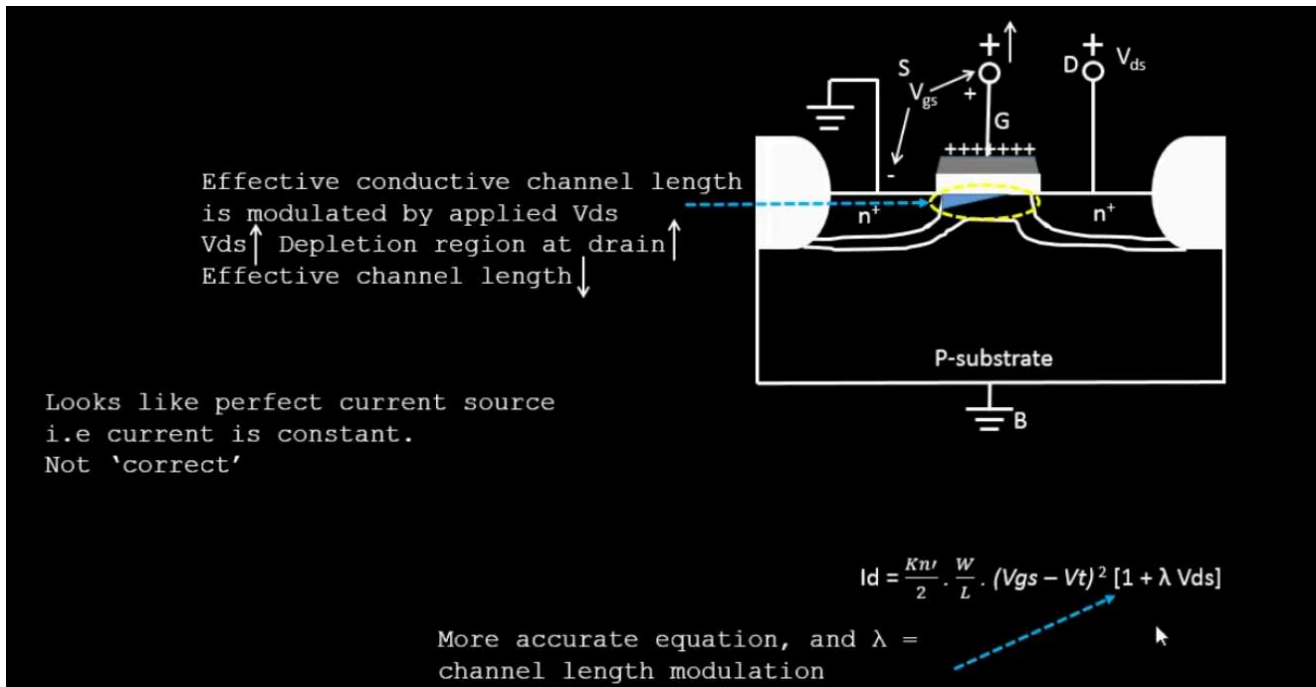
- The drain current simplifies to:

$$I_D = \frac{k_n}{2} (V_{GS} - V_T)^2$$

- Key observations:
 - Drain current is **independent of V_{DS}**
 - Drain current depends only on:
 - V_{GS}
 - V_T
 - Device parameters (W / L)
- This gives the **ideal saturation current model**.

3. Why Saturation Current Is Not Truly Constant

- The ideal model suggests NMOS behaves like a **perfect current source**.
- In reality:
 - Increasing V_{DS} increases the **drain-side depletion region**.
 - This reduces the **effective channel length**.
- As effective channel length decreases:
 - Drain current increases slightly with V_{DS} .
- This phenomenon is known as **channel length modulation**.



Reduction of effective channel length as V_{DS} increases

4. Saturation Current with Channel Length Modulation

- To model non-ideal behaviour, introduce channel length modulation factor λ .
- Modified saturation current equation:

$$I_D = \frac{k_n}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

- Where:
 - λ = channel length modulation parameter
 - Supplied by the foundry in SPICE model files
- Effects of λ :
 - Drain current increases **slightly linearly** with V_{DS}
 - Output resistance becomes finite (not infinite)

Chapter 3: Introduction to SPICE

Lecture 1: Basic SPICE Setup (Linking Device Physics to Simulation)

1. Context from Saturation Region Operation

- In the previous chapter, NMOS behaviour in the **saturation region** was established.
- Saturation occurs when:

$$V_{GS} - V_{DS} \leq V_T$$

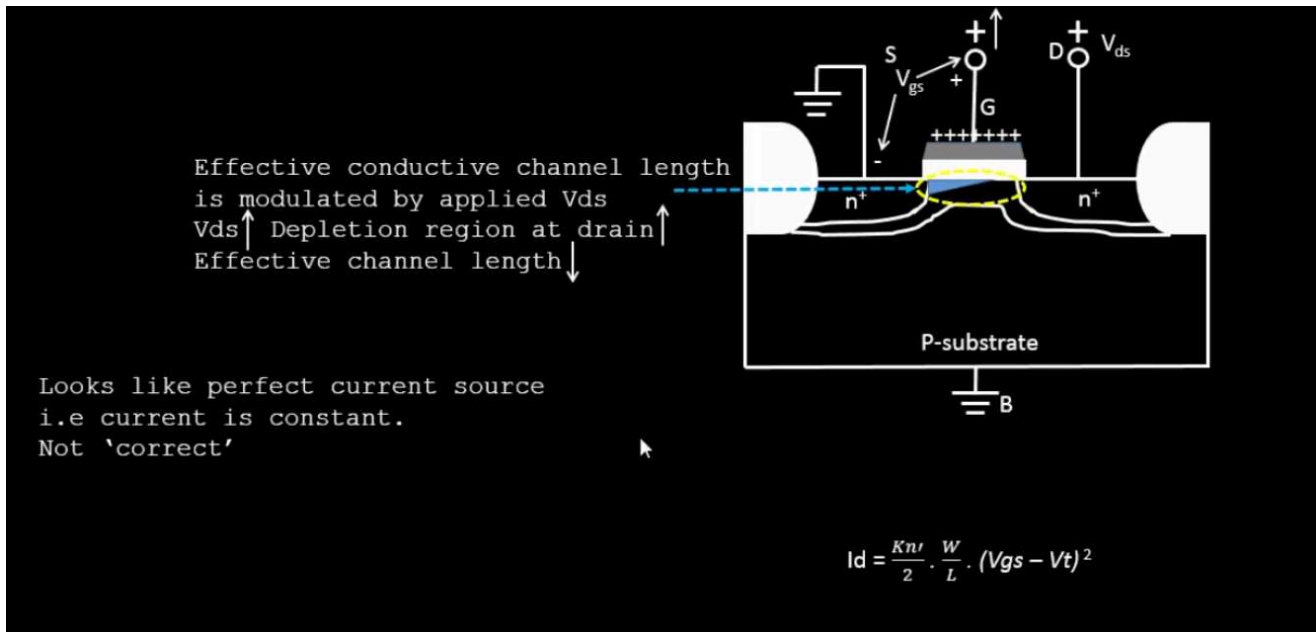
- At this condition:
 - Channel begins to disappear near the drain.
 - Channel voltage no longer follows $V_{GS} - V_{DS}$ everywhere.
- This creates an apparent contradiction:
 - Earlier, channel voltage depended on V_{DS} .
 - Now, channel voltage appears fixed.
- Resolution:
 - In saturation, the **channel voltage clamps to** $(V_{GS} - V_T)$.
 - Any extra V_{DS} drops across the drain-side depletion region.

2. Channel Voltage in Saturation Region

- In linear region:
 - Channel voltage varies along the channel length.
- In saturation region:
 - Channel voltage becomes approximately constant and equal to:

$$V_{channel} = V_{GS} - V_T$$

- Reason:
 - Drain-end inversion condition is lost.
 - Additional drain voltage no longer modulates channel charge.
- This simplifies current modelling because:
 - Drain current no longer strongly depends on V_{DS} .



Channel pinch-off near the drain and constant channel voltage

3. Saturation Region Drain Current Model (Ideal)

- Start from the linear-region drain current equation:

$$I_D = k_n \left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2} \right]$$

- At saturation boundary, substitute:

$$V_{DS} = V_{GS} - V_T$$

- Drain current simplifies to:

$$I_D = \frac{k_n}{2} (V_{GS} - V_T)^2$$

- Observations:

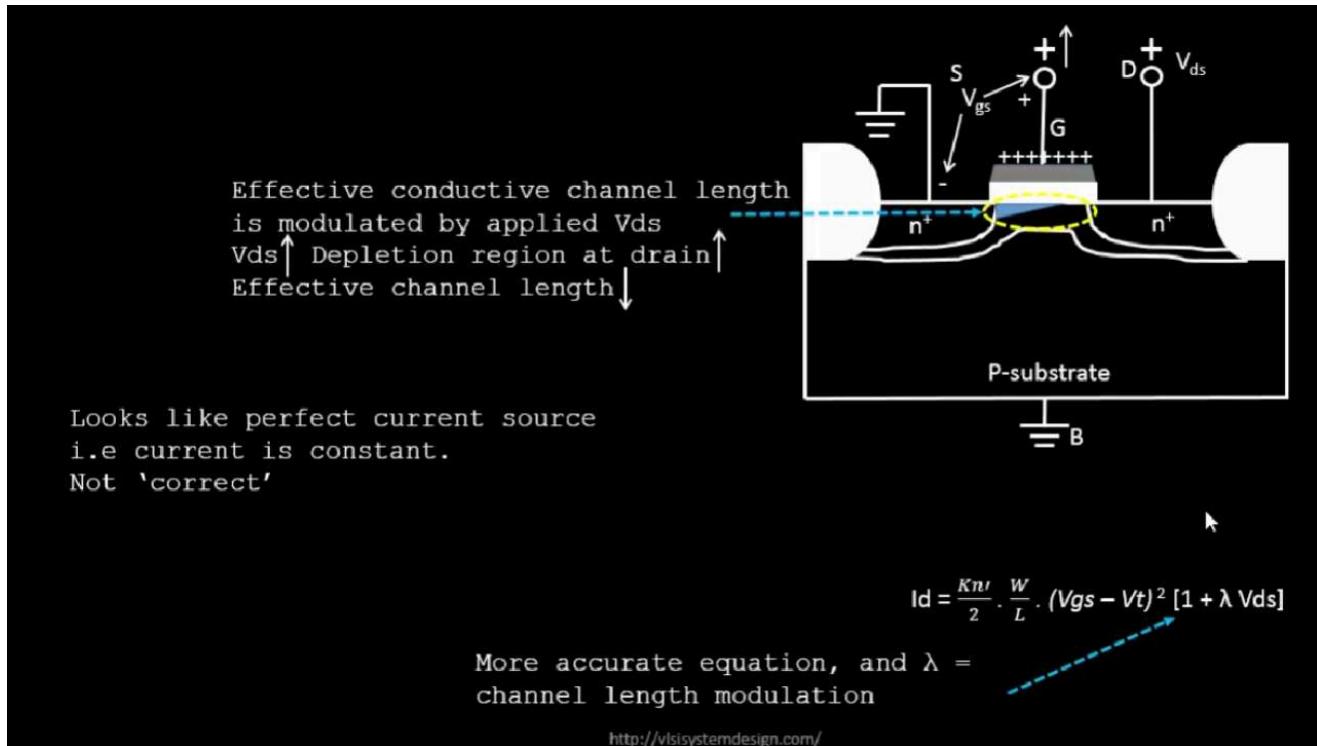
- Drain current depends only on V_{GS} and V_T .
- Drain current appears independent of V_{DS} .

- This gives the **ideal saturation current model**.

4. Why Ideal Saturation Is Not Fully Accurate

- The ideal model suggests NMOS behaves like a perfect current source.
- In practice:
 - Increasing V_{DS} enlarges the drain-side depletion region.
 - Effective conductive channel length reduces.
- This reduction causes:

- A small increase in drain current with increasing V_{DS} .
- Hence, drain current is **not truly constant** in saturation.



Reduction of effective channel length with increasing V_{DS}

5. Channel Length Modulation and Improved Model

- To account for non-ideal behaviour, introduce channel length modulation factor λ .
- Modified saturation current equation:

$$I_D = \frac{k_n}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

- Where:
 - λ models dependency of current on V_{DS} .
 - λ is provided in foundry SPICE model files.
- Effects:
 - Drain current increases slightly with V_{DS} .
 - Output resistance becomes finite.

6. Motivation for SPICE Simulations

- At this stage, three key models are available:
 - i. Threshold voltage model
 - ii. Linear-region drain current model
 - iii. Saturation-region drain current model

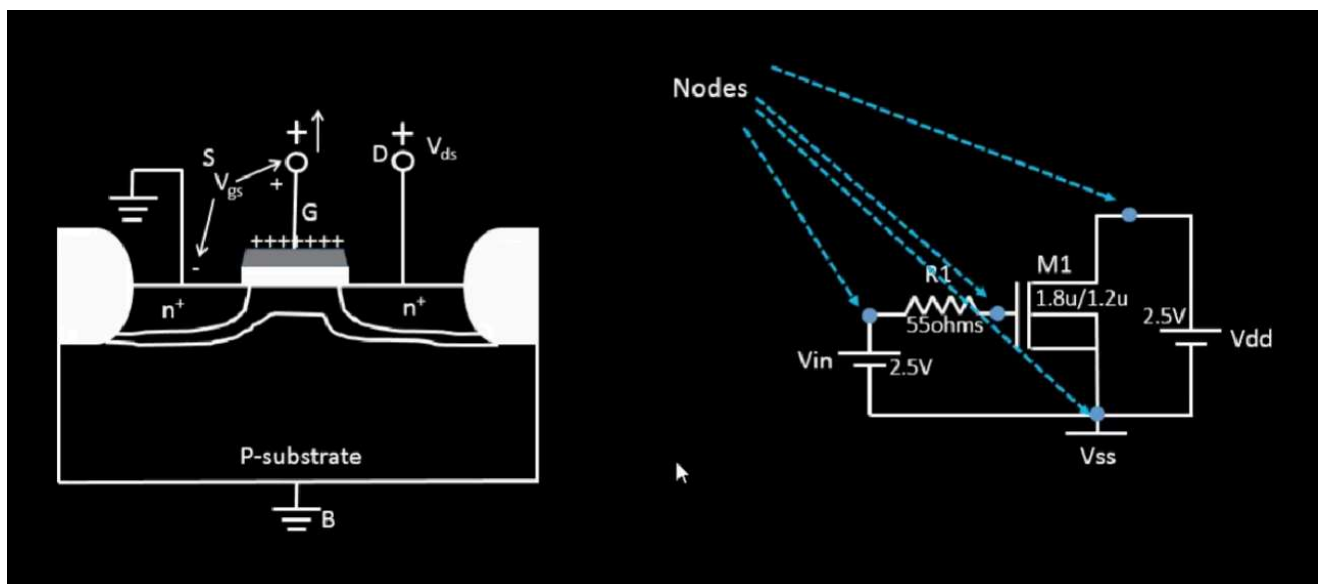
- Manually evaluating these equations for:
 - Multiple V_{GS}
 - Sweeping V_{DS}
 - Different technology nodes is impractical.
- SPICE simulators:
 - Use these equations internally
 - Take foundry-provided parameters
 - Automatically generate $I_D - V_{DS}$ curves
- Next step:
 - Build a basic SPICE setup
 - Understand required parameters
 - Run simulations for different technology nodes

Lecture 2: Circuit Description in SPICE Syntax (NMOS Example)

1. Objective of This Lecture

- Convert a simple NMOS circuit into a valid SPICE netlist.
- Understand how SPICE interprets a circuit using nodes and components.
- Learn the correct syntax and pin ordering used by SPICE for MOSFETs, resistors, and voltage sources.
- Prepare the circuit description so it can be simulated using a technology (model) file.

2. Identifying and Defining Nodes



- In SPICE, a node is a continuous electrical connection with no obstruction in between.

- Any wire segment without a component in between is treated as one node.
- Nodes are identified first before writing the netlist.

For the given circuit:

- Ground node is always named `0` in SPICE.
- The drain supply node is named `VDD`.
- The gate input node is named `IN`.
- The intermediate node between the resistor and the MOSFET drain is named `N1`.

Summary of nodes:

- `0` → Ground (VSS)
- `VDD` → Drain supply node
- `IN` → Gate input node
- `N1` → Drain node of MOSFET and resistor connection

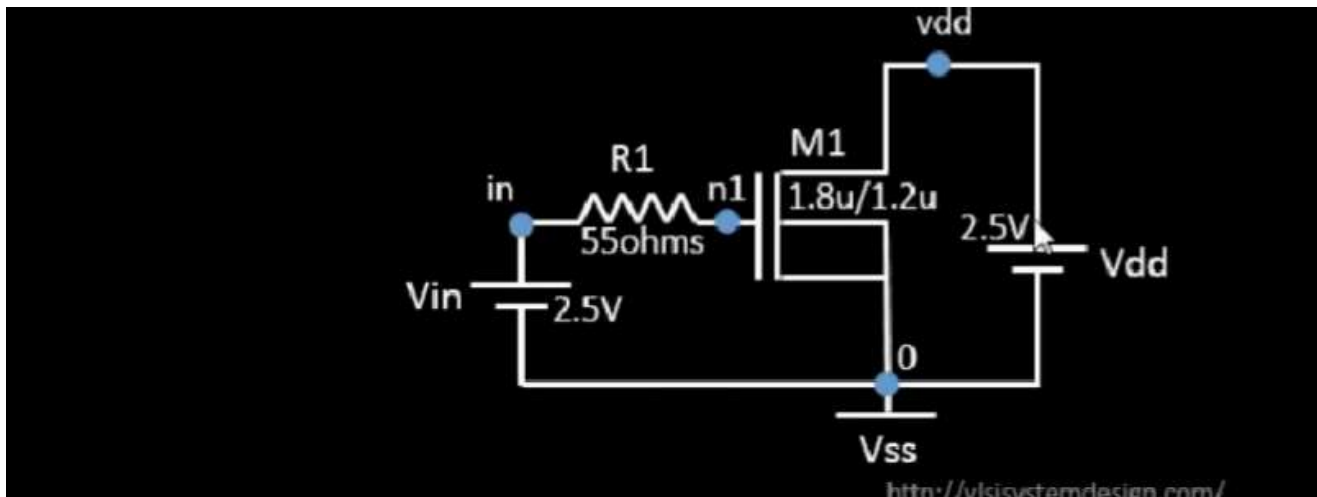
These node names can be chosen freely, but must be used consistently.

3. MOSFET Declaration in SPICE

- MOSFETs are declared using the letter `M`.
- General syntax:

`M W= L=`

- Pin order is fixed and very important:
 - Drain
 - Gate
 - Source
 - Bulk (Body)



For this circuit:

- MOSFET name: `M1`

- Drain node: N1
- Gate node: IN
- Source node: 0
- Bulk node: 0
- Model name: NMOS
- Width = 0.8 micron
- Length = 1.2 micron

SPICE line:

```
M1 N1 IN 0 0 NMOS W=0.8u L=1.2u
```



This line fully describes the NMOS device and connects it electrically to the circuit.

4. Resistor Definition

- Resistors start with the letter R .
- Syntax:

R

For the circuit:

- Resistor name: R1
- Connected between nodes VDD and N1
- Resistance value: 55 ohms

SPICE line:

```
R1 VDD N1 55
```

5. Voltage Source Definitions

- Voltage sources start with the letter V .
- Syntax:

V <positive_node> <negative_node>

Drain supply (VDD):

- Name: VDD
- Positive node: VDD
- Negative node: 0
- Voltage: 2.5 V

```
VDD VDD 0 2.5
```

Gate input voltage (VIN):

- Name: VIN
- Positive node: IN
- Negative node: 0
- Voltage: 2.5 V

VIN IN 0 2.5

6. Complete SPICE Netlist (So Far)

Putting everything together:

```
M1 N1 IN 0 0 NMOS W=0.8u L=1.2u R1 VDD N1 55 VDD VDD 0 2.5 VIN IN 0 2.5
```

- This netlist fully represents the NMOS circuit topology.
- At this stage, SPICE understands the circuit connectivity but not the device physics.
- The electrical behaviour will only be defined after adding the technology (model) file.

Lecture 3: Circuit Description in SPICE Syntax and Technology Parameters

1. Goal of This Lecture

- The objective is to translate a physical NMOS circuit into a SPICE-readable netlist.
- SPICE does not understand schematics or drawings; it only understands text-based descriptions called netlists.
- A valid simulation requires three things:
 - Correct circuit connectivity using nodes
 - Correct device declarations (NMOS, resistors, sources)
 - A technology (model) file that provides device parameters

SPICE netlist

Technology file

```
M1 vdd n1 0 0 nmos W=1.8u L=1.2u
R1 in n1 55
Vdd vdd 0 2.5
Vin in 0 2.5
```

```
.lib cmos_models
.MODEL nmos NMOS (TOX = ..
+ VTH0 = .. U0 = .. GAMMA1 = ..)
.MODEL pmos PMOS (TOX = ..
+ VTH0 = .. U0 = .. GAMMA1 = ..)
.endl
```



Package in a file

xxxx_025um_model.mod

2. Node Identification and Naming

- A node is a continuous electrical connection with no obstruction (no component in between).
- All points connected by plain wires belong to the same node and must share the same node name.
- The ground node is always named `0` in SPICE and is mandatory.

Example node identification in this circuit:

- Node `vdd` : Drain supply node connected to VDD source and NMOS drain
- Node `n1` : Gate node between resistor R1 and NMOS gate
- Node `in` : Input node connected to Vin and resistor R1
- Node `0` : Ground node connected to source and substrate

These node names are user-defined and can be numeric or symbolic, but must be used consistently.

3. Writing the SPICE Netlist (Circuit Description)

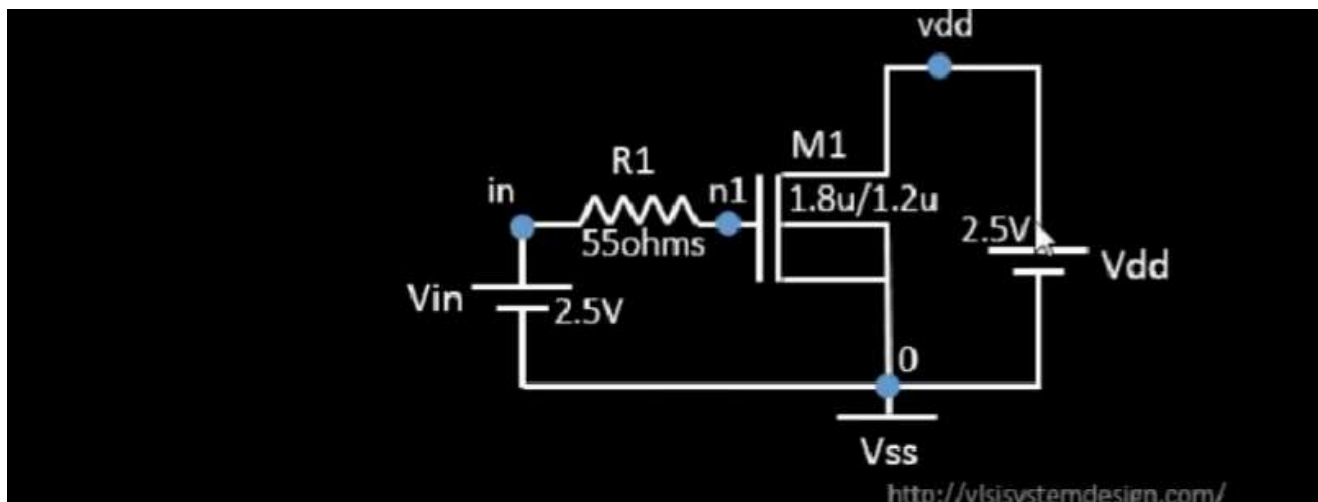
Each component is written as a single line using the format:

- Component name
- Connected nodes
- Component value or model

NMOS Declaration

- MOSFETs start with the letter `M`
- Order of terminals is strictly: Drain Gate Source Substrate (D G S B)

`M1 vdd n1 0 0 nmos W=1.8u L=1.2u`



Explanation:

- `M1` is the MOSFET name
- `vdd` is the drain
- `n1` is a gate
- `0` is the source
- `0` is substrate (body tied to ground)
- `nmos` is the model name
- `W` and `L` are width and length in microns

Resistor Declaration

- Resistors start with `R`

`R1 in n1 55`

- `R1` connects node `in` to node `n1`
- Resistance value is 55 ohms

Voltage Sources

- Voltage sources start with `v`

Vdd vdd 0 2.5 Vin in 0 2.5

- First node is a positive terminal
- The Second node is the negative terminal
- Last value is DC voltage in volts

4. Role of the Technology (Model) File

- The SPICE engine does not compute NMOS behaviour using geometry alone.
- It requires a model file containing physical and process parameters.
- These parameters come from the foundry and depend on the technology node (1.2 μm , 350 nm, etc.).

Key model parameters include:

- V_{TO} : Threshold voltage at $V_{SB} = 0$
- γ : Body effect coefficient
- K_P or k_n : Process transconductance
- λ : Channel length modulation
- t_{ox} : Oxide thickness
- μ_0 : Carrier mobility

These parameters allow SPICE to internally evaluate:

- Threshold voltage equation
- Linear-region drain current
- Saturation-region drain current

5. Defining NMOS and PMOS Models

Models are defined using the `.MODEL` statement.

```
.MODEL nmos NMOS (VTO=0.45 GAMMA=0.4 KP=120u LAMBDA=0.02) .MODEL pmos PMOS (...)
```

Important rules:

- The model name `nmos` must exactly match the name used in the MOSFET line (`M1 ... nmos`).
- Any mismatch between the model name and the netlist name will cause simulation failure or incorrect results.
- NMOS and PMOS have separate models, even if only NMOS is used in the circuit.

6. Including the Model File in the Netlist

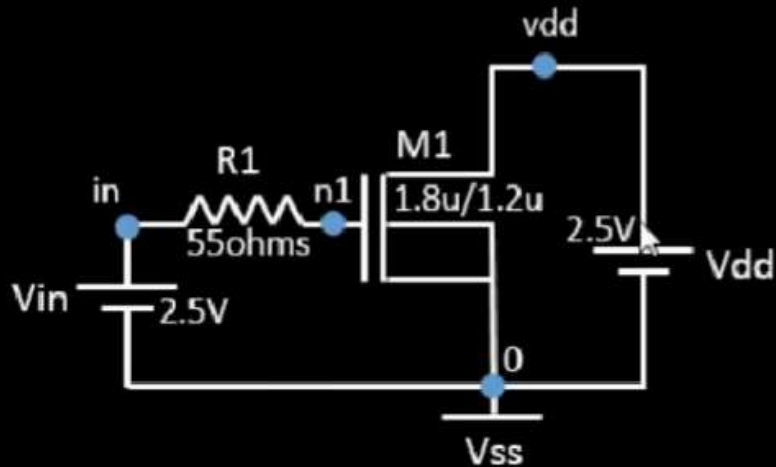
- Model parameters are usually stored in a separate `.mod` file.
- This file is included in the main netlist using `.include` or `.lib`.

SPICE netlist

Simulation Commands

```
*** NETLIST Description ***
M1 vdd n1 0 0 nmos W=1.8u L=1.2u
R1 in n1 55
Vdd vdd 0 2.5
Vin in 0 2.5

*** .include xxxx_1um_model.mod ***
.LIB "xxxx_025um_model.mod" CMOS_MODELS
```



<http://vlsisystemdesign.com/>

Example:

- include xxxx_1um_model.mod
- LIB "xxxx_025um_model.mod" CMOS_MODELS

Structure of a complete SPICE setup:

- Netlist description (devices and connections)
- Model file inclusion
- Simulation commands (to be added next)

Anything starting with * is treated as a comment and ignored by SPICE.

Lecture 4: First SPICE simulation

- You start by opening the terminal and attempting to clone the SKY130 workshop repository, but since it already exists, you move into the existing `sky130CircuitDesignWorkshop/design` directory. After listing the files to confirm the setup, you enter the `sky130_fd_pr` folder, which contains the SKY130 technology data used for simulations.

```

vmsduser@vmsduser:~$ cd
vmsduser@vmsduser:~$ git clone https://github.com/kunalg123/sky130CircuitDesignWorkshop.git
fatal: destination path 'sky130CircuitDesignWorkshop' already exists and is not an empty directory.
vmsduser@vmsduser:~$ cd sky130CircuitDesignWorkshop/design/
vmsduser@vmsduser:~/sky130CircuitDesignWorkshop/design$ ls
day1_nfet_idvds_L2_W5.spice      day3_inv_tran_Wp084_Wn036.spice  day5_inv_devicevariation_wp7_wn042.spice
day2_nfet_idvds_L015_W039.spice  day3_inv_vtc_Wp084_Wn036.spice    day5_inv_supplyvariation_Wp1_Wn036.spice
day2_nfet_idvgs_L015_W039.spice  day4_inv_noisemargin_wp1_wn036.spice sky130_fd_pr
vmsduser@vmsduser:~/sky130CircuitDesignWorkshop/design$ cd sky130_fd_pr/
vmsduser@vmsduser:~/sky130CircuitDesignWorkshop/design/sky130_fd_pr$ ls
cells  LICENSE  models  README.rst
vmsduser@vmsduser:~/sky130CircuitDesignWorkshop/design/sky130_fd_pr$ cd cells/
vmsduser@vmsduser:~/sky130CircuitDesignWorkshop/design/sky130_fd_pr/cells$ ls
nfet_01v8  pfet_01v8
vmsduser@vmsduser:~/sky130CircuitDesignWorkshop/design/sky130_fd_pr/cells$ cd nfet_01v8/
vmsduser@vmsduser:~/sky130CircuitDesignWorkshop/design/sky130_fd_pr/cells/nfet_01v8$ ls
sky130_fd_pr__nfet_01v8__ff.corner.spice  sky130_fd_pr__nfet_01v8__sf.corner.spice
sky130_fd_pr__nfet_01v8__ff.pm3.spice      sky130_fd_pr__nfet_01v8__sf.pm3.spice
sky130_fd_pr__nfet_01v8__fs.corner.spice    sky130_fd_pr__nfet_01v8__ss.corner.spice
sky130_fd_pr__nfet_01v8__fs.pm3.spice      sky130_fd_pr__nfet_01v8__ss.pm3.spice
sky130_fd_pr__nfet_01v8__mismatch.corner.spice sky130_fd_pr__nfet_01v8__tt.corner.spice
sky130_fd_pr__nfet_01v8.pm3.spice          sky130_fd_pr__nfet_01v8__tt.pm3.spice
vmsduser@vmsduser:~/sky130CircuitDesignWorkshop/design/sky130_fd_pr/cells/nfet_01v8$ less sky130_fd_pr__nfet_01v8__tt.pm3.spices

```

- From there, you navigate into the `cells` directory and then into `nfet_01v8`, where the NMOS device models are stored. Listing the contents shows multiple SPICE files corresponding to different process corners such as `tt`, `ss`, `ff`, and others. Opening the typical corner file lets you view the actual model parameters that NGSPICE uses during NMOS simulation, confirming how the device behaviour is defined in SKY130.

```

vmsworkshop [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Activities Terminal Feb 23 14:32

vsduser@vsduser: ~/sky130CircuitDesignWorkshop/design

vsduser@vsduser: ~/sky130CircuitDesignWorkshop/design/s... x vsduser@vsduser: ~/sky130CircuitDesignWorkshop/design x

vsduser@vsduser:~/sky130CircuitDesignWorkshop/design/sky130_fd_pr/cells/nfet_01v8$ cd ../../models/
vsduser@vsduser:~/sky130CircuitDesignWorkshop/design/sky130_fd_pr/models$ ls
all.spice parameters sky130.lib.spice
vsduser@vsduser:~/sky130CircuitDesignWorkshop/design/sky130_fd_pr/models$ cd ../../
vsduser@vsduser:~/sky130CircuitDesignWorkshop/design$ clear

```

From the `nfet_01v8` folder, you go up to the `models` directory and list the files. This shows `sky130.lib.spice`, which serves as the main entry file that links all NMOS and PMOS models across process corners. Including this one file in a netlist gives access to the complete SKY130 device library. After checking this, you return to the design directory and clear the terminal, confirming that the model setup is complete and ready for simulation.

```

vmsworkshop [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Activities Terminal Feb 23 14:45

vsduser@vsduser: ~/sky130CircuitDesignWorkshop/design

vsduser@vsduser: ~/sky130CircuitDesignWorkshop/design x vsduser@vsduser: ~/sky130CircuitDesignWorkshop/design x

Circuit: *model description
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver
Reference value : 0.00000e+00
No. of Data Rows : 190

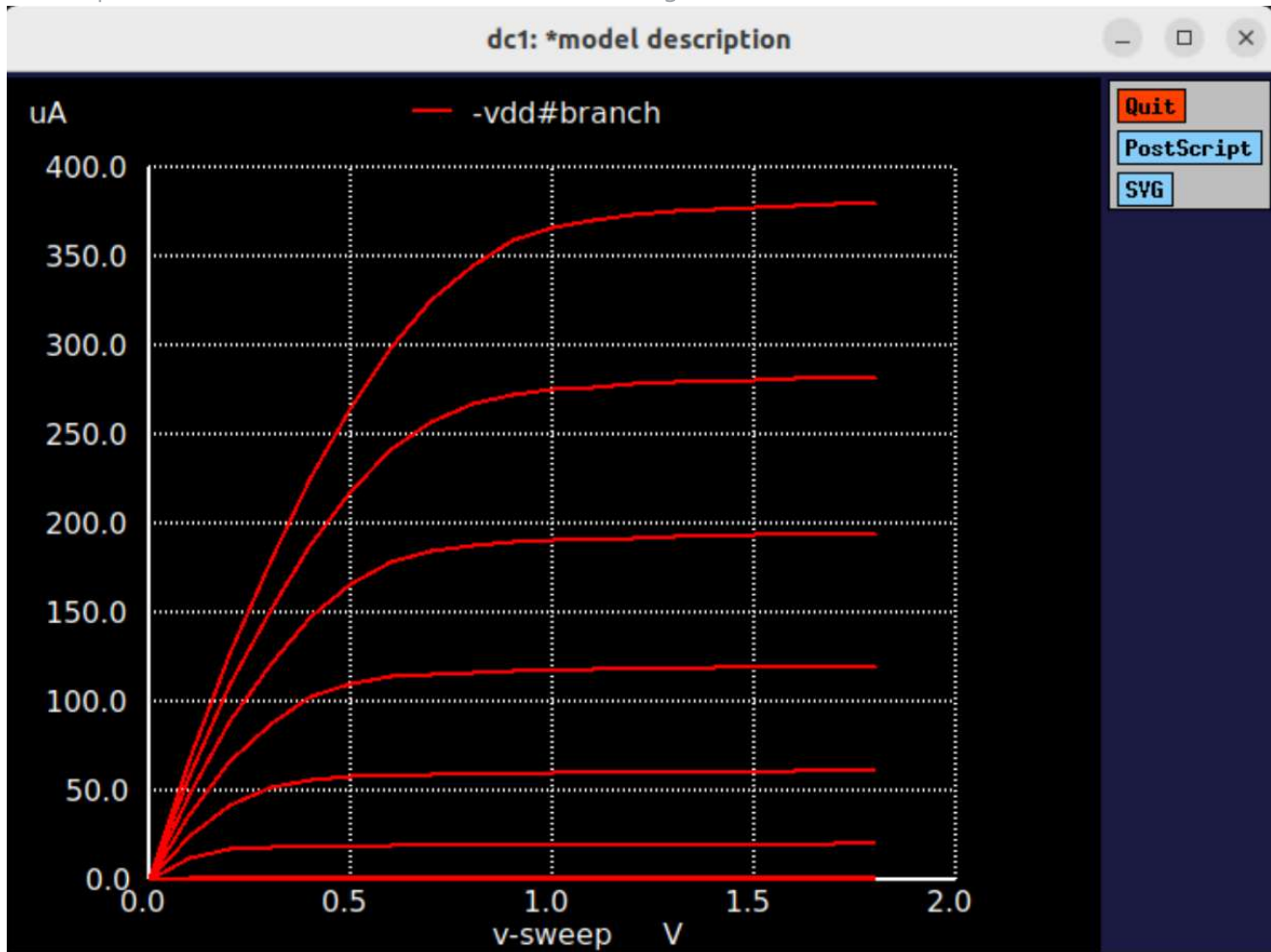
No. of Data Rows : 1
Here are the vectors currently active:

Title: *model description
Name: op1 (Operating Point)
Date: Mon Feb 23 14:44:12 2026

in : voltage, real, 1 long
m.xm1.msky130_fd_pr__nfet_01v8#body: voltage, real, 1 long
m.xm1.msky130_fd_pr__nfet_01v8#dbody: voltage, real, 1 long
m.xm1.msky130_fd_pr__nfet_01v8#sbody: voltage, real, 1 long
n1 : voltage, real, 1 long
vdd : voltage, real, 1 long [default scale]
vdd#branch : current, real, 1 long
vin#branch : current, real, 1 long
ngspice 17 ->

```

After running the simulation, NGSPICE prints the operating-point information and lists all active vectors, such as node voltages and branch currents. You can see that `vdd#branch` and `vin#branch` are available, which means the simulator is ready to report supply and input currents. Using `plot -vdd#branch` opens the DC plot window and shows the drain current flowing from VDD for different bias conditions.



The plot displays a family of I_D - V_{DS} curves as V_{DD} is swept while V_{in} is stepped. At low V_{DS} , the curves rise almost linearly; at higher V_{DS} , they flatten, indicating saturation. By clicking any point on the plot, NGSPICE reports the exact voltage and current values at the terminals, allowing you to read the drain current directly at a specific operating point and understand NMOS behaviour numerically.

```
No. of Data Rows : 190

No. of Data Rows : 1
Here are the vectors currently active:

Title: *model description
Name: op1 (Operating Point)
Date: Mon Feb 23 14:44:12 2026

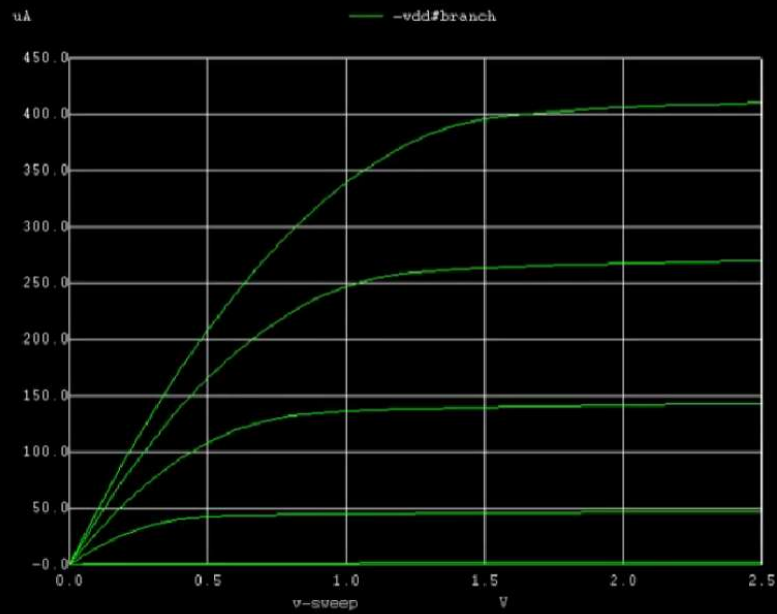
      in                : voltage, real, 1 long
m.xm1.msky130_fd_pr__nfet_01v8#body: voltage, real, 1 long
m.xm1.msky130_fd_pr__nfet_01v8#dbody: voltage, real, 1 long
m.xm1.msky130_fd_pr__nfet_01v8#sbody: voltage, real, 1 long
      n1                : voltage, real, 1 long
      vdd                : voltage, real, 1 long [default scale]
      vdd#branch         : current, real, 1 long
      vin#branch         : current, real, 1 long
ngspice 17 -> plot -vdd#branch
ngspice 18 -> plot -vdd#branch
ngspice 19 ->
x0 = 0.884944, y0 = 0.000189643    x1 = 0.891236, y1 = 0.000188304
dx = 0.00629213, dy = -1.33929e-06
dy/dx = -0.000212851    dx/dy = -4698.13
```

NgspiceSky130 - Day 2 - Velocity saturation and basics of CMOS inverter VTC

Chapter 1: SPICE simulation for lower nodes and velocity saturation effect

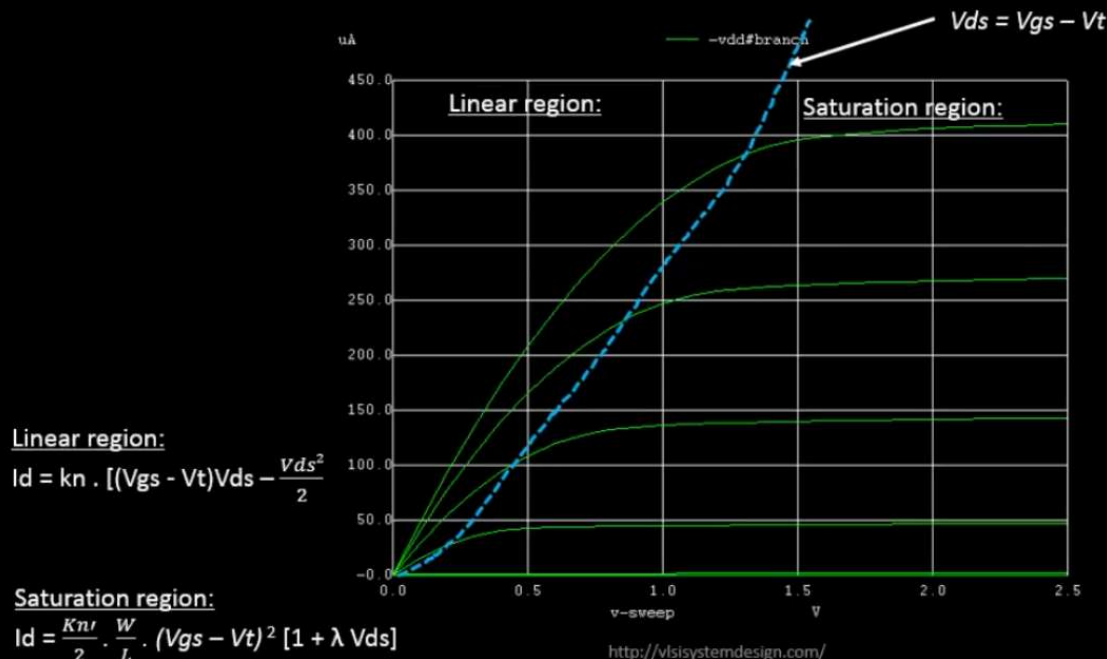
Lecture-1: SPICE simulation for lower nodes

SPICE waveform : W=1.8u, L=1.2u device (W/L = 1.5)

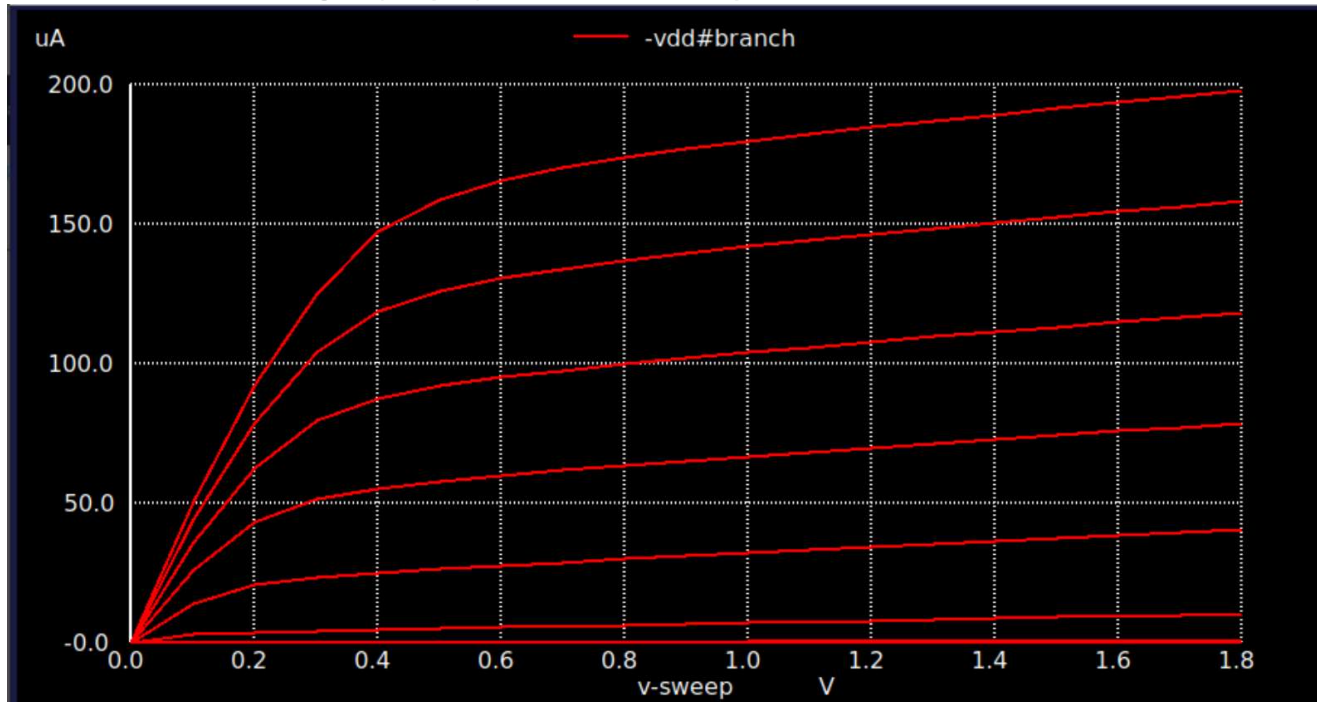


The figures show NMOS ID–VDS characteristics with drain current on the y-axis and drain-to-source voltage on the x-axis for multiple gate voltages. For $W = 1.8 \mu\text{m}$ and $L = 1.2 \mu\text{m}$ ($W/L = 1.5$), the lowest curve corresponds to $V_{GS} = 0 \text{ V}$, where the device is OFF, and the drain current is zero. As V_{GS} exceeds the threshold, a channel forms and the drain current begins to increase, producing a family of curves at higher V_{GS} values.

SPICE waveform : W=1.8u, L=1.2u device (W/L = 1.5)

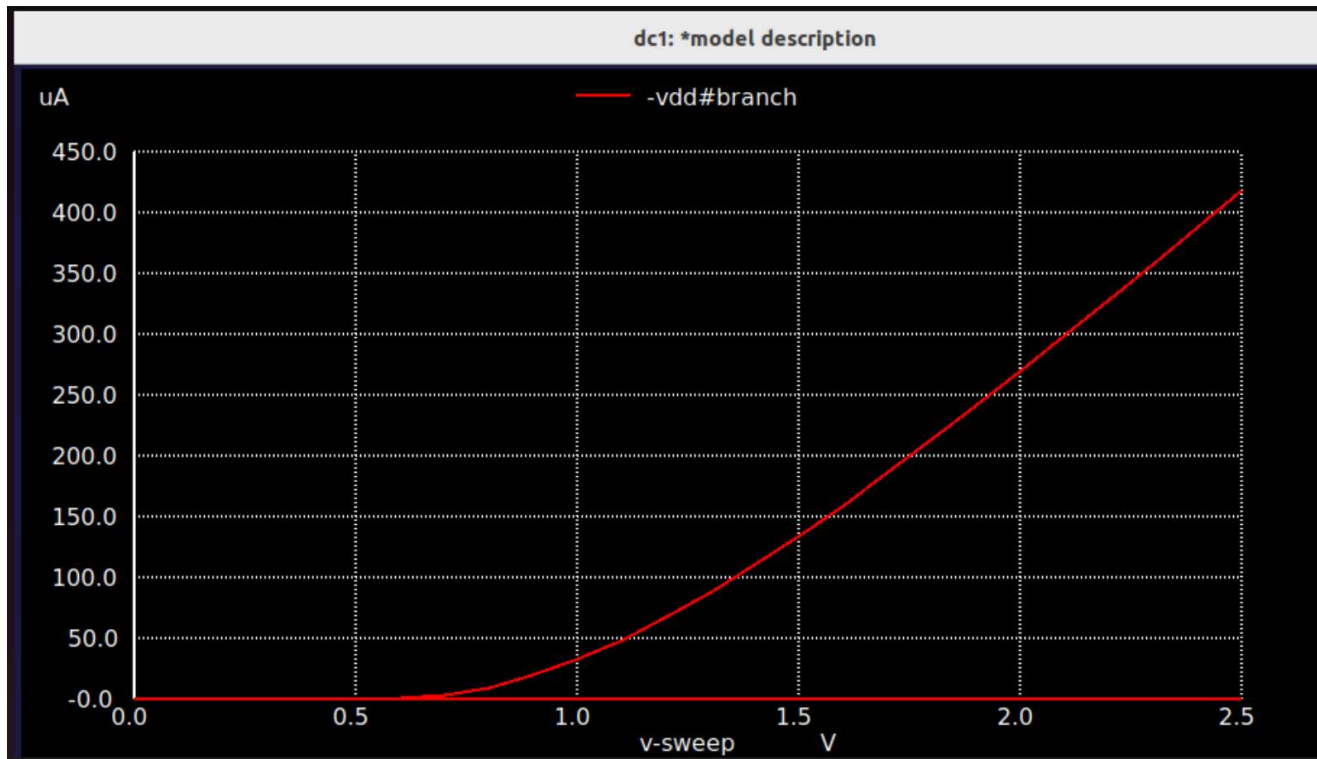


Each curve clearly splits into two regions. At low V_{DS} , the current increases almost linearly with V_{DS} , indicating the **linear (resistive) region**. When V_{DS} reaches approximately $V_{GS} - V_T$, marked in the images by the dashed line, the device enters the **saturation region**, where the channel pinches off, and the drain current flattens, increasing only slightly due to channel-length modulation.



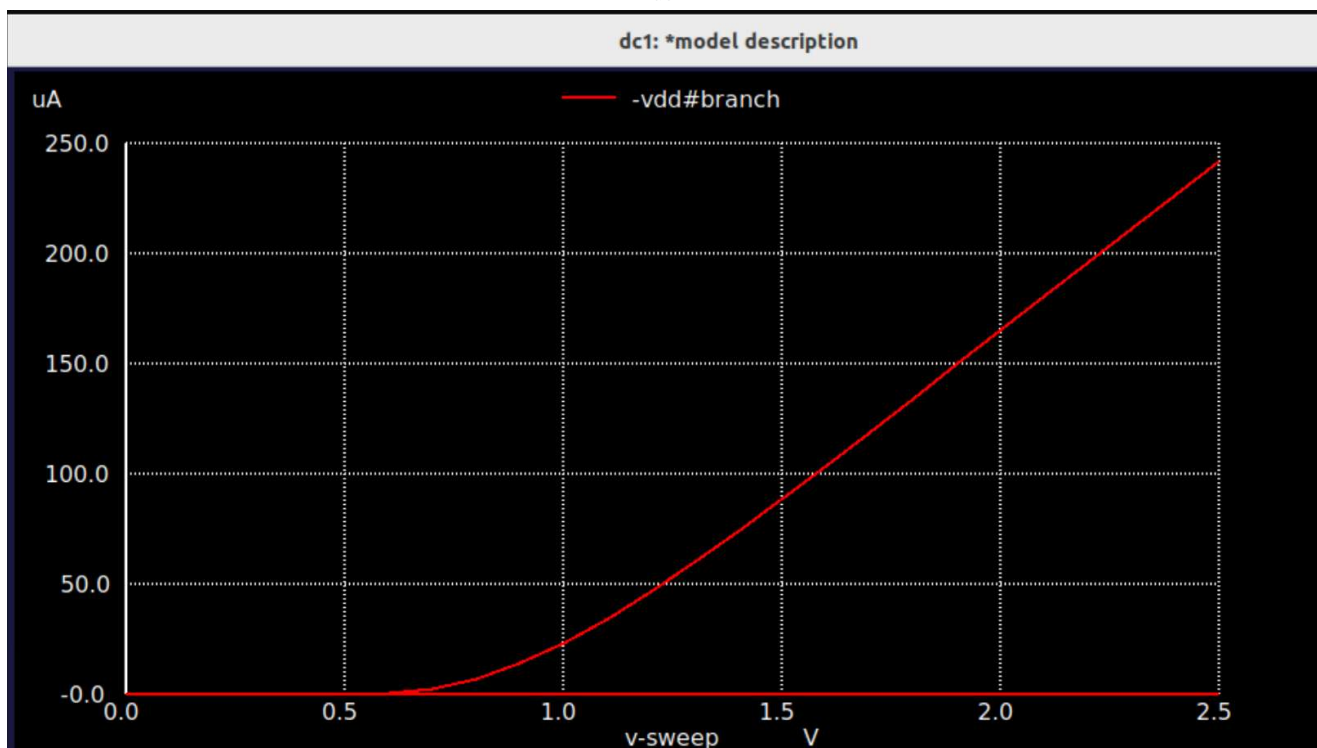
The same behaviour is then compared with a scaled device having $W = 0.375 \mu m$ and $L = 0.25 \mu m$, while keeping $W/L = 1.5$. Although theory suggests the drain current should remain constant for constant W/L , the plots show lower current levels and greater spacing between curves in the smaller device. This difference, clearly visible in the side-by-side images, highlights short-channel effects and shows why long-channel equations no longer accurately predict behaviour at lower technology nodes.

Lecture 2: Drain current vs gate voltage for long and short channel devices



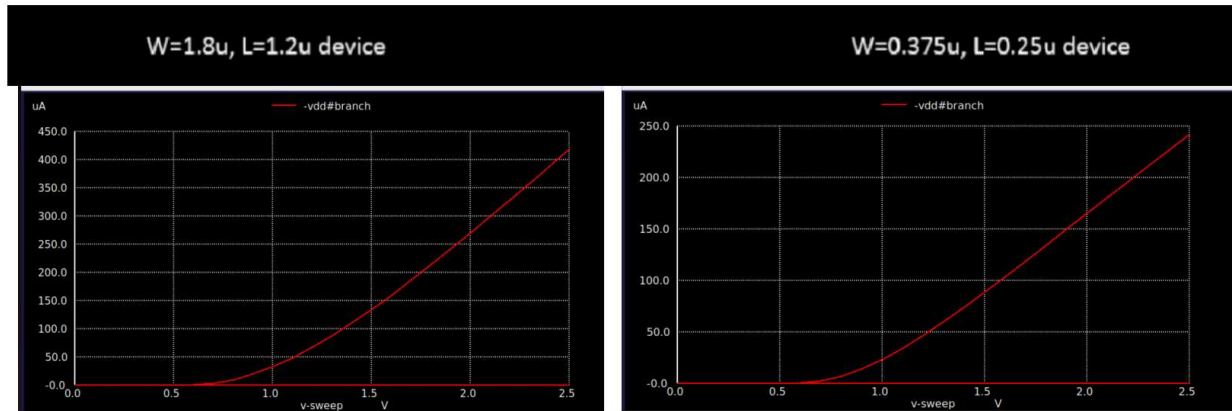
From the I_D – V_{DS} plots, if we fix the drain voltage at $V_{DS} = 2.5$ V and observe the drain current for different gate voltages, the **long-channel device** ($L = 1.2$ μm) shows a **quadratic dependence of I_D on V_{GS}** . At $V_{GS} = 0$ V, the drain current is zero because there is no channel. As V_{GS} increases to 0.5 V, 1 V, and beyond the threshold, the drain current increases nonlinearly, matching the square-law behaviour predicted by the long-channel MOSFET equation.

$$I_D \propto (V_{GS} - V_T)^2$$



For the **short-channel device** ($L = 0.25 \mu\text{m}$), the behaviour initially looks similar at low V_{GS} , where the drain current still follows a quadratic trend. However, as V_{GS} increases further, the I_D – V_{GS} curve becomes **more linear rather than quadratic**, even though V_{DS} is kept constant at 2.5 V. This deviation is clearly visible in the plotted curves and is not expected from long-channel theory.

This difference occurs because **short-channel effects**, especially **velocity saturation**, become dominant at shorter channel lengths. In short-channel devices, carrier velocity saturates before the square-law behaviour can fully develop, leading to a drain current that increases almost linearly with gate voltage at higher V_{GS} .



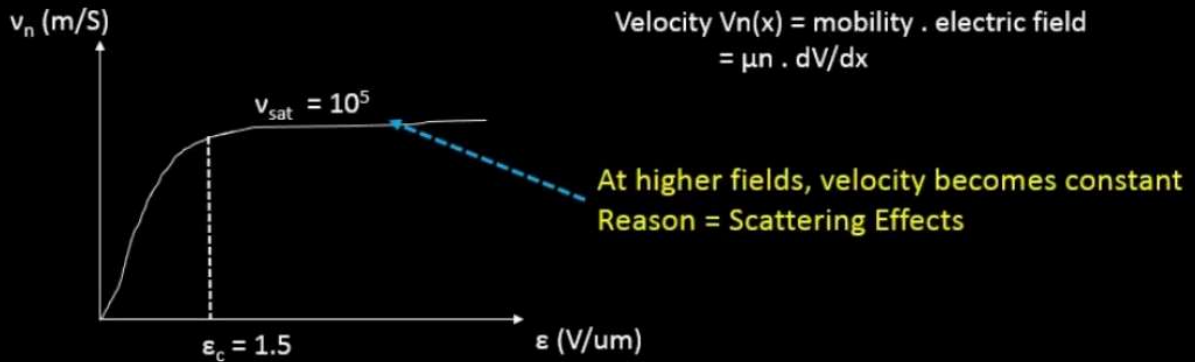
To clearly highlight this effect, the simulation is modified to keep **V_{DS} fixed at 2.5 V** and sweep **V_{GS}** , generating **I_D – V_{GS} curves** for both devices. Comparing these plots side by side confirms that long-channel devices follow quadratic behaviour, while short-channel devices deviate significantly, explaining why long-channel equations fail at advanced technology nodes.

Lecture 3: Velocity saturation at lower and higher electric fields

For long-channel devices ($\approx 1.2 \mu\text{m}$), the drain current shows a clear **quadratic dependence on gate voltage** when the drain voltage is fixed. This follows the long-channel saturation model, where $I_D \propto (V_{GS} - V_T)^2$. At $V_{GS} = 0 \text{ V}$, the device is OFF and $I_D = 0$. As V_{GS} increases beyond the threshold, the drain current rises rapidly in a curved (quadratic) manner, which is clearly visible in the I_D – V_{GS} plots.

For short-channel devices ($\approx 0.25 \mu\text{m}$ and below), the behaviour changes at higher gate voltages. At low V_{GS} , the drain current still follows a quadratic trend. However, as V_{GS} increases further, the I_D - V_{GS} curve becomes **more linear than quadratic**. This deviation shows that long-channel assumptions break down at smaller technology nodes, indicating the presence of short-channel effects.

Velocity Saturation Effect



$$v_n \text{ (m/S)} = \mu_n \cdot \epsilon / 1 + (\epsilon / \epsilon_c) \text{ for } \epsilon \leq \epsilon_c$$

$$= v_{sat} \text{ for } \epsilon \geq \epsilon_c$$

$$\epsilon_c = 2 v_{sat} / \mu_n$$

<http://vlsisystemdesign.com/>

Udemy

Velocity Saturation Effect

$$Q_i(x) = -C_{ox} ([V_{gs} - V(x)] - V_t)$$

$$\text{Velocity } V_n(x) = \text{mobility} \cdot \text{electric field}$$

$$= \mu_n \cdot dV/dx$$

Re-deriving drain current
using below boundary condition

$$I_d = -V_n(x) \cdot Q_i(x) \cdot W \rightarrow I_d = \frac{\mu_n \cdot C_{ox}}{1 + \left(\frac{V_{ds}}{\epsilon_c L}\right)} \cdot (W/L) [(V_{gs} - V_t)V_{ds} - V_{ds}^2/2]$$

Too complex

The leading cause of this behaviour is **velocity saturation**. At low electric fields, carrier velocity increases linearly with electric field ($v_n = \mu_n \cdot E$). Beyond a critical electric field, carrier velocity saturates and becomes almost constant ($v_n \approx v_{sat}$) due to scattering effects. Since drain current depends directly on carrier velocity, once velocity saturates, increasing VGS no longer produces a quadratic increase in current, leading to a nearly linear ID–VGS relationship.

Velocity Saturation Effect

Let's come up with one model

Operation modes

Let's call $(V_{gs} - V_t) = V_{gt}$

Long Channel (>250nm)	Short Channel (<250nm)
Cutoff	Cutoff
Resistive	Resistive
	Velocity Saturation
Saturation	Saturation

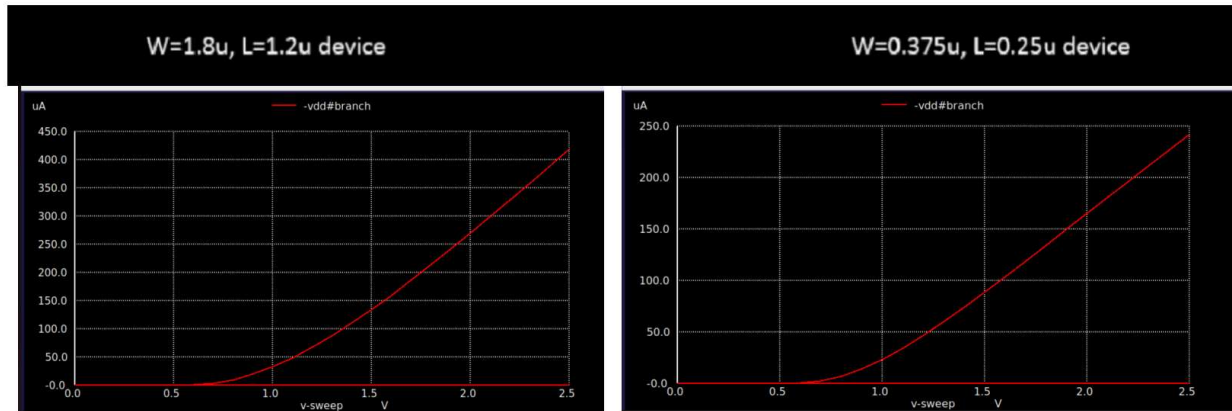
$I_d = 0$, for $V_{gt} < 0$, cutoff mode

For all other modes, let's use below model

$$I_d = k_n \cdot [(V_{gt} \cdot V_{min}) - \frac{V_{min}^2}{2}] \cdot [1 + \lambda V_{ds}]$$

Because of velocity saturation, short-channel MOSFETs introduce an **additional region of operation** beyond cutoff, linear (resistive), and saturation. This new region is called **velocity saturation**. To model all regions using one compact expression, an effective voltage is defined as $V_{min} = \min(V_{GS} - V_T, V_{DS}, V_{DSAT})$. The drain current is zero when $V_{GS} < V_T$ (cutoff), and in all other regions it follows a unified expression that smoothly captures both long-channel and velocity-saturated short-channel behaviour.

Lecture 4: Velocity saturation drain current model



From the I_D - V_{DS} plots, we observe that velocity saturation occurs earlier in short-channel MOSFETs. For the **long-channel device** ($W = 1.8 \mu m$, $L = 1.2 \mu m$), the drain current increases smoothly and reaches a higher peak current of about **410 μA** . In contrast, the **short-channel device** ($W = 0.375 \mu m$, $L = 0.25 \mu m$) saturates much earlier and reaches a lower peak current of about **210 μA** , even though the W/L ratio is kept the same. This clearly shows that current scaling does not hold at lower technology nodes.

To capture this behaviour, a **single-drain-current model** is introduced that operates across the entire range of operation. The idea is to use an effective voltage v_{min} , defined as the minimum of $(V_{GS} - V_T)$, V_{DS} , and V_{DSAT} . The drain current is zero when $V_{GS} < V_T$ (cutoff). For all other cases, the same compact expression is used, and the actual operating region is determined by which voltage becomes v_{min} .

Velocity Saturation Effect

Let's come up with one model

Operation modes Let's call $(V_{GS} - V_T) = V_{gt}$

Long Channel (>250nm)	Short Channel (<250nm)	
Cutoff	Cutoff	$I_d = 0$, for $V_{gt} < 0$, cutoff mode For all other modes, let's use below model
Resistive	Resistive	
	Velocity Saturation	
Saturation	Saturation	

$$I_d = k_n \cdot \left[(V_{gt} \cdot v_{min}) - \frac{v_{min}^2}{2} \right] \cdot [1 + \lambda V_{ds}]$$

Where $v_{min} = \min(V_{gt}, V_{ds}, V_{dsat})$

For eg., If V_{dsat} = minimum value

$$I_d = k_n \cdot \left[(V_{gt} \cdot V_{dsat}) - \frac{V_{dsat}^2}{2} \right] \cdot [1 + \lambda V_{ds}]$$

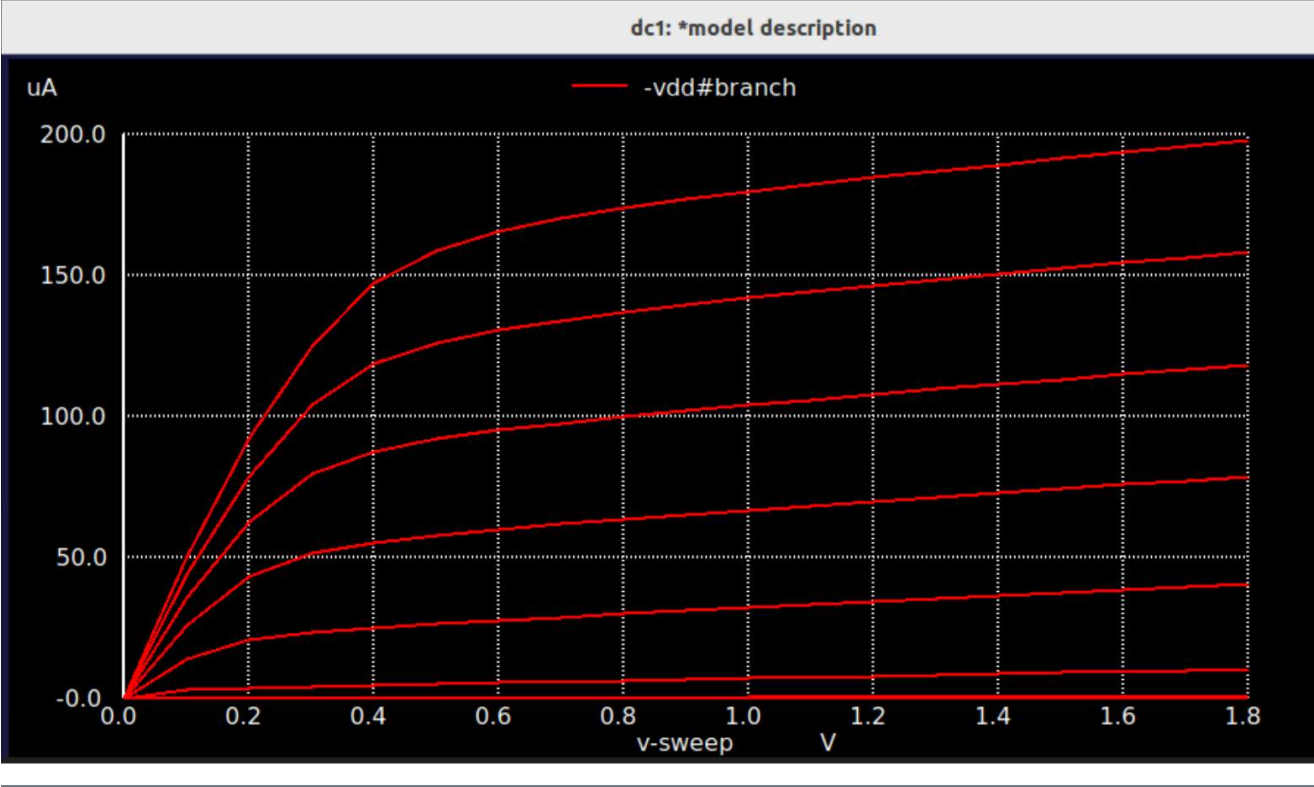
$$I_d = \mu_n \cdot C_{ox} \cdot (W/L) \cdot \left[(V_{gt} \cdot V_{dsat}) - \frac{V_{dsat}^2}{2} \right] \cdot [1 + \lambda V_{ds}]$$

<http://vlsi-systemdesign.com/>

When $(V_{GS} - V_T)$ is at its minimum, the device operates in the **saturation region**, and the model reduces to the familiar long-channel square-law behaviour. When V_{DS} is the minimum, the device operates in the **linear (resistive) region**, and the drain current becomes proportional to V_{DS} . These two cases apply to both long-channel and short-channel devices.

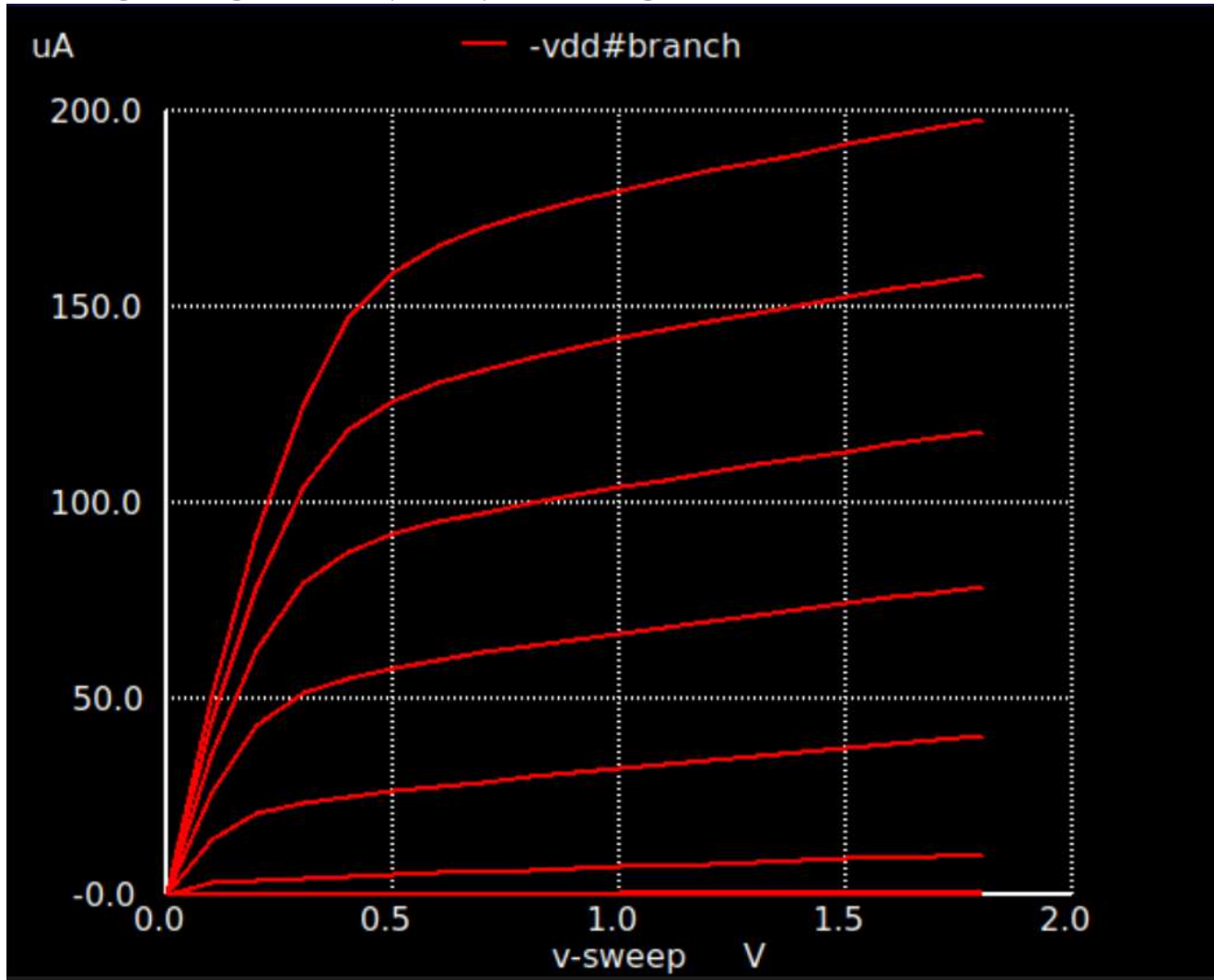
For short-channel devices only, when V_{DSAT} is at its minimum, the device enters the velocity-saturation region. Here, the carrier velocity has already saturated due to the high electric fields, so increasing the gate voltage no longer produces a quadratic increase in drain current. This explains why short-channel devices exhibit lower peak current and earlier saturation than long-channel devices, even at the same W/L ratio. Velocity saturation is therefore a key short-channel effect that must be included in modern MOSFET models and directly impacts current drive and delay in CMOS

circuits.

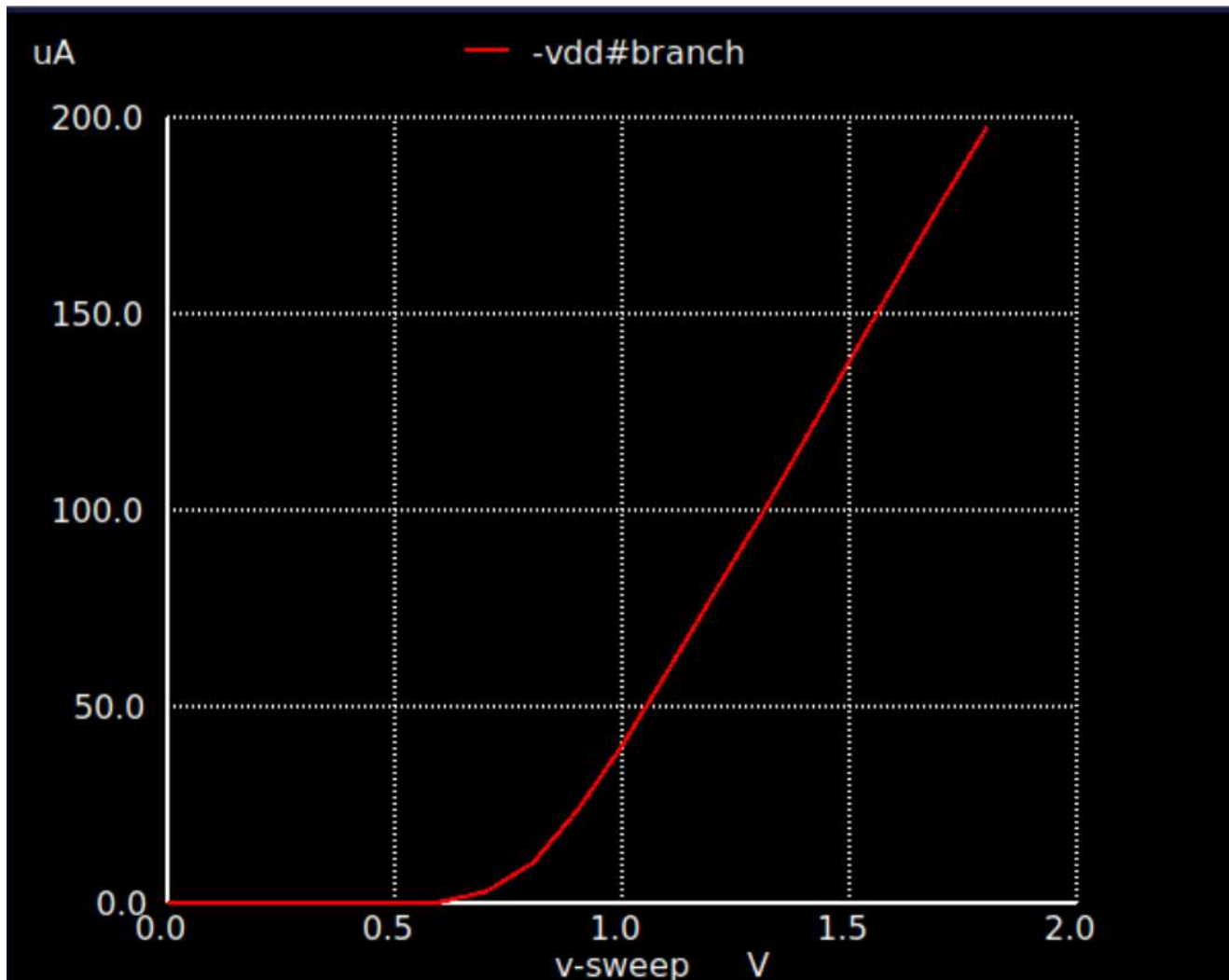


Lecture 5 & 6: Labs Sky130 Id-Vgs

In this SPICE simulation, a short-channel NMOS device with a channel length of around $0.12\text{--}0.15\text{ }\mu\text{m}$ is analysed at a supply voltage of 1.8 V . The drain voltage V_{DS} is swept from 0 to 1.8 V with a step of 0.1 V , while the gate voltage V_{GS} is swept in steps of 0.2 V to generate the $I_D\text{--}V_{DS}$ characteristics.



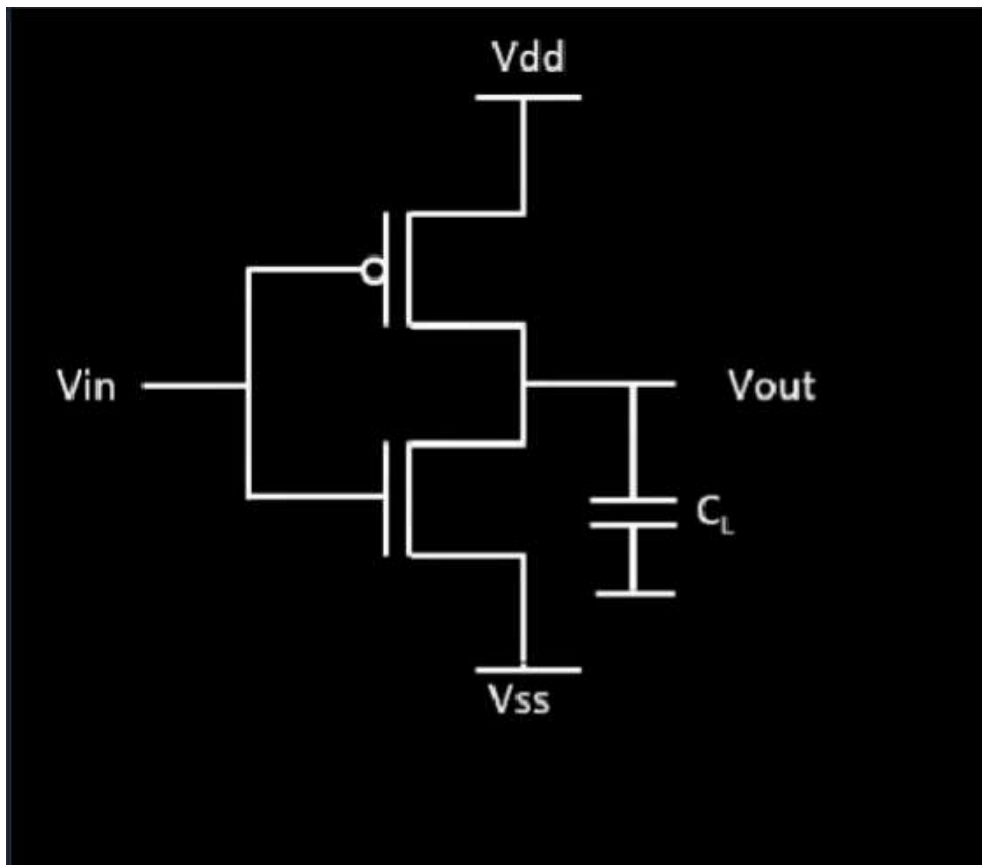
From the ID–VDS plots, the device shows normal behaviour at lower VDS, but at higher VDS, the curves become more linear and saturate early. At $V_{DS} = 1.8$ V, the peak drain current observed from the plot is approximately 0.196 mA (196 μ A). This early saturation and reduced peak current confirm velocity saturation in short-channel devices.



When the simulation is modified to keep $V_{DS} = 1.8$ V constant and sweep V_{GS} from 0 to 1.8 V with a step of 0.1 V, the resulting ID– V_{GS} curve shows a nearly linear behaviour. Since the channel length is around 0.12–0.15 μ m, the quadratic dependence of drain current is no longer valid, and velocity saturation dominates the device operation.

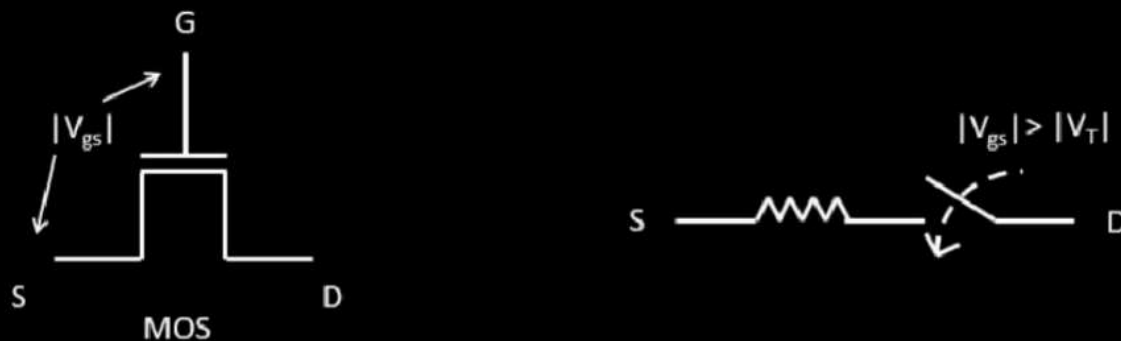
Chapter 2: CMOS voltage transfer characteristics (VTC)

Lecture 1:MOSFET as a switch



To understand **CMOS Voltage Transfer Characteristics (VTC)**, we shift our view of MOSFETs from a pure device-physics perspective to a **switch-level perspective**. A MOS transistor (NMOS or PMOS) turns **ON** when the **magnitude of VGS exceeds the threshold voltage V_T** , and turns **OFF** otherwise. Using $|V_{GS}|$ allows the same rule to apply to both NMOS (positive VGS) and PMOS (negative VGS).

MOS device characteristics

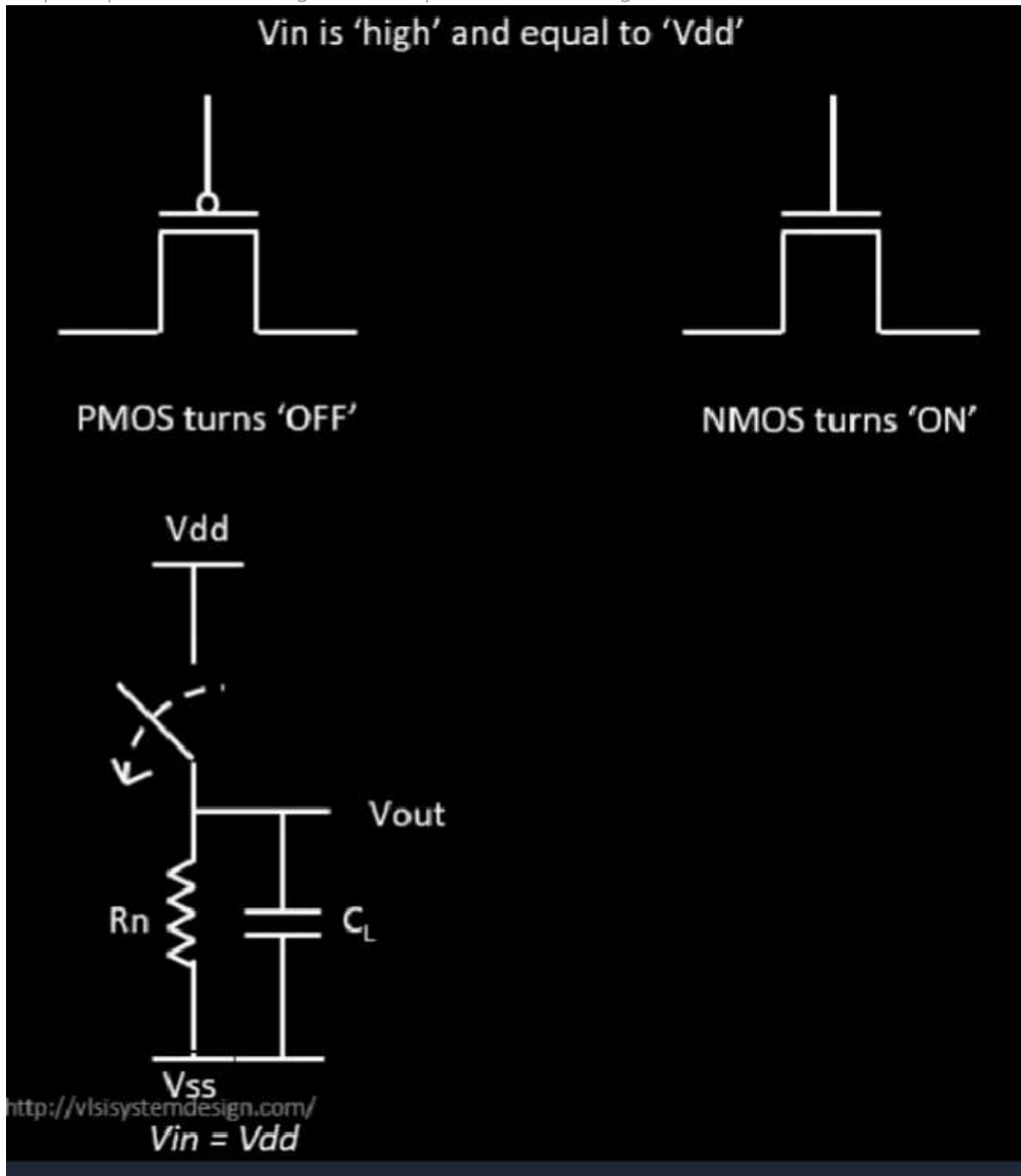


Transistor → Switch

- With infinite OFF resistance when $|V_{gs}| < |V_T|$
- With finite ON resistance when $|V_{gs}| > |V_T|$

From a switch viewpoint, when $|v_{GS}| < V_T$, the MOSFET behaves like an **open switch** with very high resistance, so no current flows between source and drain. When $|v_{GS}| \geq V_T$, the switch **closes**, creating a finite (nonlinear) resistance between source and drain. This simple ON–OFF behaviour underlies treating MOSFETs as digital switches.

A CMOS inverter is built using a PMOS at the top connected to V_{DD} and an NMOS at the bottom connected to ground (V_{SS}). The **gates of both transistors are tied together** and driven by the input voltage V_{IN} , while their drains are connected to form the output node. A load capacitance C_L at the output represents the wiring and the input of the next stage.

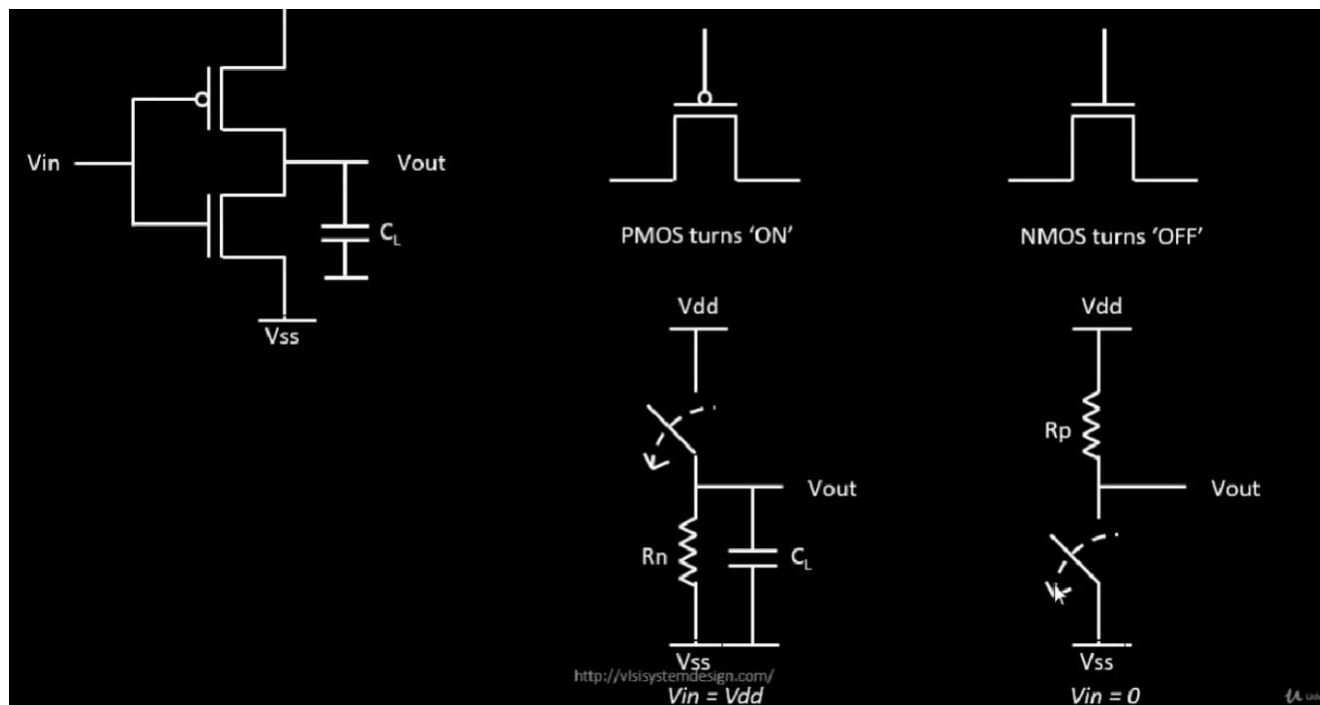


When $V_{IN} = V_{DD}$, the NMOS gate-to-source voltage becomes $V_{GS} = V_{DD} - 0 = V_{DD}$, so the NMOS turns ON. At the same time, the PMOS gate-to-source voltage becomes $V_{GS} = V_{DD} - V_{DD} = 0$,

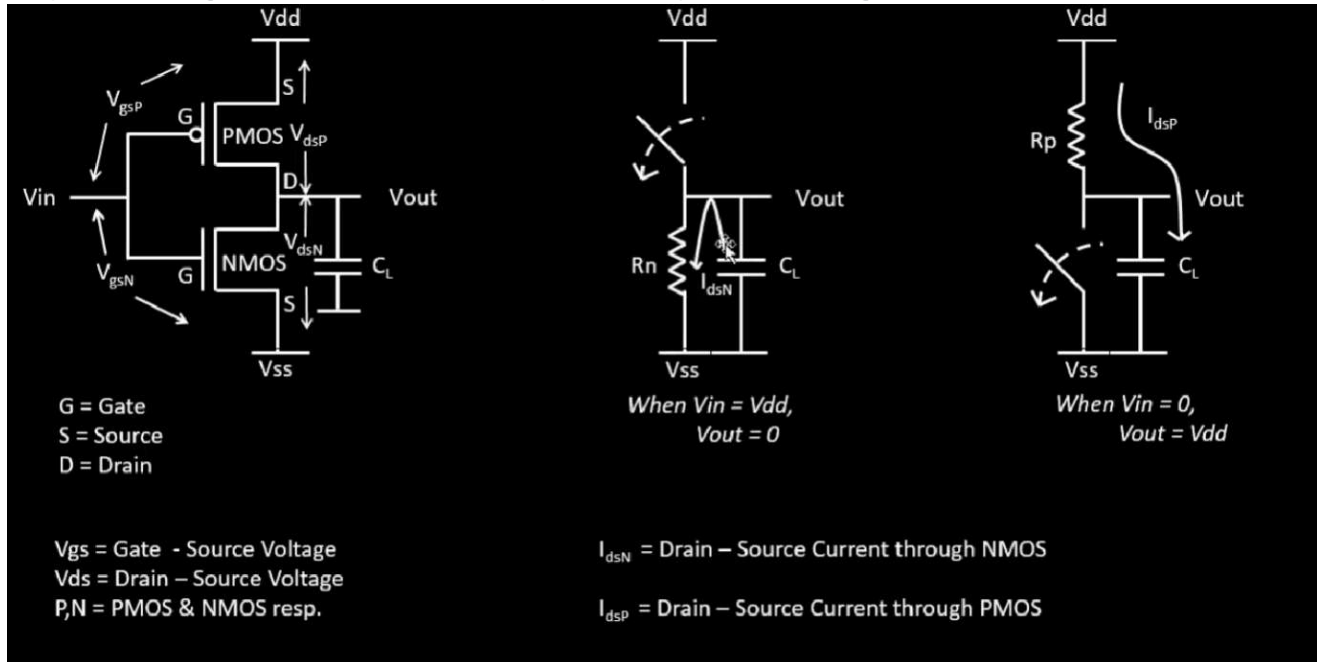
so the PMOS turns OFF. This pulls the output node to ground. In the opposite case (to be analysed next), when $V_{IN} = 0$, the NMOS turns OFF, and the PMOS turns ON, pulling the output to V_{DD} . Combining these two cases yields the CMOS voltage transfer characteristic, which describes how the output voltage changes with the input voltage.

Lecture 2: Introduction to standard MOS voltage current parameters

To derive the **CMOS Voltage Transfer Characteristic (VTC)**, we analyse the CMOS inverter under different **input boundary conditions** and replace the transistors with their **equivalent circuits**. This step is essential because the VTC is later used to compute **cell delay**, not only for inverters but also for logic gates like AND and OR.



When $V_{IN} = V_{DD}$ (assume $V_{DD} = 5\text{ V}$), the **NMOS turns ON** because its gate-to-source voltage $V_{GSN} = V_{DD} - 0 = V_{DD}$, which is greater than V_T . At the same time, the **PMOS turns OFF** because its gate-to-source voltage $V_{GSP} = V_{DD} - V_{DD} = 0$, which is below its threshold. In this case, the NMOS is modelled as a finite resistance R_N , the PMOS as an open switch, and there exists a direct discharge path from the output node to ground. As a result, the output load capacitor discharges and $V_{OUT} = 0$.



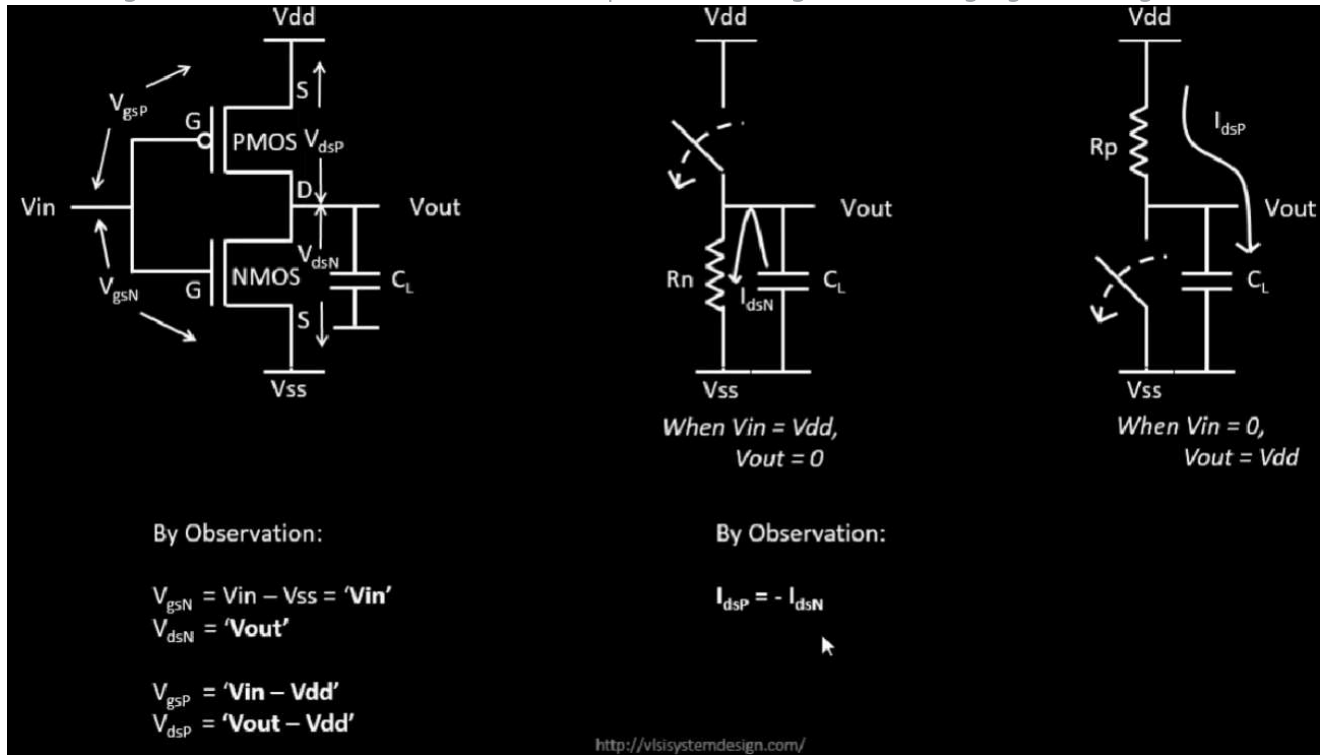
When $V_{IN} = 0\text{ V}$, the situation reverses. The **PMOS turns ON** because $V_{GSP} = 0 - V_{DD} = -V_{DD}$, whose magnitude exceeds V_T , while the **NMOS turns OFF** since $V_{GSN} = 0$. The PMOS is modelled as a finite resistance R_P , the NMOS as an open switch, and a direct charging path from V_{DD} to the output node exists. The output capacitor fully charges, resulting in $V_{OUT} = V_{DD}$.

To systematically derive equations, clear **naming conventions** are introduced. V_{GSN} and V_{GSP} denote gate-to-source voltages of NMOS and PMOS, respectively, while V_{dsN} and V_{dsP} represent their drain-to-source voltages. Similarly, I_{dsN} and I_{dsP} denote the drain-to-source currents through NMOS and PMOS. These definitions allow both devices to be analysed together while accounting for polarity differences.

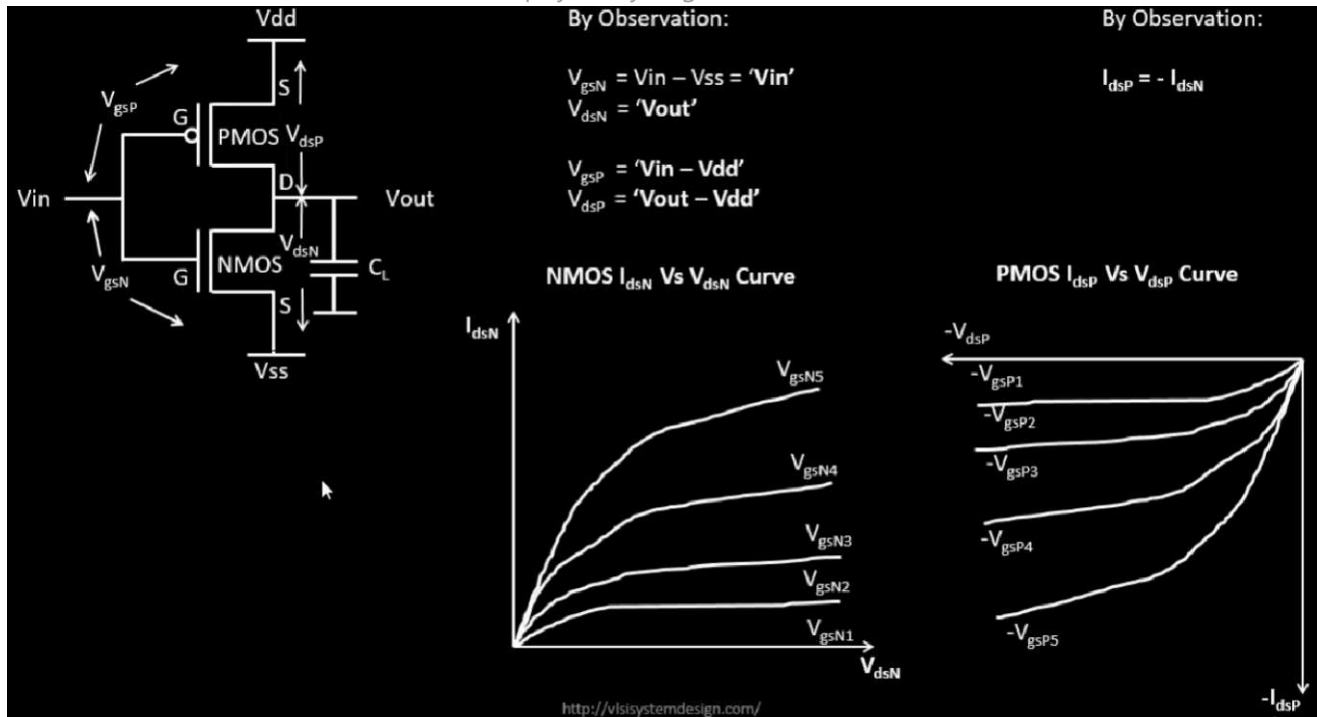
With both extreme input cases ($V_{IN} = 0$ and $V_{IN} = V_{DD}$) understood in terms of voltages, currents, and equivalent circuits, the next step is to merge these conditions and derive a continuous relationship between V_{IN} and V_{OUT} . This combined analysis leads directly to the CMOS voltage transfer characteristic, which forms the foundation for delay and timing analysis in digital circuits.

Lecture 3: PMOS/NMOS drain current v/s drain voltage

To derive the CMOS Voltage Transfer Characteristic (VTC), we first establish clear voltage and current naming conventions for NMOS and PMOS. Gate-to-source voltages are defined as $V_{GSN} = V_{IN} - V_{SS}$ for NMOS and $V_{GSP} = V_{IN} - V_{DD}$ for PMOS. Similarly, drain-to-source voltages are $V_{DSN} = V_{OUT}$ and $V_{DSP} = V_{OUT} - V_{DD}$. These definitions are important because NMOS calculations are simpler because the source is tied to ground, whereas PMOS calculations require subtracting V_{DD} , making sign handling critical.



An important observation is that the current directions are opposite for NMOS and PMOS. When the output capacitor charges, current flows from V_{DD} through the PMOS; when it discharges, current flows from the capacitor through the NMOS to ground. Because of this, the drain current of PMOS is taken as the **negative of NMOS drain current**, meaning $I_{dsP} = -I_{dsN}$. This sign difference is purely due to current direction, not because the current itself is physically negative.



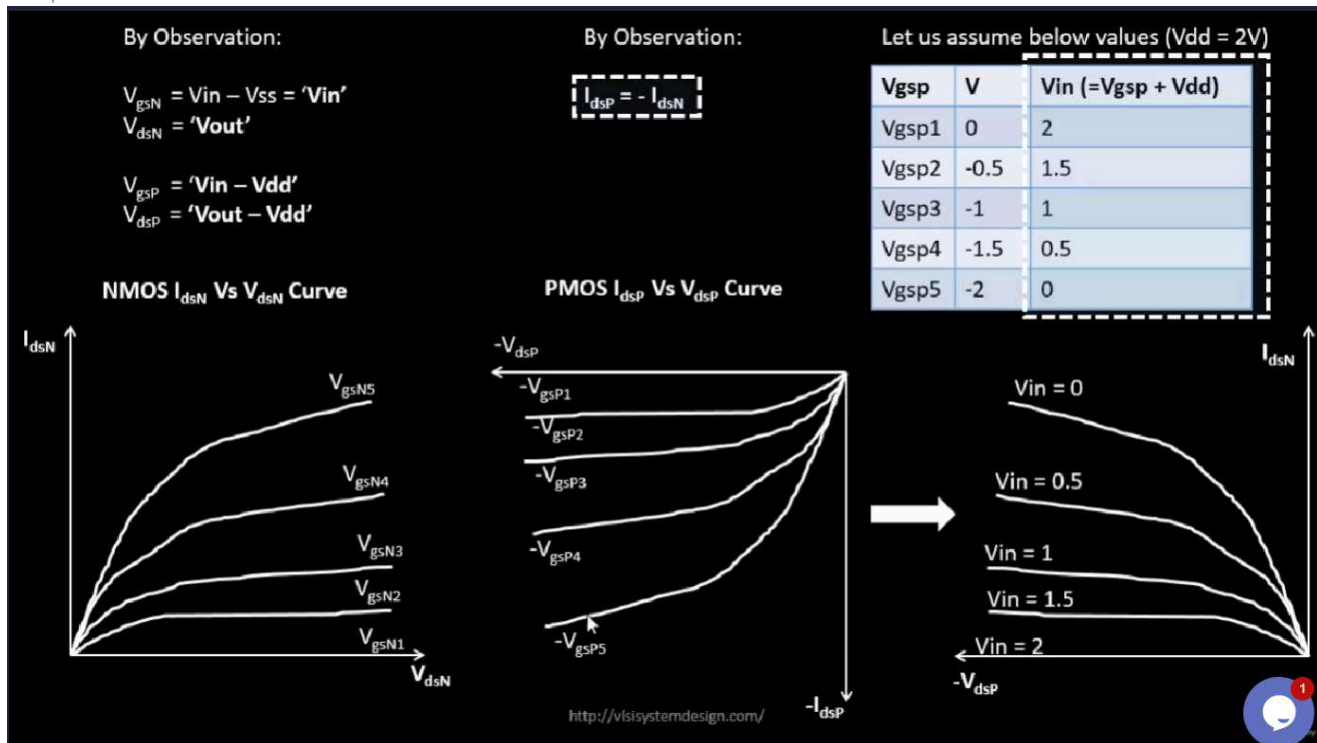
Using prior SPICE simulations, we already know the **NMOS ID–VDS characteristics**. For $V_{gsN} < V_T$, the NMOS is OFF, and the drain current is zero. As V_{gsN} exceeds the threshold, the NMOS turns ON, and the drain current increases with V_{ds} , eventually reaching saturation. Multiple curves exist for different V_{gsN} values, with higher V_{gsN} producing higher drain current.

The **PMOS characteristics** are similar in shape but mirror each other. The I_{dsP} – V_{dsP} curves appear in the opposite quadrant because PMOS requires a **negative gate-to-source voltage** to turn ON. When $|V_{gsP}| < |V_T|$, the PMOS is OFF and the current is zero. As V_{gsP} becomes more negative and exceeds the threshold in magnitude, the PMOS conducts and drain current increases, reaching its maximum at the highest $|V_{gsP}|$.

These **NMOS and PMOS load curves** are the key building blocks for CMOS analysis. By combining them under the constraint that both devices share the same output node voltage, we can merge their behaviours and derive the **CMOS voltage transfer characteristic**, which directly relates V_{out} to V_{in} and forms the foundation for delay and timing analysis.

Lecture 4: Step1 – Convert PMOS gate-source-voltage to V_{in}

To derive the **CMOS Voltage Transfer Characteristic (VTC)**, we must express device behaviour using only **V_{in} and V_{out}** , since internal voltages like V_{GSN} , V_{GSP} , V_{DSN} , and V_{DSP} are not visible at the logic level. Although NMOS and PMOS have separate device voltages, a digital cell is defined only by its input and output.



The first step is to eliminate **V_{GSP}** by rewriting it in terms of input voltage. Since $V_{GSP} = V_{in} - V_{DD}$, this becomes $V_{in} = V_{GSP} + V_{DD}$. Assuming a long-channel inverter with $V_{DD} = 2V$, each PMOS gate voltage maps directly to a V_{in} value (for example, $V_{GSP} = 0 \rightarrow V_{in} = 2V$, $V_{GSP} = -0.5 \rightarrow V_{in} = 1.5V$). This allows PMOS characteristics to be referenced using V_{in} instead of V_{GSP} .

Next, PMOS current is converted into NMOS current form. Because the PMOS current flows in the opposite direction, **I_{DSP} is the negative of I_{DSN}** . By flipping the PMOS current axis, all currents can be represented using a single current variable, I_{DSN} , without changing their physical meaning.

After these steps, PMOS current curves become **I_{DSN} versus V_{in} curves**, and V_{GSP} is fully removed from the analysis. The next step (done later) is to eliminate V_{DSP} and V_{DSN} in favour of **V_{out}** , after which NMOS and PMOS load curves can be combined to obtain the **CMOS voltage transfer characteristic**.

Lecture 5: Step2 & Step3 – Convert PMOS and NMOS drain-source-voltage to vout

By Observation:

$$\begin{aligned} V_{gsN} &= V_{in} - V_{ss} = 'V_{in}' \\ V_{dsN} &= 'V_{out}' \end{aligned}$$

$$\begin{aligned} V_{gsp} &= 'V_{in} - V_{dd}' \\ V_{dsp} &= 'V_{out} - V_{dd}' \end{aligned}$$

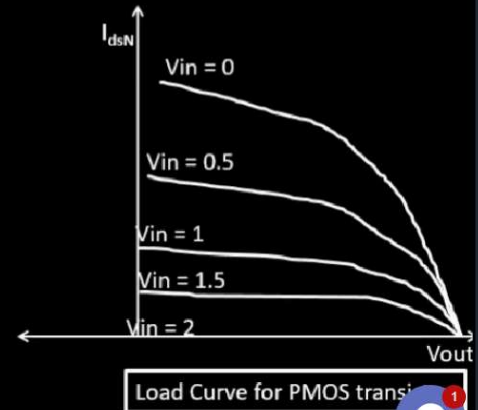
By Observation:

$$I_{dsp} = -I_{dsN}$$

Let us assume below values ($V_{dd} = 2V$)

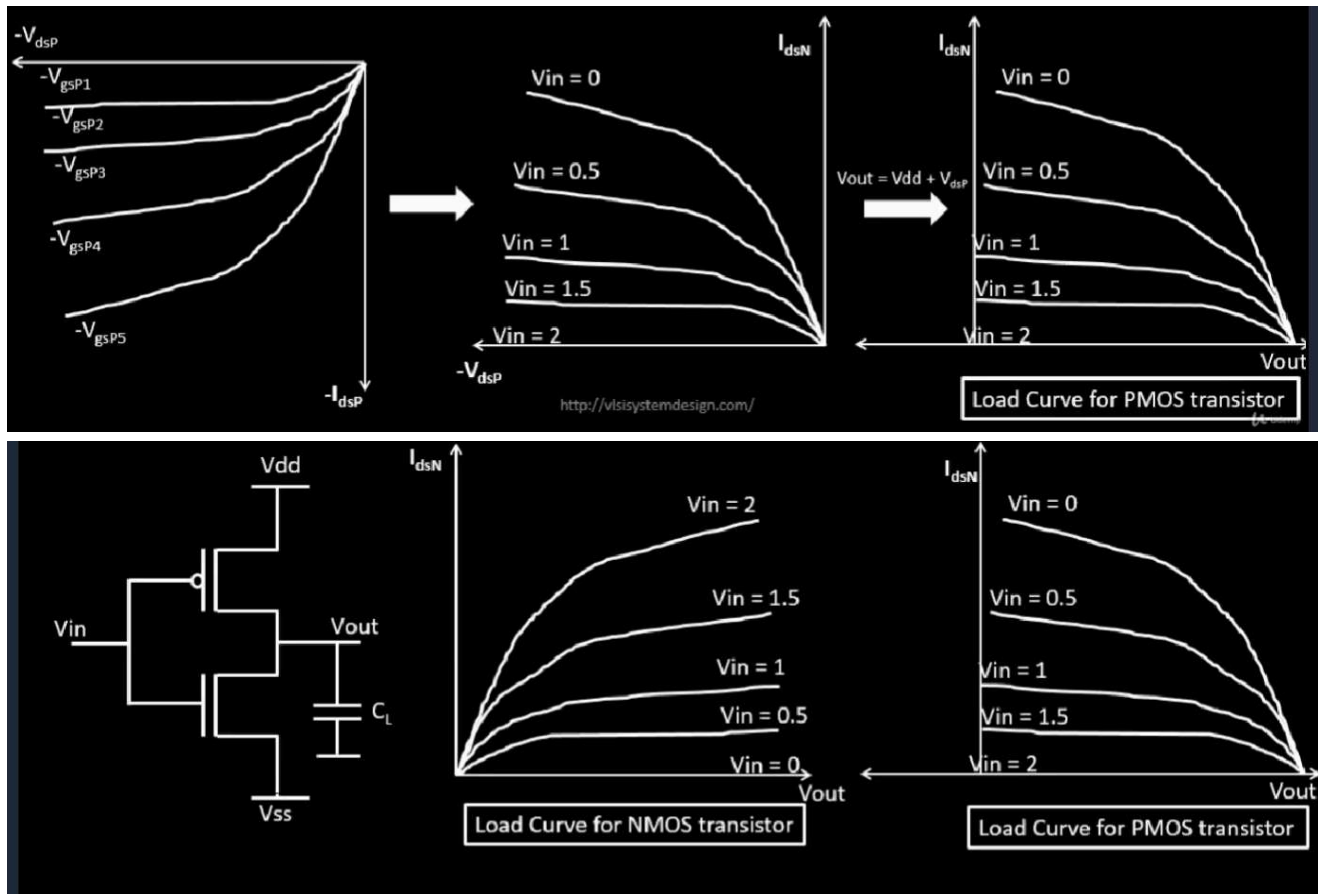
Vgsp	V	Vin (=Vgsp + Vdd)
Vgsp1	0	2
Vgsp2	-0.5	1.5
Vgsp3	-1	1
Vgsp4	-1.5	0.5
Vgsp5	-2	0

Let us obtain the Load Curve for NMOS transistor using the above equations



<http://visisystemdesign.com/>

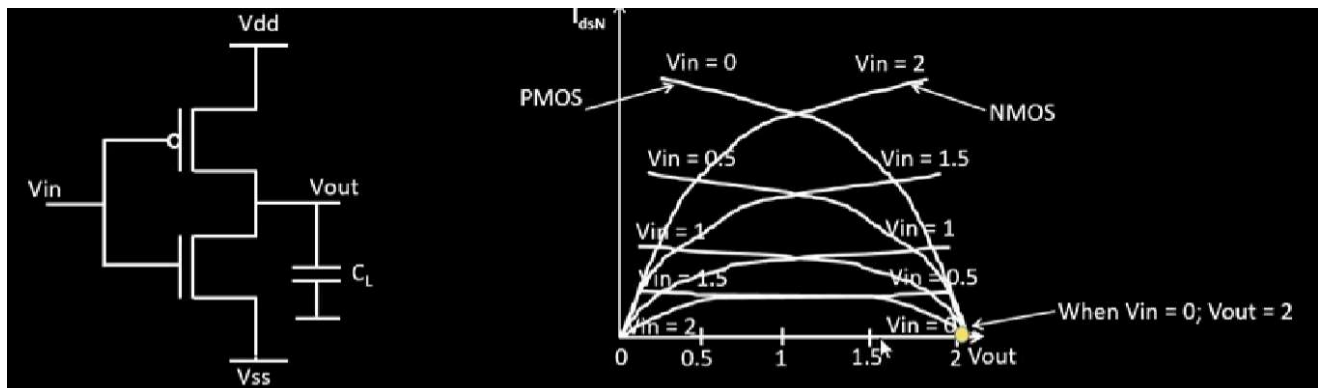
- To derive the **PMOS load curve**, we next eliminate v_{dsp} by expressing it in terms of the **output voltage**. Since $v_{out} = v_{dd} + v_{dsp}$, the PMOS I_D - v_{dsp} curves can be shifted horizontally by $+v_{dd}$ and reinterpreted as **I_{DSN} versus V_{OUT}** curves. For example, when $v_{dsp} = -2V$ and $v_{dd} = 2V$, $v_{out} = 0V$. At this point, a finite current flows because the output capacitor is fully discharged and must be charged by the PMOS. When $v_{dsp} = 0$, $v_{out} = v_{dd}$, the output capacitor is fully charged, and the current becomes zero. This shift converts the PMOS characteristics into a load curve expressed solely in terms of V_{IN} , V_{OUT} , and I_{DSN} .



- After this conversion, the PMOS behaviour is no longer described using V_{gsP} , V_{dsP} , or I_{dsP} . Instead, everything is expressed using **VIN**, **VOUT**, and a single current variable (**IDSN**). This curve is called the ****PMOS load curve**** in a CMOS inverter context, where PMOS behaviour corresponds to the charging current of the output capacitor.
- The **NMOS load curve** is simpler to derive. Since the NMOS source is connected to ground, $V_{gsN} = V_{in}$ and $V_{dsN} = V_{out}$ directly. Therefore, the NMOS I_D – V_{DS} curves obtained from SPICE simulations can be converted into load curves simply by **renaming the axes**: replace V_{gsN} with V_{in} and V_{dsN} with V_{out} . No shifting or sign changes are required.
- At this point, both **PMOS** and **NMOS** load curves are expressed in the same variables: **VIN**, **VOUT**, and **IDSN**. The final step is to merge these two load curves, which yields the **CMOS Voltage Transfer Characteristic (VTC)**. This VTC directly relates the input voltage to the output voltage and provides the foundation for understanding switching behaviour and delay in CMOS inverters.

Lecture 6: Step 4 – Merge PMOS – NMOS load curves and plot VTC

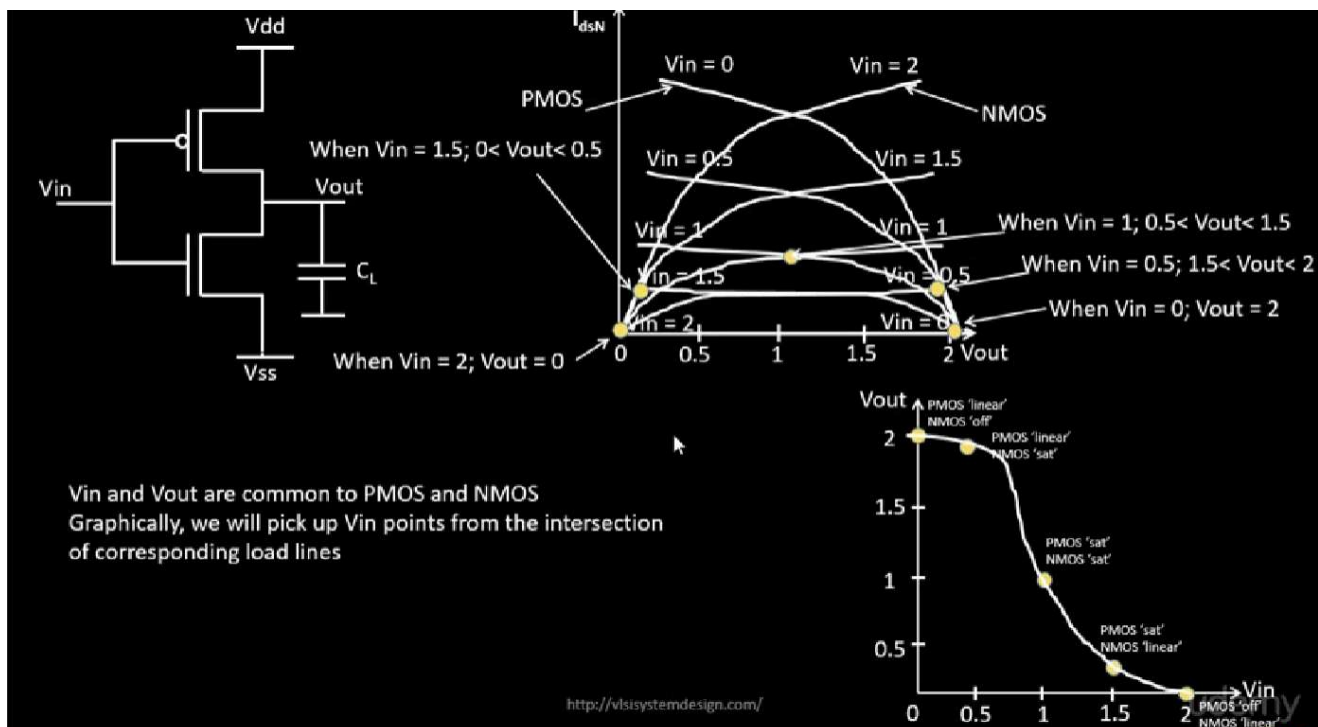
CMOS Voltage Transfer Characteristics (VTC) using Load Curves



To derive the CMOS Voltage Transfer Characteristic (VTC), we superimpose the NMOS and PMOS load curves on a common graph. This works because V_{IN} and V_{OUT} are common to both transistors, while the operating point of the inverter is determined by the intersection of their load curves for a given V_{IN} .

Key idea

For each input voltage V_{IN} , the intersection point of the NMOS and PMOS load curves gives the corresponding output voltage V_{OUT} . Plotting these (V_{IN}, V_{OUT}) pairs produces the VTC.



Extreme operating points

- $V_{IN} = 0\text{ V}$
 - NMOS: cutoff (OFF)
 - PMOS: linear region

- Intersection gives $V_{OUT} = V_{DD} = 2\text{ V}$
- $V_{IN} = 2\text{ V}$
 - NMOS: linear region
 - PMOS: cutoff (OFF)
 - Intersection gives $V_{OUT} = 0\text{ V}$

These two points define the inverter's logical behaviour.

Intermediate input voltages

- $V_{IN} = 0.5\text{ V}$
 - NMOS: saturation
 - PMOS: linear
 - Intersection gives $1.5\text{ V} < V_{OUT} < 2\text{ V}$
- $V_{IN} = 1\text{ V}$
 - NMOS: saturation
 - PMOS: saturation
 - Intersection gives $0.5\text{ V} < V_{OUT} < 1.5\text{ V}$
 - This is the **high-gain transition region**, where a small change in V_{IN} causes a large change in V_{OUT} .
- $V_{IN} = 1.5\text{ V}$
 - NMOS: linear
 - PMOS: saturation
 - Intersection gives $0\text{ V} < V_{OUT} < 0.5\text{ V}$

Final VTC construction

By plotting all intersection points:

- $(0, 2)$
- $(0.5, \sim 1.7)$
- $(1, \sim 1)$
- $(1.5, \sim 0.3)$
- $(2, 0)$

and connecting them smoothly, we obtain the **CMOS inverter VTC**.

Important observations

- The **middle region** (both NMOS and PMOS in saturation) has **very high gain**.
- Digital circuits rely on the **steep slope** of this region for fast switching.

- The load-curve method also clearly shows which region (cutoff / linear / saturation) each transistor operates in for a given V_{IN} .

Next step

Validate the analytical VTC by performing a SPICE-based DC sweep simulation of V_{IN} and directly plotting V_{OUT} .

NgspiceSky130-Day 3: CMOS Switching threshold and dynamic simulations

Chapter 1: Voltage transfer characteristics – SPICE simulations

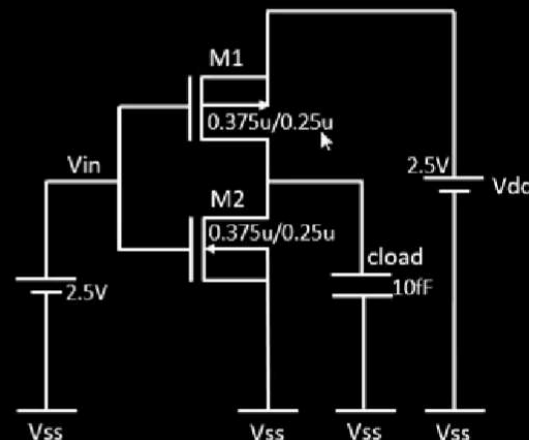
Lecture 1: SPICE deck creation for CMOS inverter

CMOS Inverter SPICE Deck

- Goal: Prepare a SPICE netlist for a CMOS inverter to simulate VTC (Voltage Transfer Characteristics).
- Technology assumed: $L = 0.25 \mu\text{m}$, $V_{DD} = 2.5 \text{ V}$ (long-channel assumption).

SPICE deck

- Component connectivity
- Component values



Circuit Components and Values

- PMOS (M1)

- Type: PMOS
- Drain = out
- Gate = in
- Source = vdd
- Bulk = vdd
- Width $w = 0.375 \mu\text{m}$, Length $L = 0.25 \mu\text{m}$

- **NMOS (M2)**

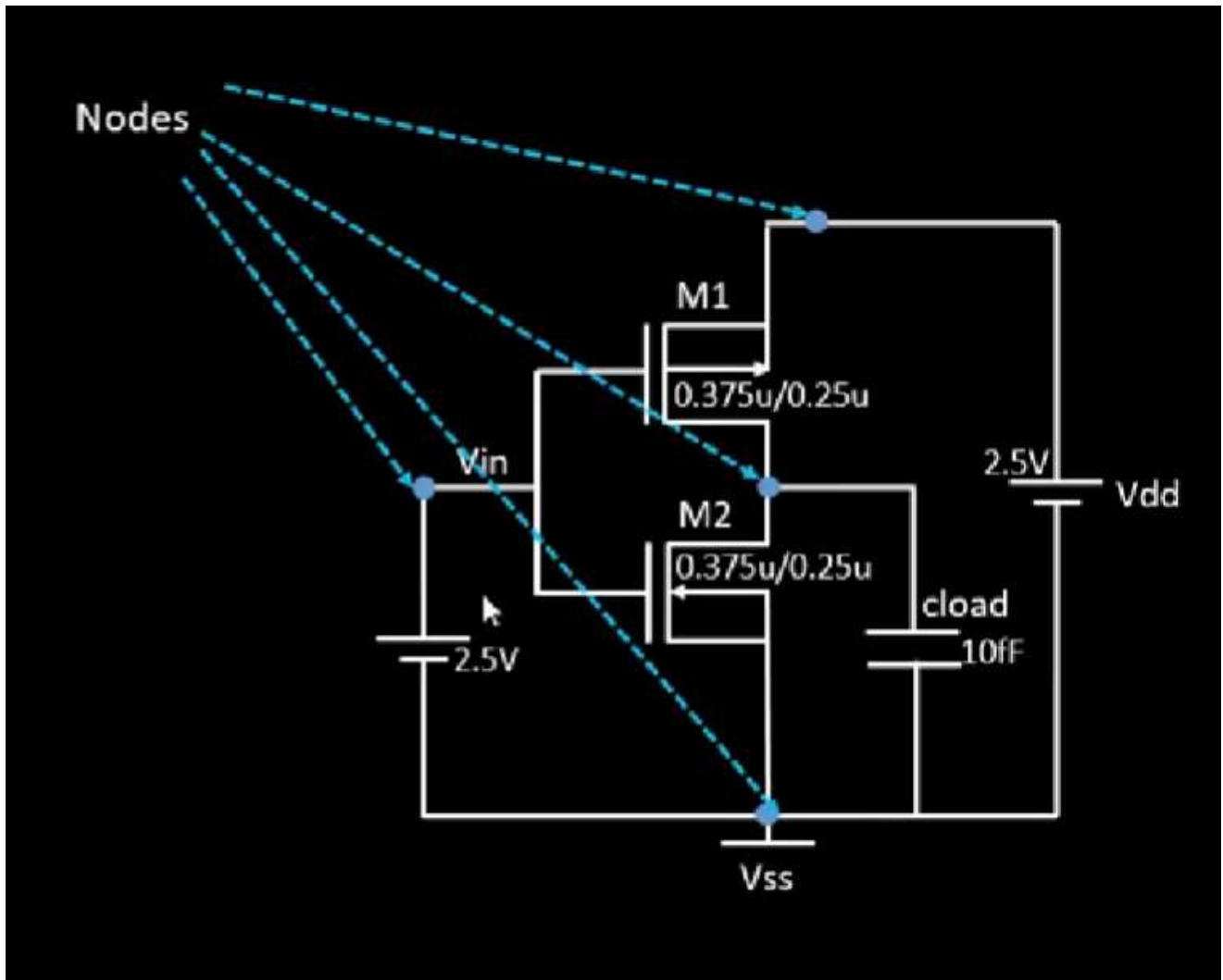
- Type: NMOS
- Drain = out
- Gate = in
- Source = 0 (VSS)
- Bulk = 0
- Width $w = 0.375 \mu\text{m}$, Length $L = 0.25 \mu\text{m}$

- **Load Capacitance**

- $C_{\text{load}} = 10 \text{ fF}$ between out and 0
- Represents lumped capacitance of interconnect + next-stage gates

- **Supplies**

- $V_{\text{DD}} = 2.5 \text{ V}$
- $V_{\text{SS}} = 0 \text{ V}$
- Input voltage v_{in} swept from 0 $\rightarrow$ 2.5 V

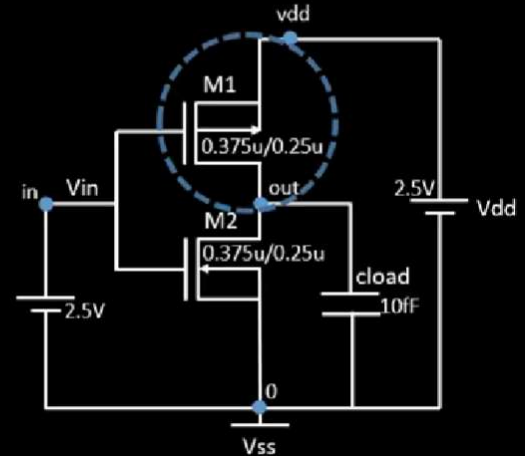


Node Definitions

- in → common gate node of PMOS and NMOS
- out → common drain node (output)
- vdd → supply node
- $\emptyset$ → ground (V_{SS})

SPICE deck

```
*** MODEL Descriptions ***
*** NETLIST Description ***
M1 out in vdd vdd pmos W=0.375u L=0.25u
M2 out in 0 0 nmos W=0.375u L=0.25u
```



SPICE Netlist (Core)

```
*** MODEL Descriptions *** *** NETLIST Description ***
```

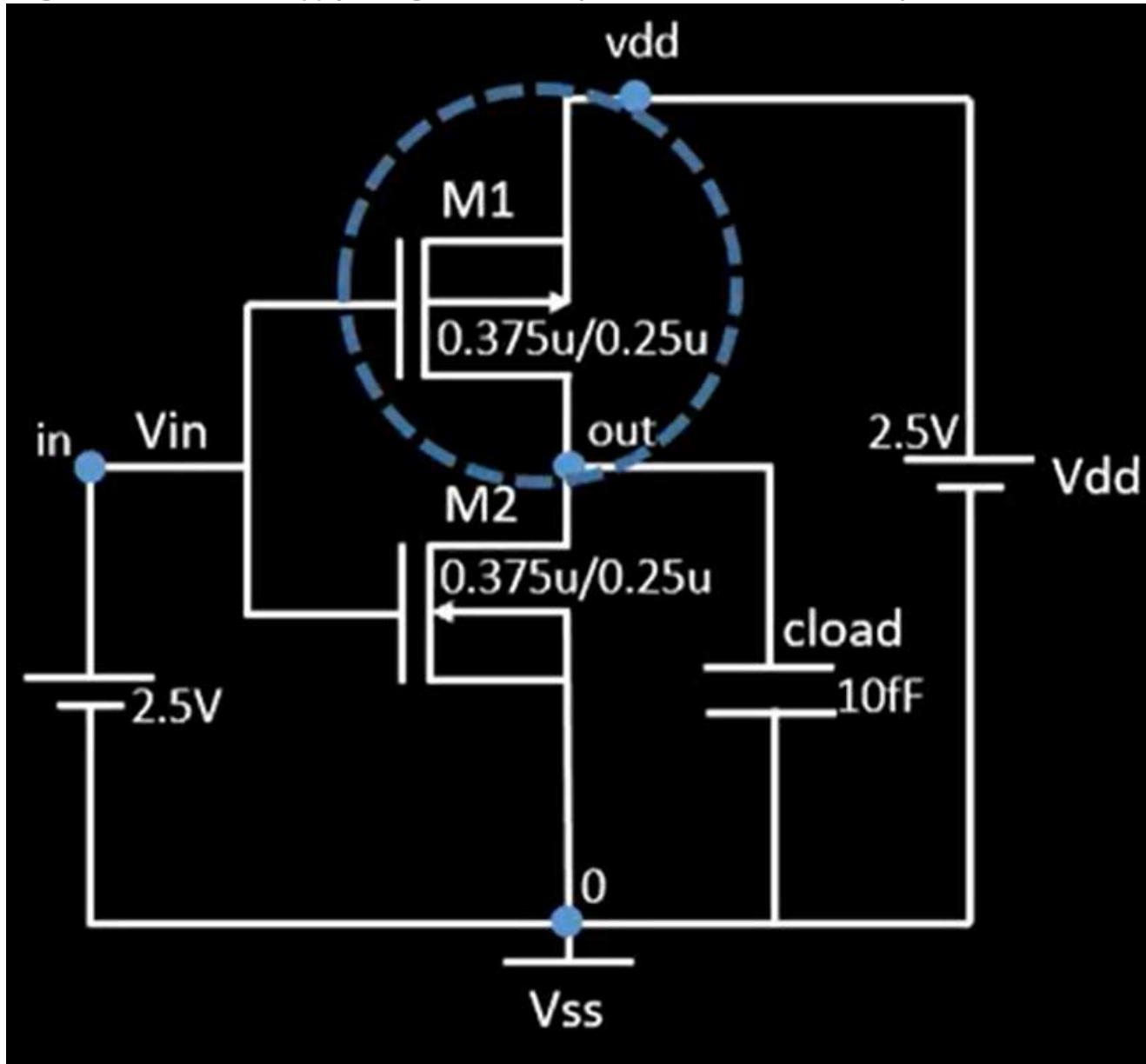
- M1 out in vdd vdd pmos W=0.375u L=0.25u
- M2 out in 0 0 nmos W=0.375u L=0.25u

Important Observations

- Substrate (bulk) must be explicitly connected:
 - PMOS bulk → vdd
 - NMOS bulk → vss
- Equal PMOS and NMOS sizing used **only for illustration**.
 - In practice, PMOS is usually 2–3× wider to balance mobility.
- This setup captures **static CMOS behavior**; dynamic delay analysis comes later.

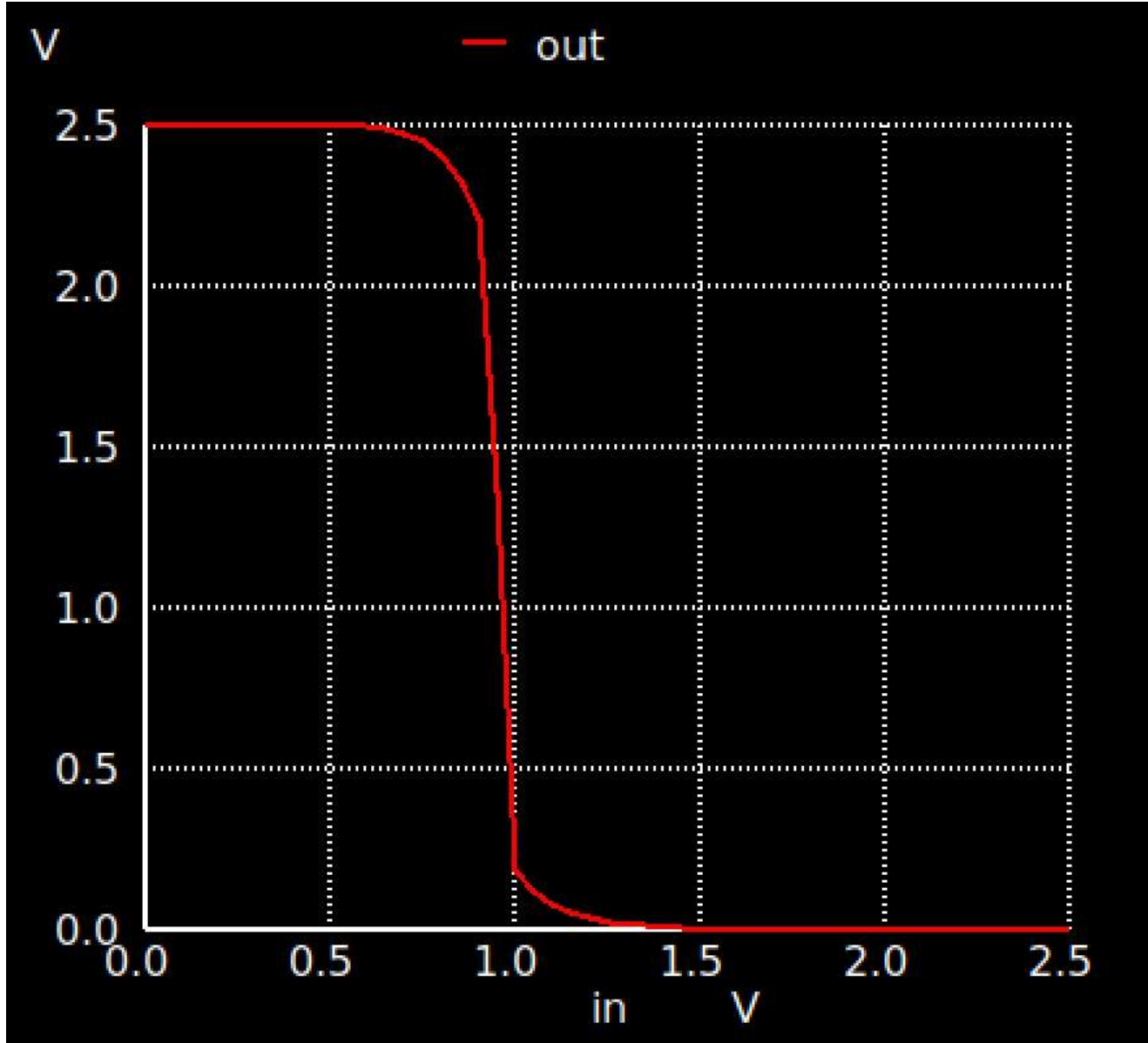
Lecture 2: SPICE simulation for CMOS inverter

We perform a SPICE DC simulation of a CMOS inverter to obtain its Voltage Transfer Characteristics (VTC) and verify the theory derived from NMOS–PMOS load curves. The technology uses a channel length of $0.25\text{ }\mu\text{m}$ and a supply voltage of 2.5 V . Only static (DC) behaviour is analysed.



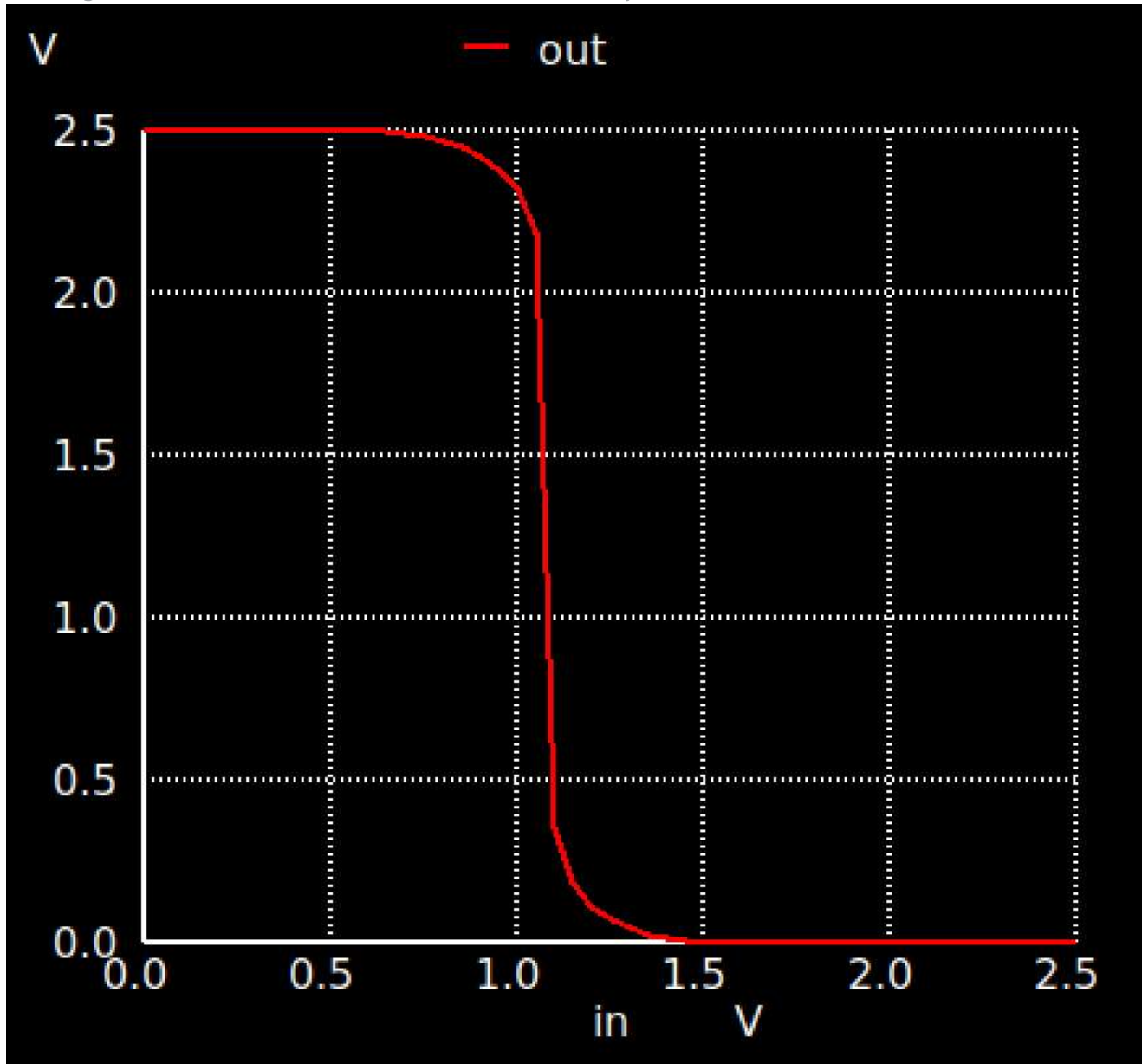
The inverter consists of one PMOS (M1) and one NMOS (M2). For PMOS (M1), the drain is connected to out, the gate to in, and both source and substrate to vdd. Its dimensions are $W = 0.375\text{ }\mu\text{m}$, $L = 0.25\text{ }\mu\text{m}$. For NMOS (M2), the drain is connected to out, the gate to in, and both source and substrate to ground (0). Its dimensions are $W = 0.375\text{ }\mu\text{m}$, $L = 0.25\text{ }\mu\text{m}$.

An **output load capacitor of 10 fF** is connected between `out` and ground, representing interconnect and fan-out loading. The **supply voltage $V_{DD} = 2.5$ V** is applied between `vdd` and ground. The **input voltage V_{in}** is applied between `in` and ground.



To generate the VTC, a **DC sweep** is performed on `vin` from 0 V to 2.5 V with a step of 0.05 V, and the output voltage `v(out)` is plotted against `v(in)`. The transistor behaviour is defined using a **model file** containing all NMOS and PMOS technology parameters for the 250 nm node.

In the first simulation, PMOS and NMOS have equal widths ($0.375\text{ }\mu\text{m}$). The resulting VTC is left-shifted, showing an unbalanced inverter due to lower hole mobility in PMOS devices.



In the second simulation, the PMOS width is increased to $0.9375\text{ }\mu\text{m}$ ($2.5\times$ NMOS width) while keeping $L = 0.25\text{ }\mu\text{m}$. This produces a more symmetric VTC, with the switching point closer to $V_{DD}/2$, indicating balanced pull-up and pull-down strengths.

This confirms that PMOS devices must be wider than NMOS devices (typically $2\text{--}3\times$) to achieve a symmetric VTC and improved noise margins.

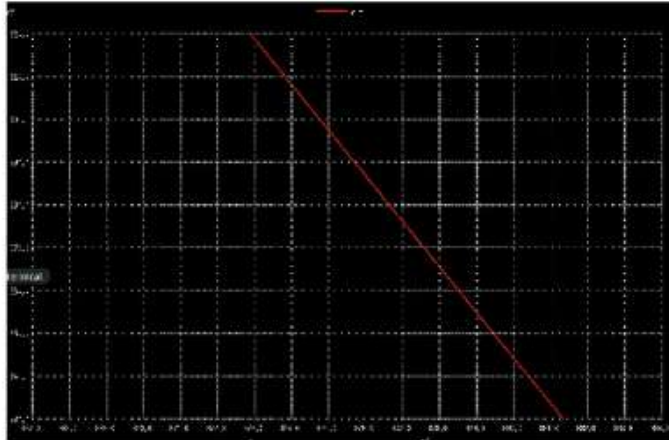
Lecture 3: Labs Sky130 SPICE simulation for CMOS

CMOS Inverter – VTC and Transient Analysis (NGSPICE)

VTC (DC Sweep)

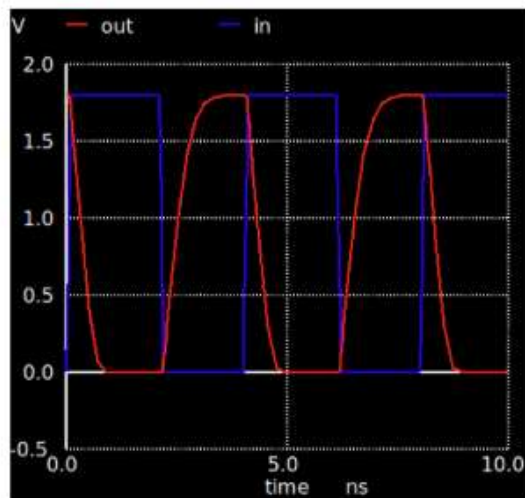
The CMOS inverter is simulated in the **typical corner** with $V_{DD} = 1.8 \text{ V}$. The NMOS width is $0.36 \mu\text{m}$, the PMOS width is $0.84 \mu\text{m}$, giving $W_p/W_n \approx 2.33$. The output load capacitance is 50 fF .

The input voltage is swept from 0 V to 1.8 V with a step of 0.01 V , and $v(\text{out})$ is plotted versus $v(\text{in})$ to obtain the VTC. The inverter switches correctly from 1.8 V to 0 V with a sharp transition region.



The switching threshold (V_M), defined where $V_{in} = V_{out}$, is found by zooming into the intersection region. It lies between 0.87 V and 0.88 V , with a measured value of approximately 0.876 V . Since $V_{DD}/2 = 0.9 \text{ V}$, this confirms that $W_p/W_n \approx 2.33$ centers the switching point close to mid-supply.

Transient Analysis



Using the same sizing and $C_{load} = 50 \text{ fF}$, a pulse input from 0 V to 1.8 V is applied with:

- Rise time = 0.1 ns
- Fall time = 0.1 ns
- Pulse width = 2 ns
- Period = 4 ns

Delays are measured at 50% of $V_{DD} = 0.9 \text{ V}$.

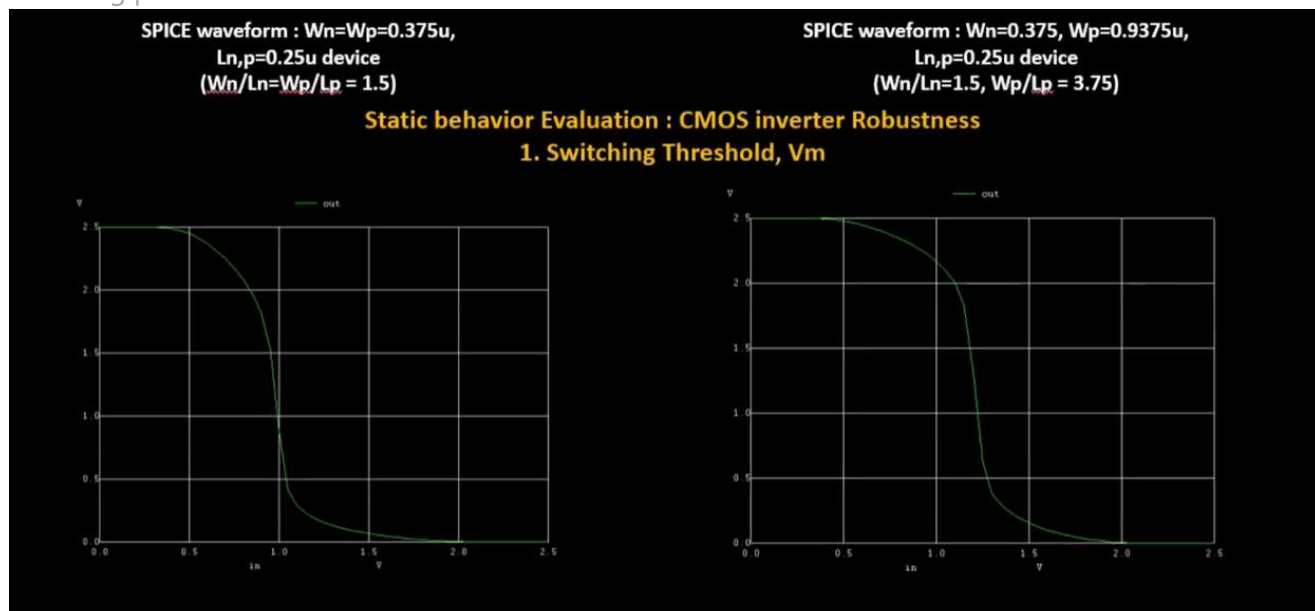
- Rise delay (t_{pLH}): Output crosses 0.9 V at 2.482 ns, Input crosses 0.9 V at 2.15 ns, $\rightarrow t_{pLH} = 0.332 \text{ ns}$
- Fall delay (t_{pHL}): Output crosses 0.9 V at 4.3349 ns, Input crosses 0.9 V at 4.049 ns, $\rightarrow t_{pHL} = 0.285 \text{ ns}$

Chapter 2: Static behavior evaluation – CMOS inverter robustness – Switching Threshold

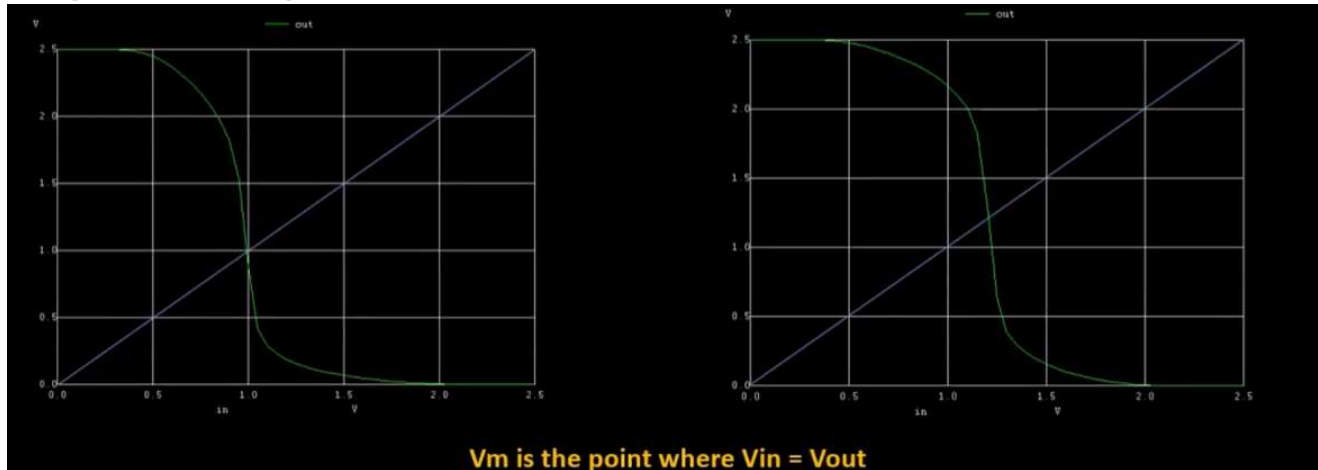
Lecture 1: Switching Threshold, V_m

Comparison of CMOS Inverters with Different W/L Ratios

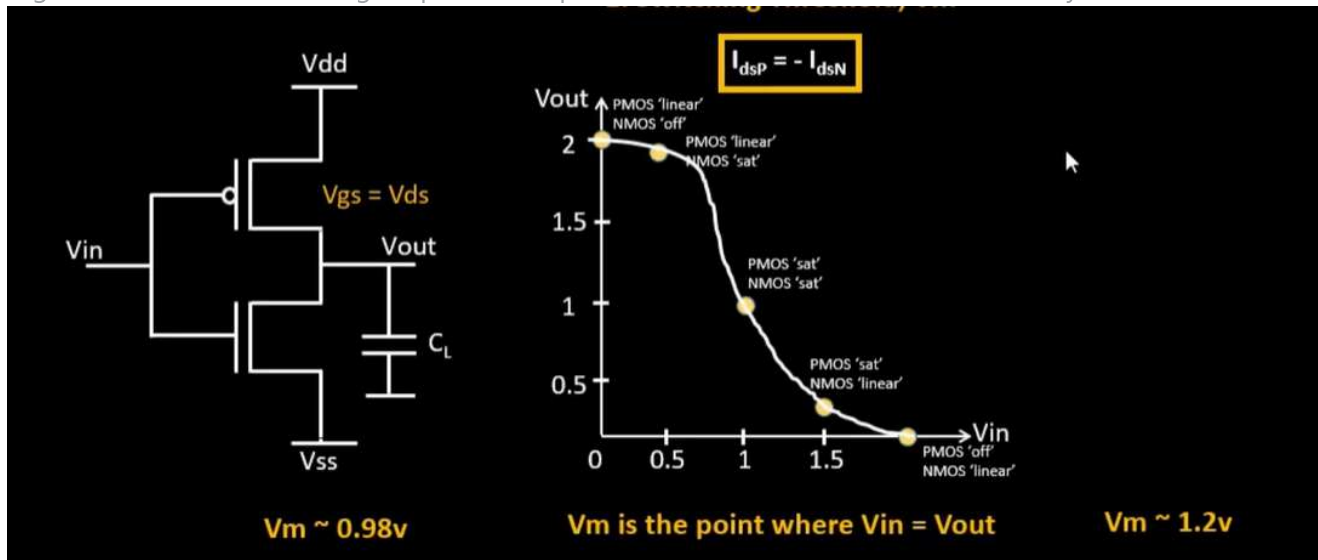
When we compare two CMOS inverters designed with different PMOS and NMOS W/L ratios, the overall shape of the Voltage Transfer Characteristic (VTC) remains the same. In both cases, the inverter clearly transitions from high output to low output with a sharp switching region. This confirms the structural robustness of the CMOS inverter — resizing devices does not distort the VTC shape, but it does shift the switching point.



To determine the switching threshold voltage (V_m), a 45° line ($V_{out} = V_{in}$) is drawn on the VTC curve. The point where this line intersects the VTC gives V_m . In the first case, V_m is approximately 0.9 V, while in the second case, V_m shifts to around 1.2 V. This shift occurs because changing the W/L ratio alters the relative strength of the PMOS and NMOS devices. A stronger PMOS shifts the switching threshold toward higher voltages, while a stronger NMOS shifts it lower.



The region around V_m is particularly important. At this operating point, both PMOS and NMOS are in saturation and conduct simultaneously. This creates a direct current path from VDD to ground, resulting in maximum current flow. Although this condition exists only during switching, it represents a critical region because it leads to higher power dissipation and determines the inverter's delay characteristics.



Overall, while device sizing shifts the switching threshold, the fundamental behavior of the CMOS inverter remains intact. This demonstrates both the robustness of CMOS logic and the sensitivity of its switching point to transistor sizing.

Lecture 2: Analytical expression of V_m as a function of $(W/L)_n$ and $(W/L)_p$

Analytical Expression of Switching Threshold (V_m)

Condition at Switching Threshold

At the switching threshold:

$$V_{in} = V_{out} = V_m$$

At this point, the pull-up and pull-down currents are equal in magnitude:

$$I_{dsp} = -I_{dsn}$$

or

$$I_{dsp} + I_{dsn} = 0$$

This means both transistors conduct equal current.

Drain Current Expressions

Define:

$$k_n = \mu_n * C_{ox} * (W_n/L_n)$$

$$k_p = \mu_p * C_{ox} * (W_p/L_p)$$

NMOS current at V_m

$$I_{dsn} = k_n * [(V_m - V_{tn}) * V_{dsatn} - (V_{dsatn}^2 / 2)]$$

PMOS current at V_m

$$I_{dsp} = k_p * [(V_m - V_{DD} - V_{tp}) * V_{dsatp} - (V_{dsatp}^2 / 2)]$$

Current Balance Equation

At switching threshold:

$$k_p * [(V_m - V_{DD} - V_t) * V_{dsatp} - (V_{dsatp}^2 / 2)]$$

+

$$k_n * [(V_m - V_t) * V_{dsatn} - (V_{dsatn}^2 / 2)]$$

$$= 0$$

Solving this equation for V_m gives:

Final Expression

$$V_m = (R * V_{DD}) / (1 + R)$$

where

$$R = (k_p * V_{dsatp}) / (k_n * V_{dsatn})$$

Substituting k_n and k_p :

$$R = [(W_p/L_p) * K_p' * V_{dsatp}] / [(W_n/L_n) * K_n' * V_{dsatn}]$$

Interpretation

If $R = 1$

→ $V_m = V_{DD} / 2$ (perfectly symmetric inverter)

If $R > 1$

→ $V_m > V_{DD} / 2$ (PMOS stronger)

If $R < 1$

→ $V_m < V_{DD} / 2$ (NMOS stronger)

Key Insight

The switching threshold V_m is directly controlled by:

$$(W_p/L_p) / (W_n/L_n)$$

Thus, transistor sizing shifts V_m without changing the overall VTC shape.

Lecture 3: Analytical expression of $(W/L)_n$ and $(W/L)_p$ as a function of V_m

Reverse Design of CMOS Inverter: Finding $(W/L)_p$ and $(W/L)_n$ for a Given V_m

Objective

Given a desired switching threshold V_m , we want to determine the required (W/L) ratio of PMOS and NMOS such that:

$$V_m = V_{DD} / 2$$

For this case:

$$V_{DD} = 2.5 \text{ V}$$

$$\text{Required } V_m = 1.25 \text{ V}$$

Instead of finding V_m from W/L , we now move in **reverse direction**: We calculate W/L ratios such that V_m is fixed.

Starting Condition at Switching Threshold

At V_m :

$$V_{in} = V_{out} = V_m$$

The current balance condition is:

$$I_{dsn} = -I_{dsp}$$

or

$$I_{dsp} + I_{dsn} = 0$$

Both transistors conduct equal magnitude current.

Expanded Current Equation

Using velocity saturation current model:

$$I_{dsP} = -I_{dsN}$$

$$I_{dsP} + I_{dsN} = 0$$

$$k_p \cdot \left[([V_m - V_{dd} - V_t] \cdot V_{dsatp}) - \frac{V_{dsatp}^2}{2} \right] + k_n \cdot \left[([V_m - V_t] \cdot V_{dsatn}) - \frac{V_{dsatn}^2}{2} \right] = 0$$

Alternatively, the required ratio of PMOS v/s NMOS transistor size can be derived, such that V_m is set

$$\frac{K_p \cdot V_{dsatp}}{K_n \cdot V_{dsatn}} = \frac{([V_m - V_t]) - \frac{V_{dsatn}}{2}}{([-V_m + V_{dd} + V_t]) + \frac{V_{dsatp}}{2}}$$

where:

$$k_n = K_n' \cdot (W_n / L_n)$$

$$k_p = K_p' \cdot (W_p / L_p)$$

K_n' and K_p' are technology gain parameters obtained from model files.

Required Ratio for Setting V_m

Rearranging the equation gives:

$$(k_p \cdot V_{dsatp}) / (k_n \cdot V_{dsatn}) =$$

$$[(V_m - V_t) - (V_{dsatn} / 2)]$$

$$[(-V_m + V_{DD} + V_t) + (V_{dsatp} / 2)]$$

Substituting gain factors:

$$\frac{\left(\frac{W_p}{L_p}\right) K_{p'} \cdot V_{dsatp}}{\left(\frac{W_n}{L_n}\right) K_{n'} \cdot V_{dsatn}} = \frac{([V_m - V_t]) - \frac{V_{dsatn}}{2}}{([-V_m + V_{dd} + V_t]) + \frac{V_{dsatp}}{2}}$$

Final Expression for Sizing Ratio

$$\frac{\left(\frac{W_p}{L_p}\right)}{\left(\frac{W_n}{L_n}\right)} = \frac{K_{n'} \cdot V_{dsatn}([V_m - V_t]) - \frac{V_{dsatn}}{2}}{K_{p'} \cdot V_{dsatp}([-V_m + V_{dd} + V_t]) + \frac{V_{dsatp}}{2}}$$

Important Insight

- All parameters on RHS are constants from model file.
- Only V_m is user-defined.
- Once V_m is fixed (e.g., 1.25 V), required $(W_p/L_p)/(W_n/L_n)$ can be computed.
- This tells how much wider PMOS must be than NMOS.

Design Strategy

To study inverter behavior, we vary W/L ratios such as:

W_p/L_p	$x \cdot W_n/L_n$
W_p/L_p	W_n/L_n
W_p/L_p	$2W_n/L_n$
W_p/L_p	$3W_n/L_n$
W_p/L_p	$4W_n/L_n$
W_p/L_p	$5W_n/L_n$

By simulating these ratios, we observe:

- Shift in switching threshold
- Change in VTC symmetry
- Variation in rise/fall delays
- Power consumption differences

Lecture 4: Static and Dynamic Simulation of CMOS Inverter

Static Behavior – VTC Analysis

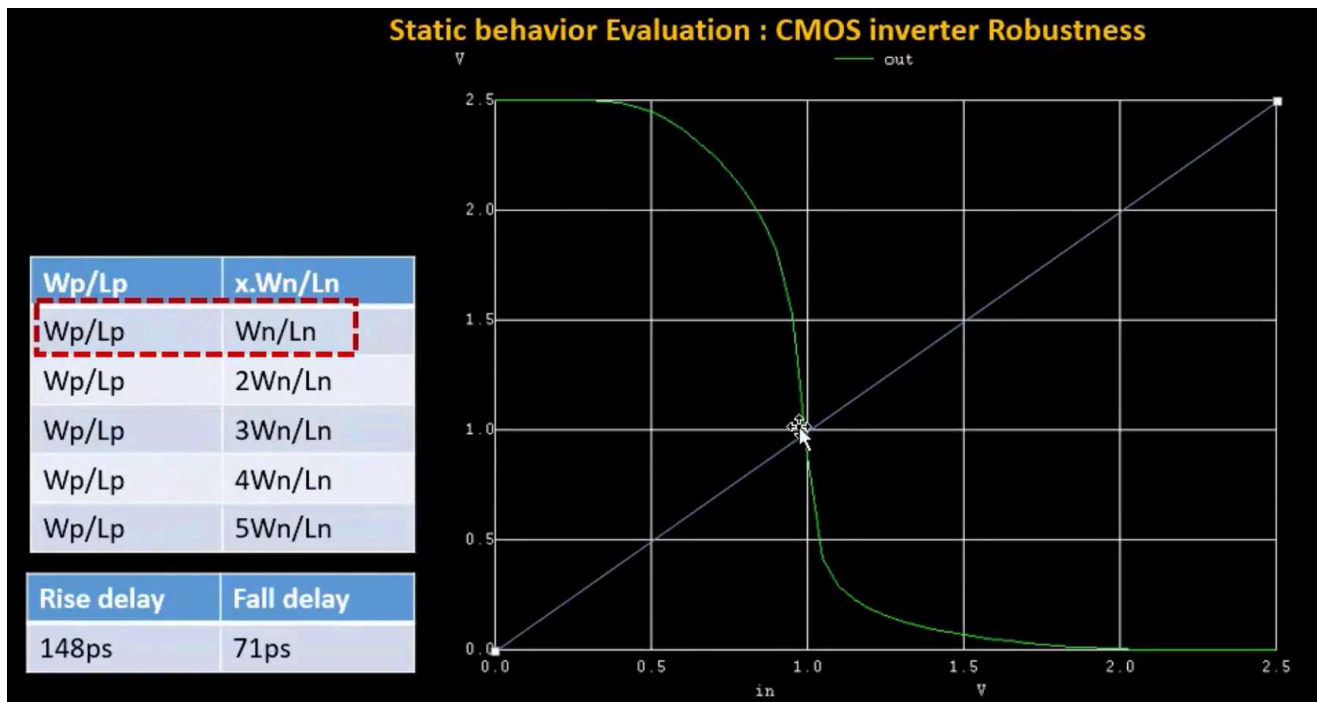
For this case, both transistors are sized equally:

$$(W/L)_n = (W/L)_p = 1.5$$

The Voltage Transfer Characteristic (VTC) is plotted along with a 45° reference line ($V_{out} = V_{in}$). The intersection of the VTC curve with the 45° line gives the switching threshold voltage (V_m).

With equal sizing, the switching threshold is observed close to the mid-supply region.

However, equal W/L does not necessarily guarantee perfectly balanced delay, since mobility differences still exist.



The VTC maintains the typical CMOS shape:

- Output $\approx V_{DD}$ for low input
- Output ≈ 0 for high input
- Sharp transition in the switching region

This confirms the static robustness of the CMOS inverter.

Dynamic Behavior – Transient Analysis

Transient simulation is used to evaluate propagation delays.

A pulse input is applied, and output response is observed.

Using the 50% VDD method:

Rise delay (t_{pLH}) = 148 ps

Fall delay (t_{pHL}) = 71 ps

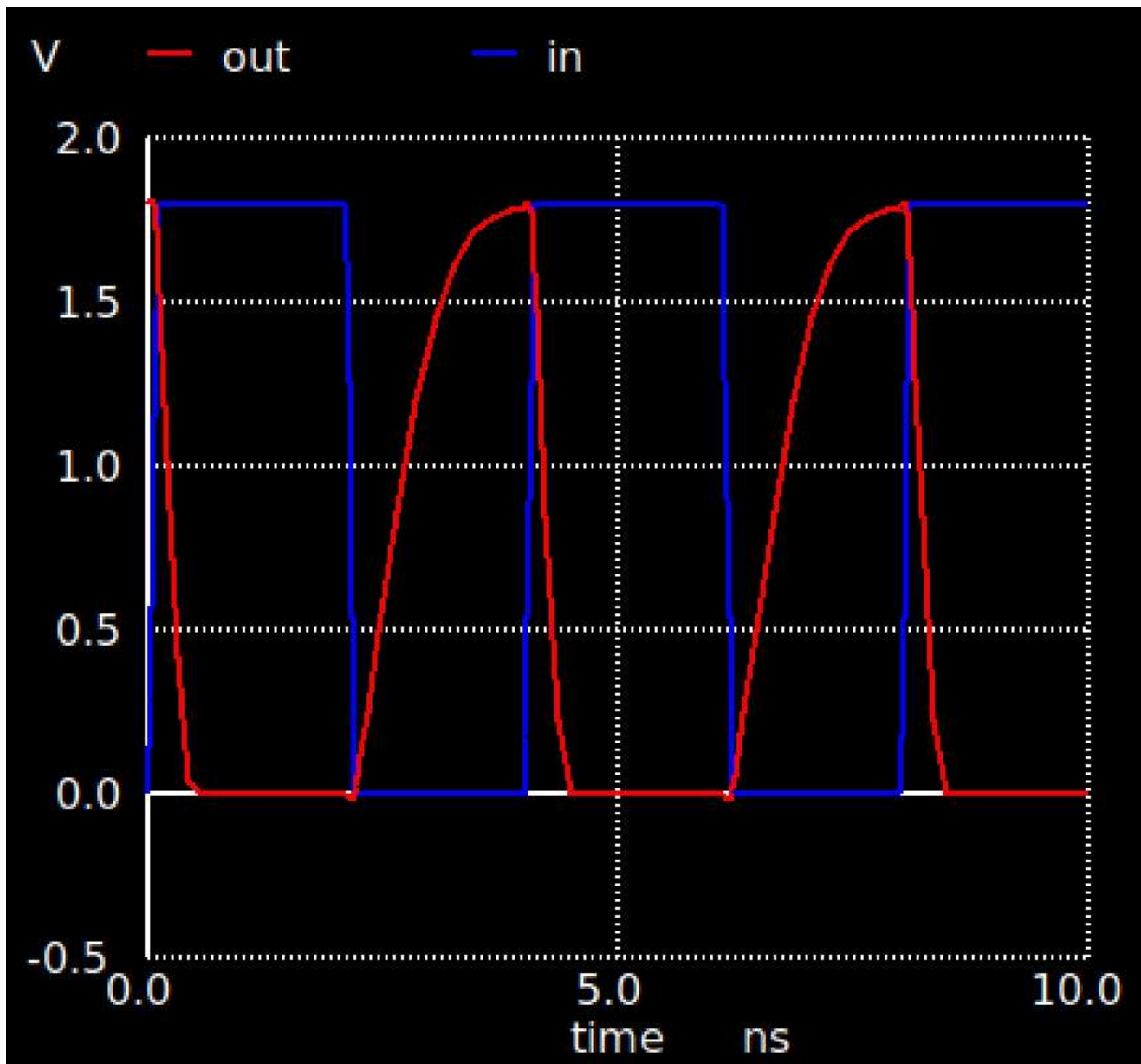
The rise delay is larger than the fall delay, which indicates that even though $(W/L)_n = (W/L)_p = 1.5$, the inverter is not dynamically symmetric. This happens due to mobility differences between electrons and holes.

Lecture 5: Static and Dynamic simulation of CMOS inverter with increased PMOS width

Effect of PMOS Width Scaling on CMOS Inverter Performance

($L_p = L_n = 0.25 \mu\text{m}$, $W_n = 1 \mu\text{m}$)

Case 1: $W_p = 1 \mu\text{m}$

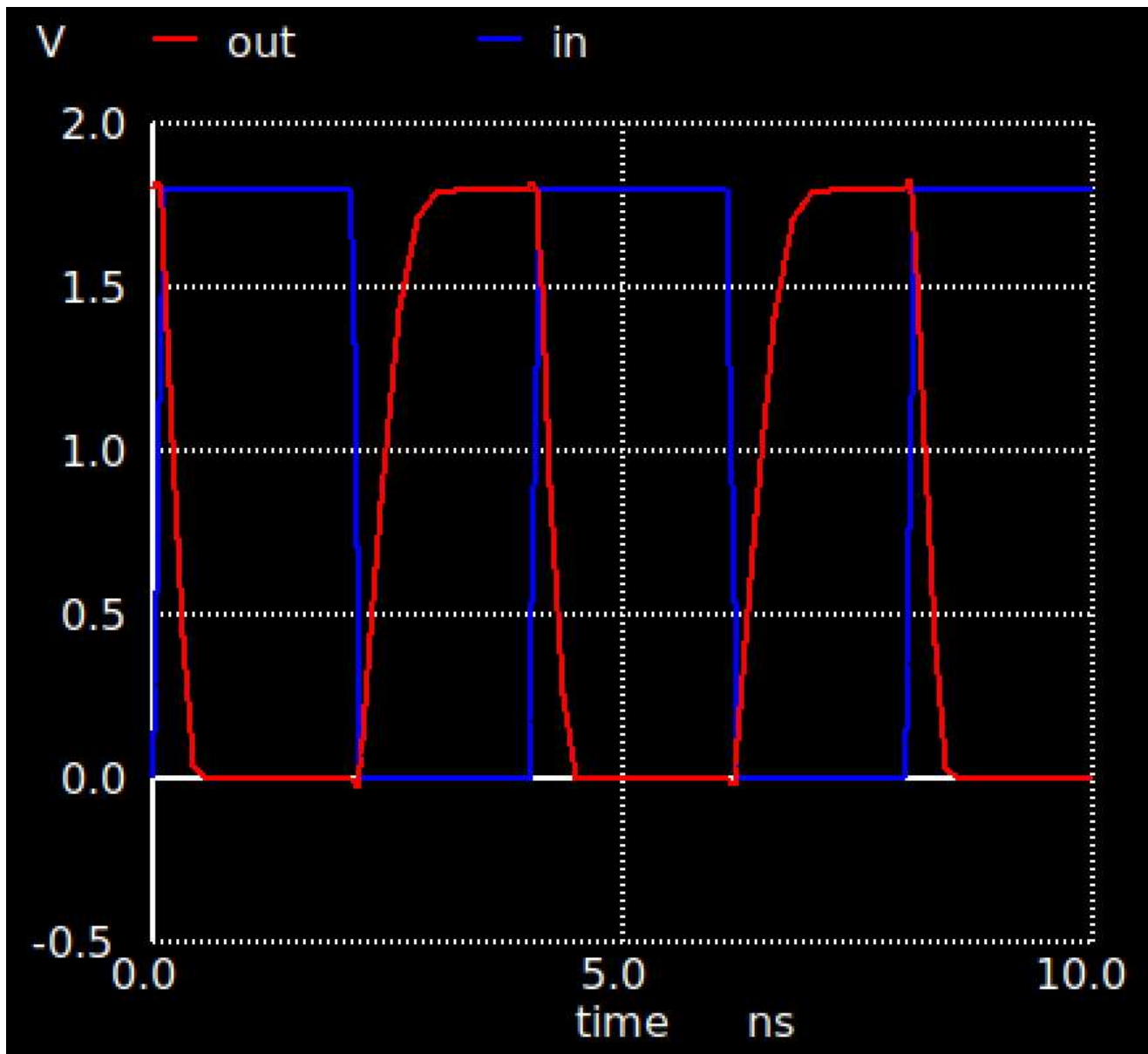


Rise delay = $2.6694 - 2.1506 = 0.5188$ ns

Fall delay = $4.2116 - 4.0502 = 0.1614$ ns

$V_m = 0.65$ V

Case 2: $W_p = 2 \mu\text{m}$

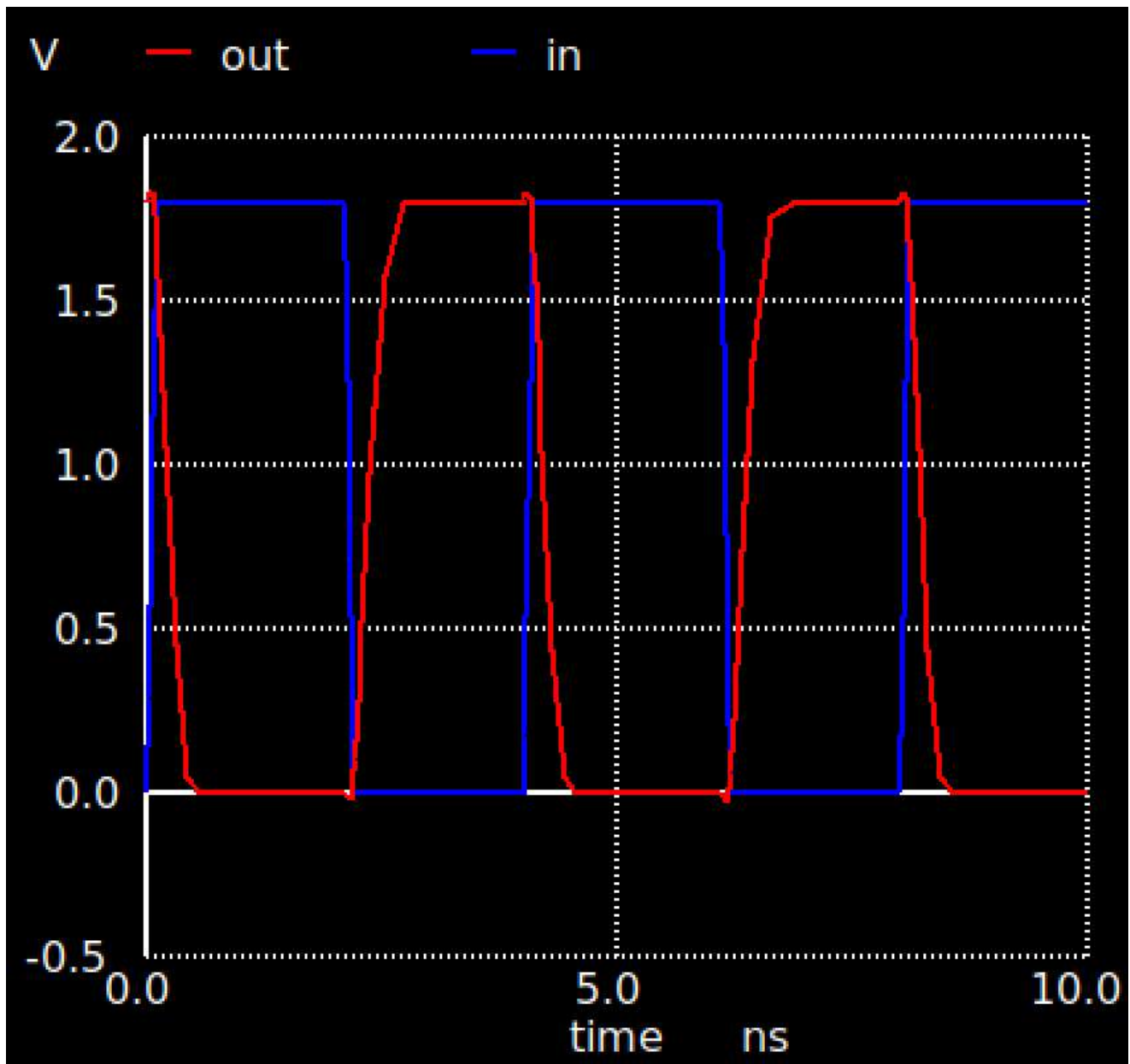


Rise delay = $2.4270 - 2.1495 = 0.2775$ ns

Fall delay = $4.2163 - 4.0494 = 0.1669$ ns

$V_m = 0.684$ V

Case 3: $W_p = 3 \mu\text{m}$

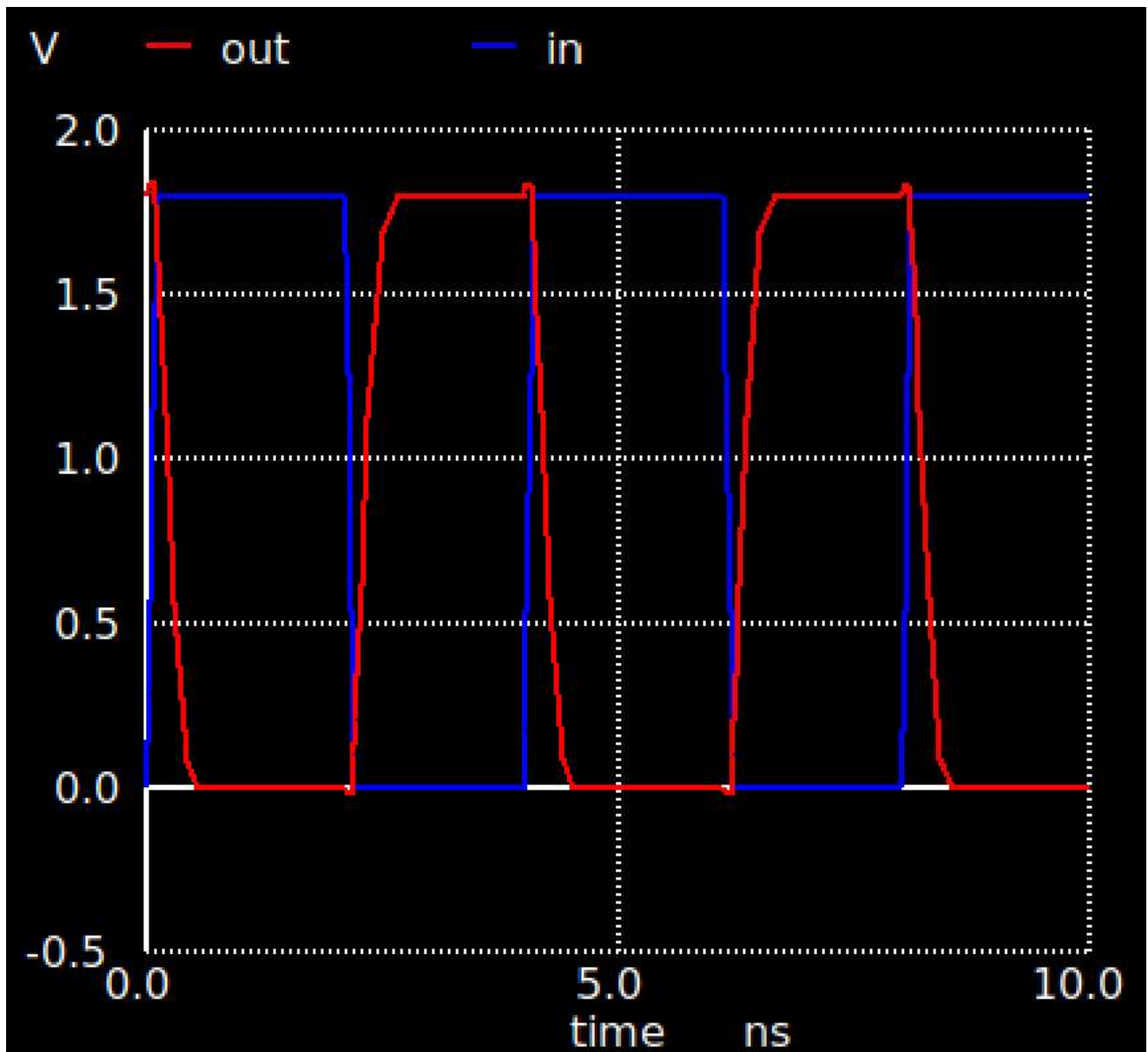


Rise delay = 2.3473 – 2.1493 = 0.198 ns

Fall delay = 4.2219 – 4.0489 = 0.173 ns

$V_m = 0.71$ V

Case 4: $W_p = 4 \mu\text{m}$

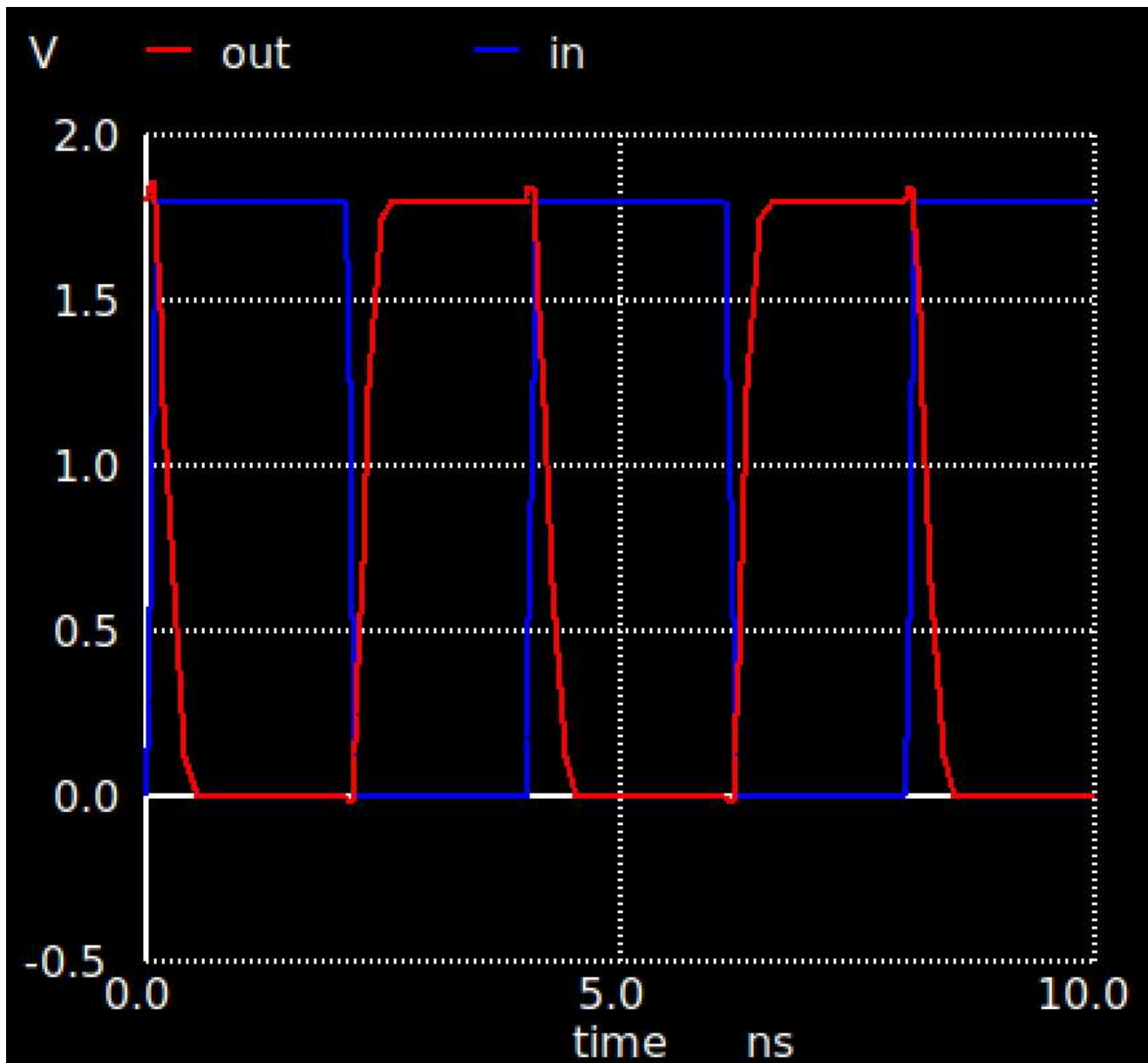


Rise delay = 2.3090 – 2.1494 = 0.1596 ns

Fall delay = 4.2236 – 4.0493 = 0.1743 ns

$V_m = 0.722$ V

Case 5: $W_p = 5 \mu\text{m}$



Rise delay = 2.2854 – 2.1505 = 0.1349 ns

Fall delay = 4.2264 – 4.0505 = 0.1759 ns

$V_m = 0.73$ V

Observation

- Increasing W_p reduces rise delay significantly (0.5188 ns → 0.1349 ns).
- Fall delay remains nearly constant (~0.16–0.1759 ns).
- V_m shifts upward (0.65 V → 0.73 V).
- Larger PMOS improves pull-up strength and balances delays.

Lecture 6: Applications of CMOS inverter in clock network and STA

Final Observations from CMOS Inverter Experiment

From the experimental data collected for different PMOS widths, we observe that varying the $(W/L)_p$ while keeping $(W/L)_n$ constant changes the switching threshold (V_m) and delays, but the overall VTC shape remains intact.

Robustness of CMOS Inverter

During fabrication, slight variations in transistor dimensions are unavoidable. However, the CMOS inverter demonstrates strong robustness:

- Even with changes in PMOS width, V_m shifts only moderately.
- The inverter continues to function correctly.
- The VTC maintains its characteristic shape.

This confirms that CMOS logic is tolerant to small process variations.

Symmetry Condition

W_p/L_p	$x.W_n/L_n$	Rise delay	Fall delay	V_m
W_p/L_p	W_n/L_n	148ps	71ps	0.99v
W_p/L_p	$2W_n/L_n$	80ps	76ps	1.2v
W_p/L_p	$3W_n/L_n$	57ps	80ps	1.25v
W_p/L_p	$4W_n/L_n$	45ps	84ps	1.35v
W_p/L_p	$5W_n/L_n$	37ps	88ps	1.4v

When:

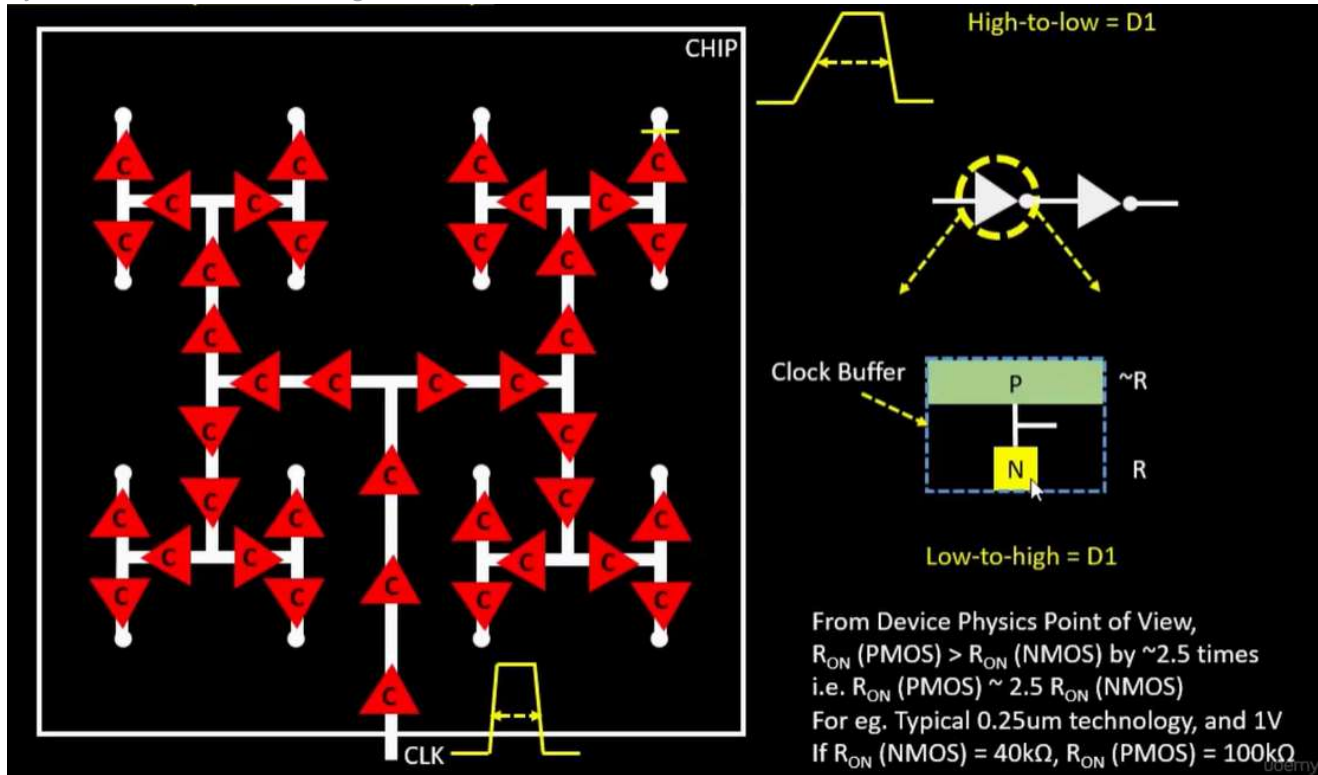
$$(W/L)_p \approx 2 \times (W/L)_n$$

the rise delay and fall delay become approximately equal.

This indicates balanced pull-up and pull-down strengths, resulting in:

- Equal propagation delays
- Symmetric switching behavior

By simulation, an exact sizing ratio can be determined where:



Rise delay = Fall delay

This condition defines the **symmetry point** of the CMOS inverter.

Design Insight

- V_m shifts gradually with sizing.
- Delay symmetry depends strongly on PMOS/NMOS ratio.
- CMOS inverter behavior remains stable under moderate sizing variation.
- Device sizing can be tuned depending on datapath requirements.

Different standard cells may use different sizing strategies based on speed, power, or robustness needs.

NgspiceSky130-Day4-CMOS Noise Margin robustness evaluation

Chapter 1: Static behaviour evaluation-CMOS inverter robustness-Noise Margin

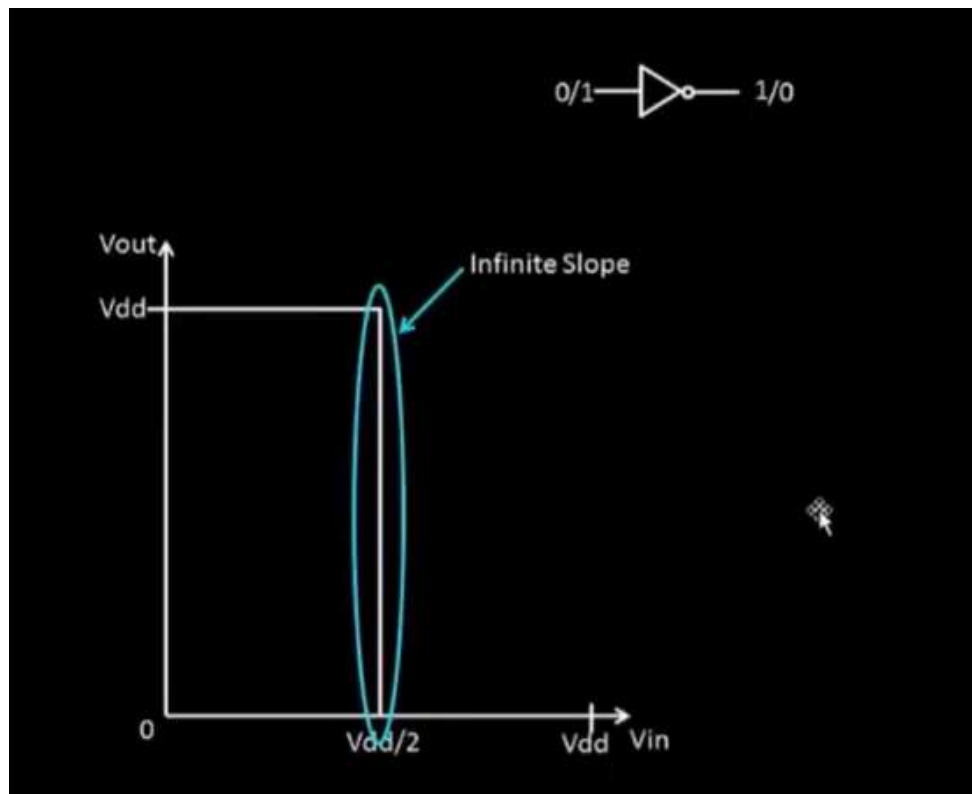
Lecture 1: Introduction to Noise Margin.

Noise Margin – Definition

Noise Margin is the measure of how much unwanted electrical noise a logic circuit can tolerate at its input without producing an incorrect output.

It indicates the robustness of a CMOS inverter against disturbances.

Ideal Inverter Behavior

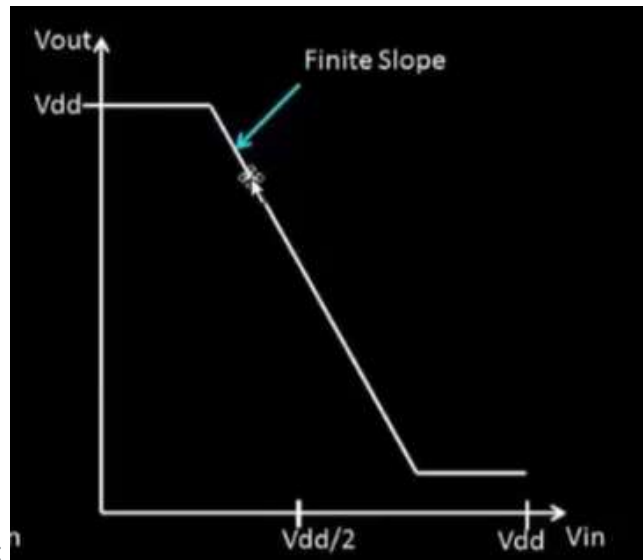


In an ideal inverter:

- Input 0 → Output 1
- Input 1 → Output 0
- The switching transition occurs exactly at $V_{DD}/2$
- The slope at switching point is infinite

This means the output changes instantly with a small change in input.

Practical Inverter Behavior

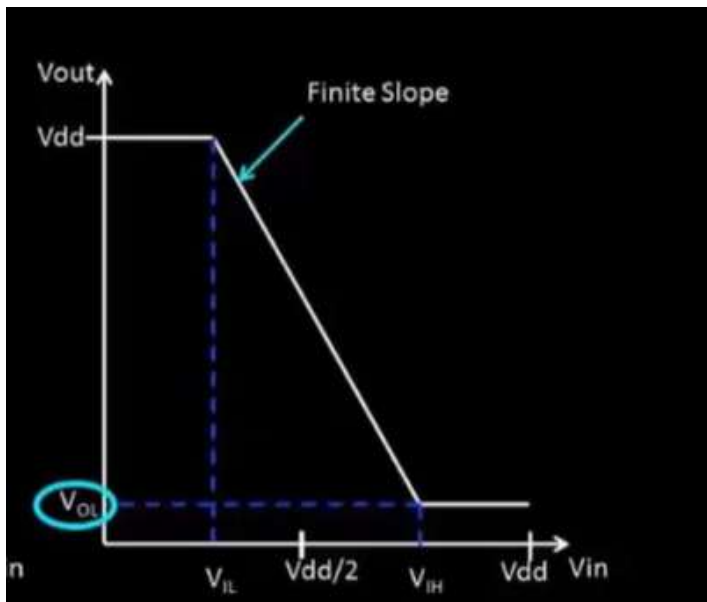


In real CMOS inverters:

- The slope at the switching region is finite.
- Due to parasitic resistances and capacitances, switching takes time.
- The VTC transition region has a finite slope instead of a vertical line.

This finite slope defines the noise tolerance limits.

Important Voltage Levels



From the VTC curve:

- $V_{OH} \rightarrow$ Output High Voltage
- $V_{OL} \rightarrow$ Output Low Voltage
- $V_{IL} \rightarrow$ Maximum input voltage recognized as LOW
- $V_{IH} \rightarrow$ Minimum input voltage recognized as HIGH

Behavior:

- If input $\in [0, V_{IL}] \rightarrow$ Output = V_{OH}
- If input $\in [V_{IH}, V_{DD}] \rightarrow$ Output = V_{OL}
- If input $\in (V_{IL}, V_{IH}) \rightarrow$ Transition region (both devices conduct)

Key Insight

Noise margin depends on how far:

- V_{OH} is from V_{IH}
- V_{IL} is from V_{OL}

A steeper VTC (higher gain near switching) results in better noise immunity.

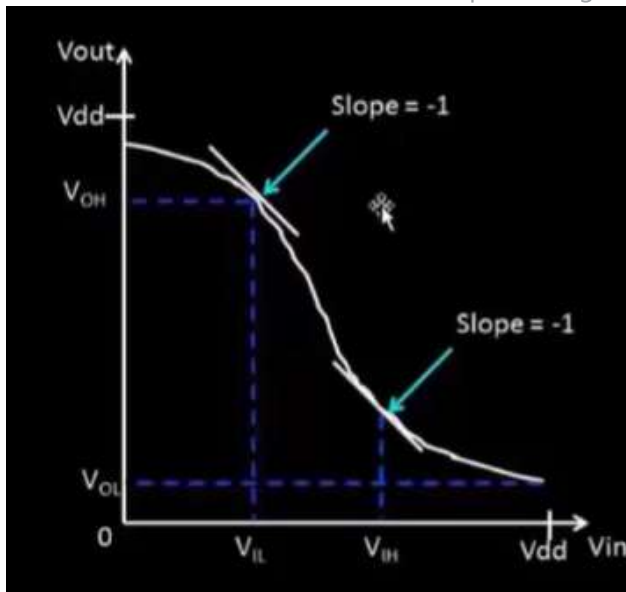
Noise margin determines the reliability of digital logic under noise conditions.

Lecture 2: Noise Margin voltage paramters

CMOS Inverter – Noise Margin & VTC Regions

1. Voltage Regions from VTC

The CMOS inverter VTC divides the input voltage into three regions:



Logic 0 Region

- Input range: $0 \rightarrow V_{IL}$
- Output: V_{OH} (Output High)

Transition Region

- Input range: $V_{IL} \rightarrow V_{IH}$
- Output is switching
- Both PMOS and NMOS conduct

- Region of high gain

Logic 1 Region

- Input range: $V_{IH} \rightarrow V_{DD}$
- Output: V_{OL} (Output Low)

2. Important Voltage Levels

- $V_{OH} \rightarrow$ Output High voltage
- $V_{OL} \rightarrow$ Output Low voltage
- $V_{IL} \rightarrow$ Maximum input voltage recognized as LOW
- $V_{IH} \rightarrow$ Minimum input voltage recognized as HIGH

For proper cascading of logic gates:

- $V_{OL} < V_{IL}$
- $V_{OH} > V_{IH}$

These conditions ensure correct logic detection in the next stage.

3. Practical VTC Behavior

In real CMOS circuits:

- V_{OH} is not exactly V_{DD}
- V_{OL} is not exactly 0 V
- The transition curve is smooth (finite slope)
- Non-idealities arise due to parasitic resistance and capacitance

Unlike an ideal inverter (vertical transition), practical VTC has finite slope.

4. Slope in Transition Region

- Slope is negative because increasing input decreases output.
- Around switching point, magnitude of slope ≈ -1 .
- This region determines noise margin and gain.

Higher magnitude of slope $\rightarrow$ Better noise immunity. CMOS inverter robustness depends on how sharply the transition occurs and how well V_{OH} , V_{OL} , V_{IL} , and V_{IH} are separated.

Lecture 3: Noise margin equation and summary

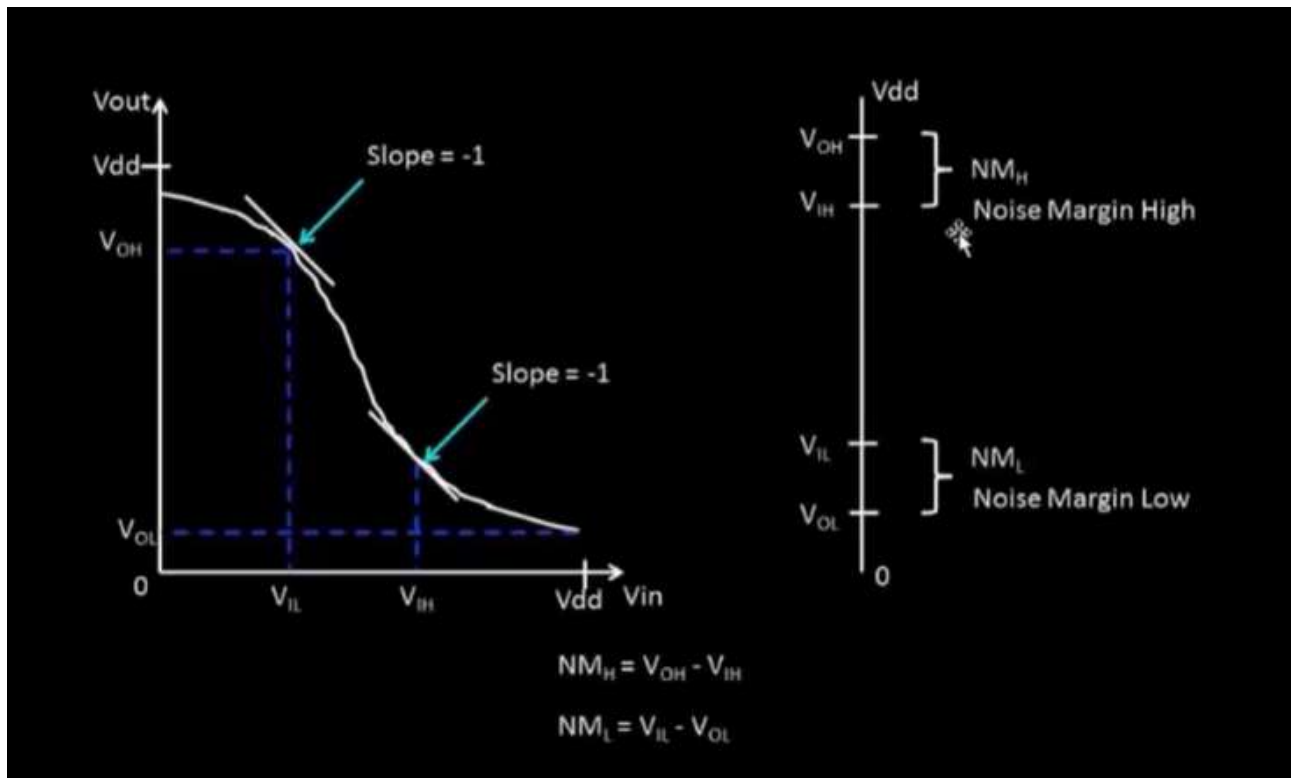
Noise Margin – CMOS Inverter

1. Noise Margin Concept

Noise margin represents the tolerance of a CMOS inverter against unwanted voltage disturbances (noise).

It defines how much noise can be added to a logic signal without causing incorrect interpretation at the next stage.

2. Important Voltage Levels from VTC



From the Voltage Transfer Characteristic (VTC):

- $V_{OH} \rightarrow$ Output High voltage
- $V_{OL} \rightarrow$ Output Low voltage
- $V_{IL} \rightarrow$ Maximum input voltage recognized as logic 0
- $V_{IH} \rightarrow$ Minimum input voltage recognized as logic 1

The slope of the VTC in the transition region is negative. At V_{IL} and V_{IH} , the slope magnitude is approximately:

$$dV_{out}/dV_{in} = -1$$

These points define the noise boundaries.

3. Noise Margin High (NMH)

Noise Margin High is the tolerance when the signal is at logic 1.

$$NMH = V_{OH} - V_{IH}$$

Meaning: If the output is logic 1, any noise added up to NMH will still be correctly interpreted as logic 1 at the next stage.

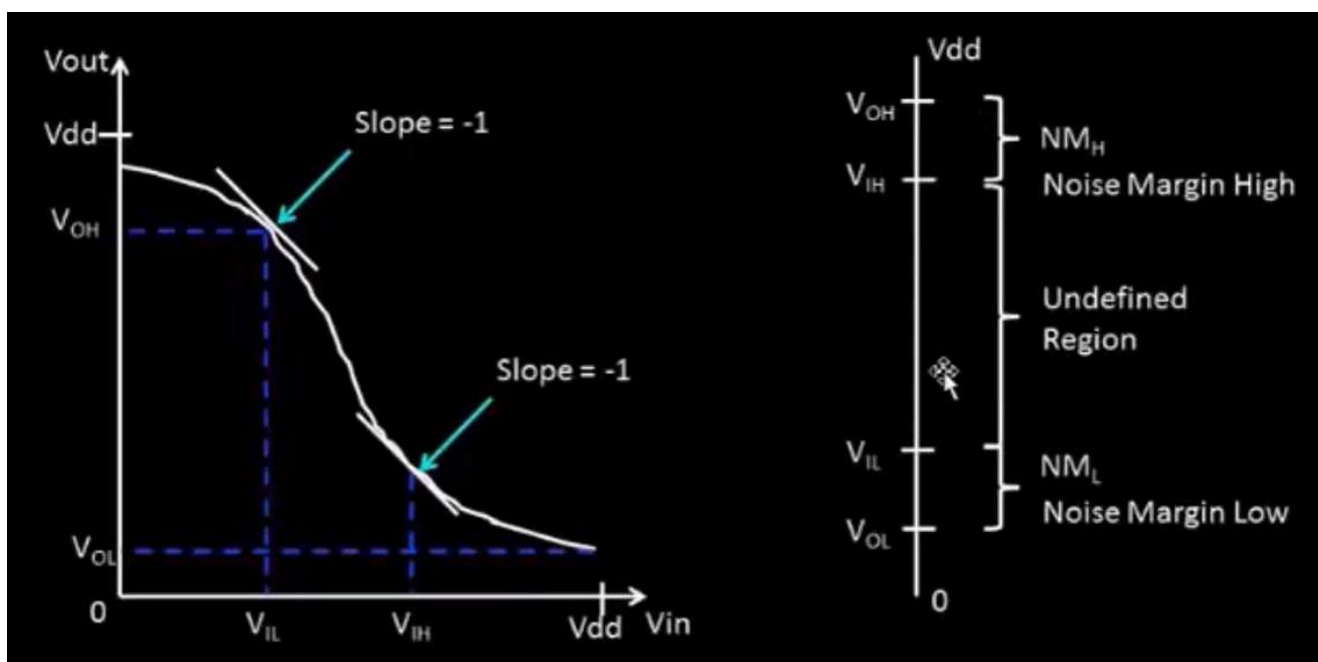
4. Noise Margin Low (NML)

Noise Margin Low is the tolerance when the signal is at logic 0.

$$NML = V_{IL} - V_{OL}$$

Meaning: If the output is logic 0, any noise added up to NML will still be correctly interpreted as logic 0.

5. Undefined Region

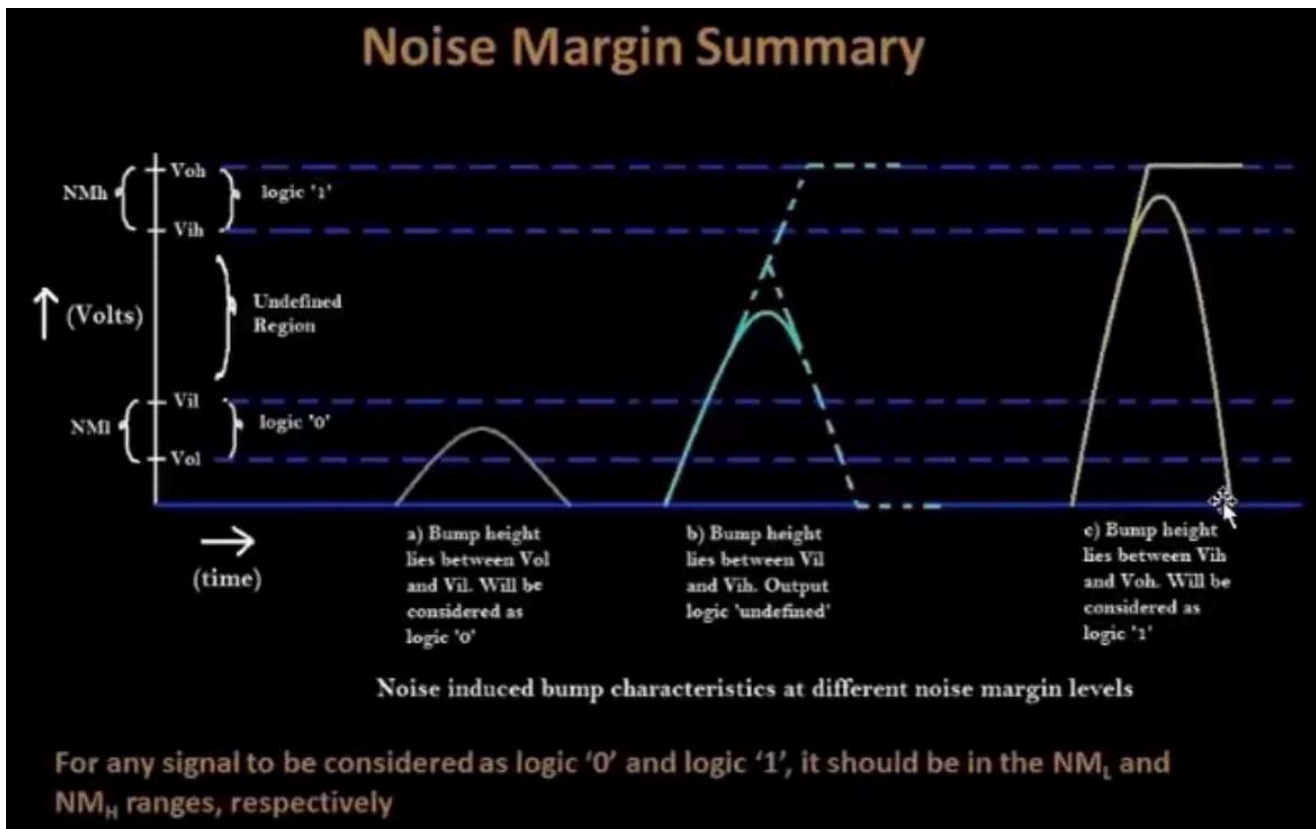


The region between V_{IL} and V_{IH} is called the undefined region.

- Any signal in this region is indeterminate.
- It may be interpreted as logic 0 or logic 1.
- Circuits must avoid operating in this region.

This is the transition zone where both PMOS and NMOS conduct.

6. Noise Bump Interpretation



From the 1D noise margin plot:

Case 1: Noise bump between V_{OL} and V_{IL}

→ Still interpreted as logic 0 (safe)

Case 2: Noise bump between V_{IL} and V_{IH}

→ Falls in undefined region (unsafe)

Case 3: Noise bump between V_{IH} and V_{OH}

→ Still interpreted as logic 1 (safe)

Thus, for reliable digital operation:

- Logic 0 must remain within NM_L range
- Logic 1 must remain within NM_H range

Lecture 4: Noise margin variation with respect to PMOS width

Noise Margin Evaluation using PMOS/NMOS Size Scaling

Noise margins of a CMOS inverter are evaluated using SPICE DC transfer simulations.

From the VTC curve, the slope ($dV_{out}/dV_{in} = -1$) points are identified to extract:

- V_{IL}
- V_{IH}

- V_{OH}
- V_{OL}

Using these values:

$$NMH = V_{OH} - V_{IH}$$

$$NML = V_{IL} - V_{OL}$$

Effect of PMOS Size Scaling

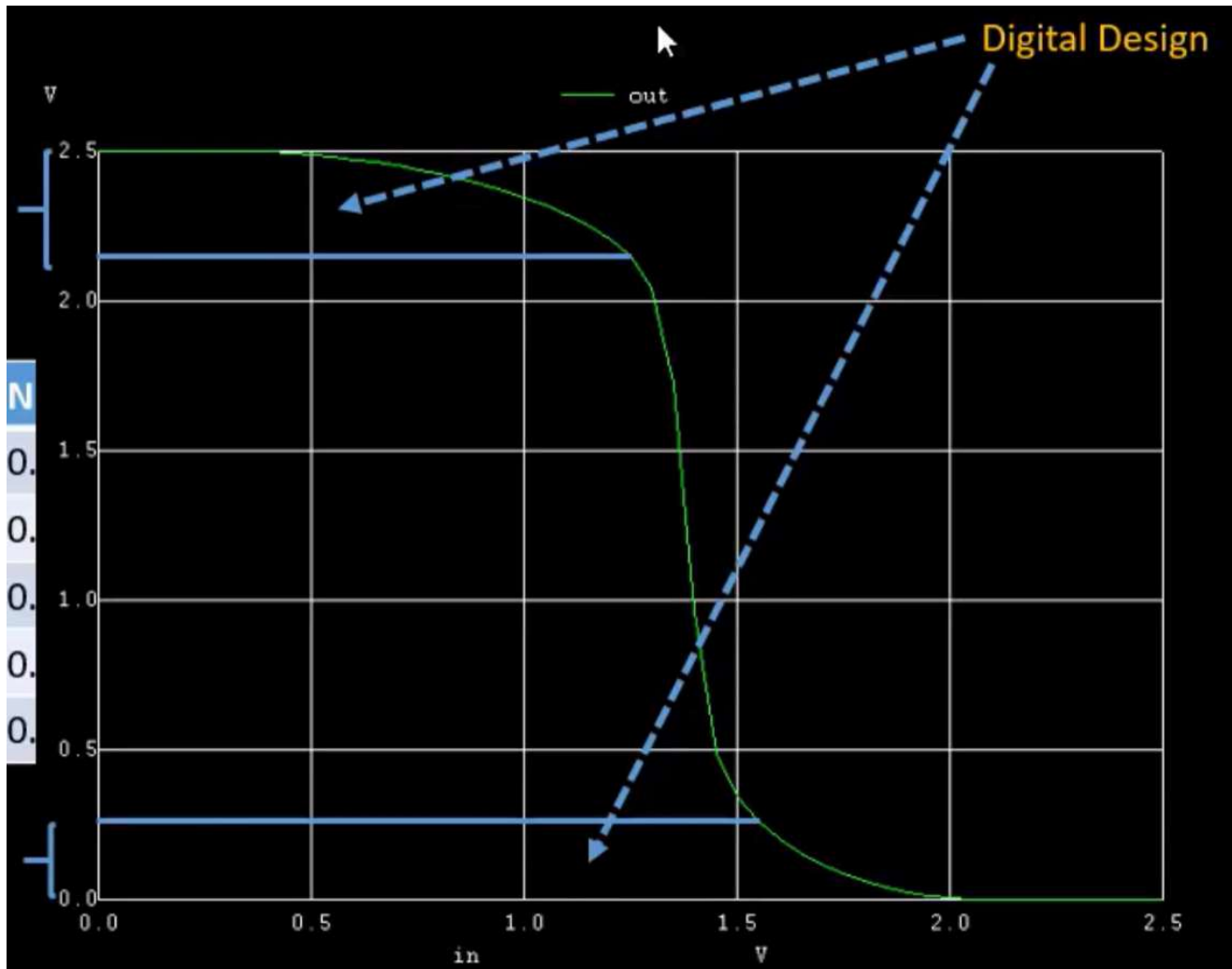
When PMOS width is increased relative to NMOS:

- NMH (Noise Margin High) increases.
- Example: NMH increases from 0.3 V $\rightarrow$ 0.42 V when PMOS = $5 \times (W/L)_n$.
- NML slightly decreases beyond $3 \times (W/L)_n$.

Thus, increasing PMOS strength shifts the switching point and improves high-level noise immunity, but slightly affects low-level margin at higher scaling.

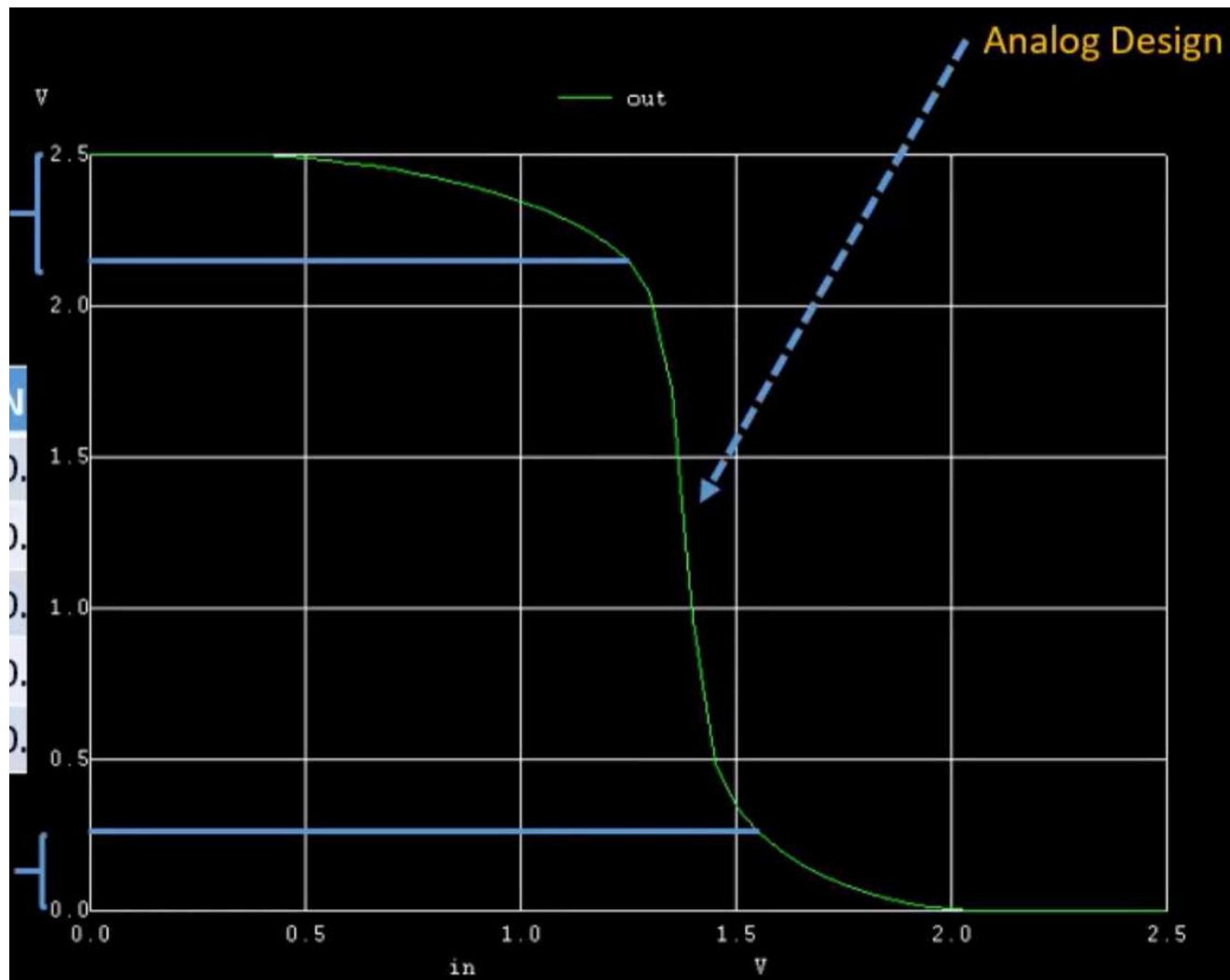
Digital vs Analog Perspective

Digital Design:



- Operates in strong logic regions (VOH and VOL). - Tolerates small variations in noise margin. - Robust even with $\pm 5\%$ fabrication variations.

Analog Design:



- Operates in transition region. - Sensitive to slope and switching point shifts.

Fabrication Impact

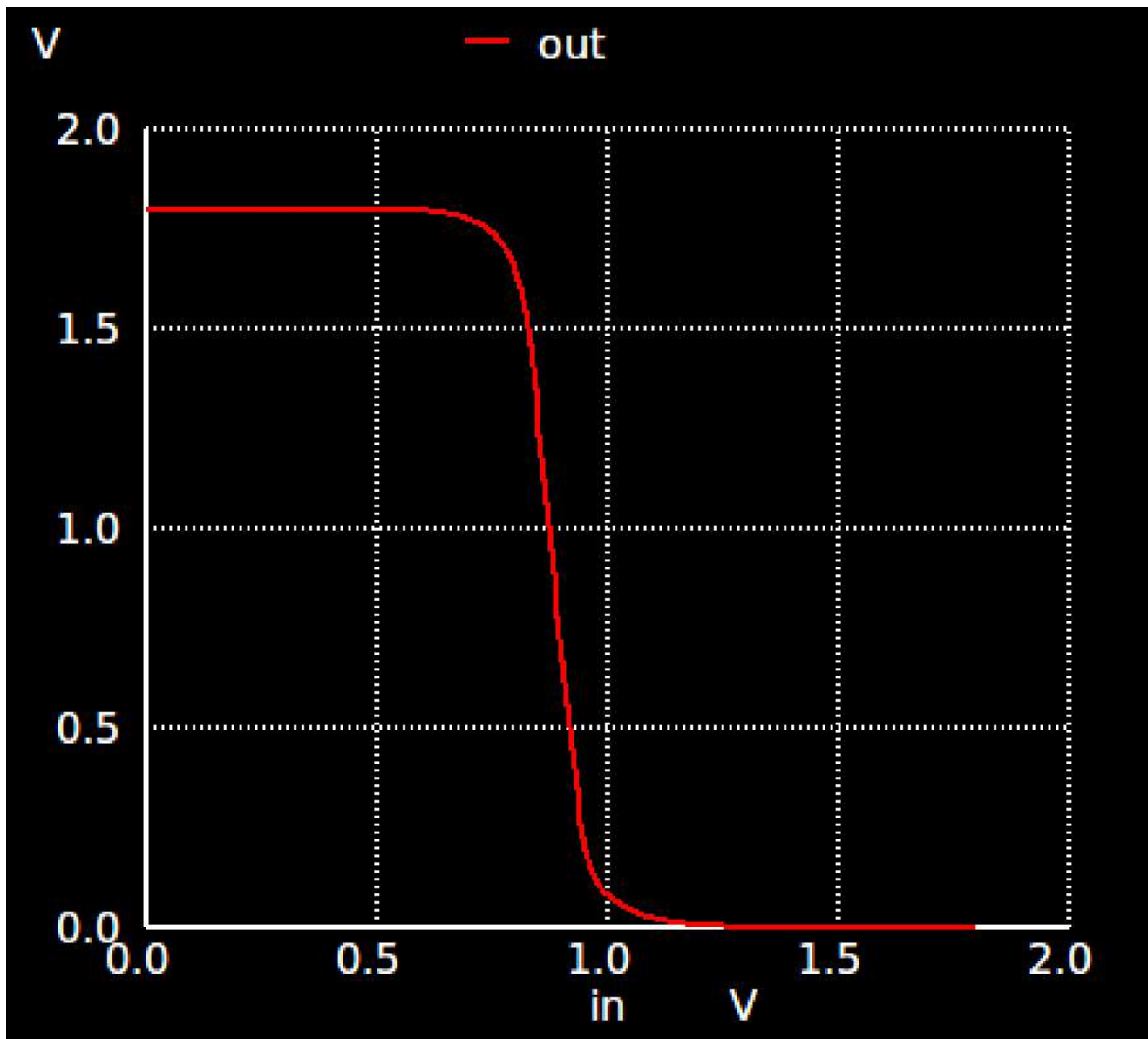
Process variations during fabrication cause:

- Slight changes in W/L ratios
- Small shifts in V_{IL} , V_{IH}
- Variation in NMH and NML

However, CMOS inverter remains robust within optimal sizing ratios.

Lecture 5: Sky130 Noise margin labs

Noise Margin Calculation (SPICE)



W/L ratio: $(W/L)_p / (W/L)_n = 2.77$

DC sweep: V_{in} from 0 → 1.8 V

Step size = 0.01 V

From VTC curve, locate points where slope $(dV_{out}/dV_{in}) = -1$:

- X-axis → V_{IL} , V_{IH}
- Y-axis → V_{OH} , V_{OL}

Extracted values: $V_{OH} = 1.70952$ V

$V_{IH} = 0.98778$ V

$V_{IL} = 0.7733$ V

$V_{OL} = 0.09523$ V

Noise Margins:

$NMH = V_{OH} - V_{IH}$

$NMH = 1.70952 - 0.98778 = 0.72$ V

$$NML = V_{IL} - V_{OL}$$

$$NML = 0.7733 - 0.09523 = 0.68 \text{ V}$$

NgspiceSky130-Day5-CMOS power supply and device variation robustness evaluation

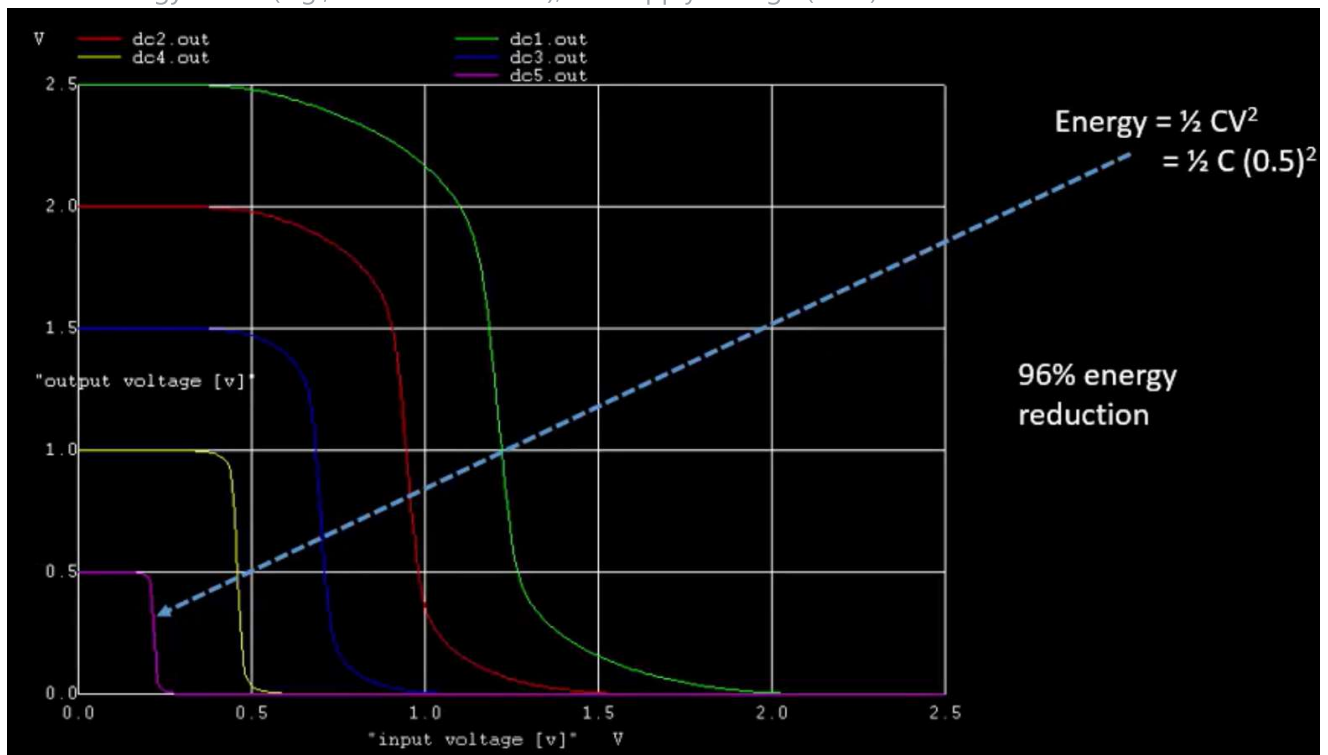
Chapter 1: Static behaviour evaluation-CMOS inverter robustness-Power supply variation

Lecture 1: Smart SPICE simulations for power supply variations

CMOS Inverter Robustness under Power Supply Scaling

To evaluate inverter robustness, power supply scaling must be considered.

As technology scales (e.g., 250 nm → 20 nm), the supply voltage (VDD) also reduces:



- Older nodes: 2.5 V
- Scaled nodes: 1 V, 0.9 V, 0.7 V

The goal is to verify that CMOS inverter behavior remains consistent under reduced VDD.

Experimental Setup (SPICE)

CMOS inverter sizing:

$(W/L)_p = 0.9375 \mu\text{m} / 0.25 \mu\text{m}$

$(W/L)_n = 0.375 \mu\text{m} / 0.25 \mu\text{m}$

PMOS is wider to balance pull-up and pull-down resistances.

Load capacitance = 10 fF

Initial VDD = 2.5 V

Power Supply Sweep

Using SPICE scripting (`.control` block):

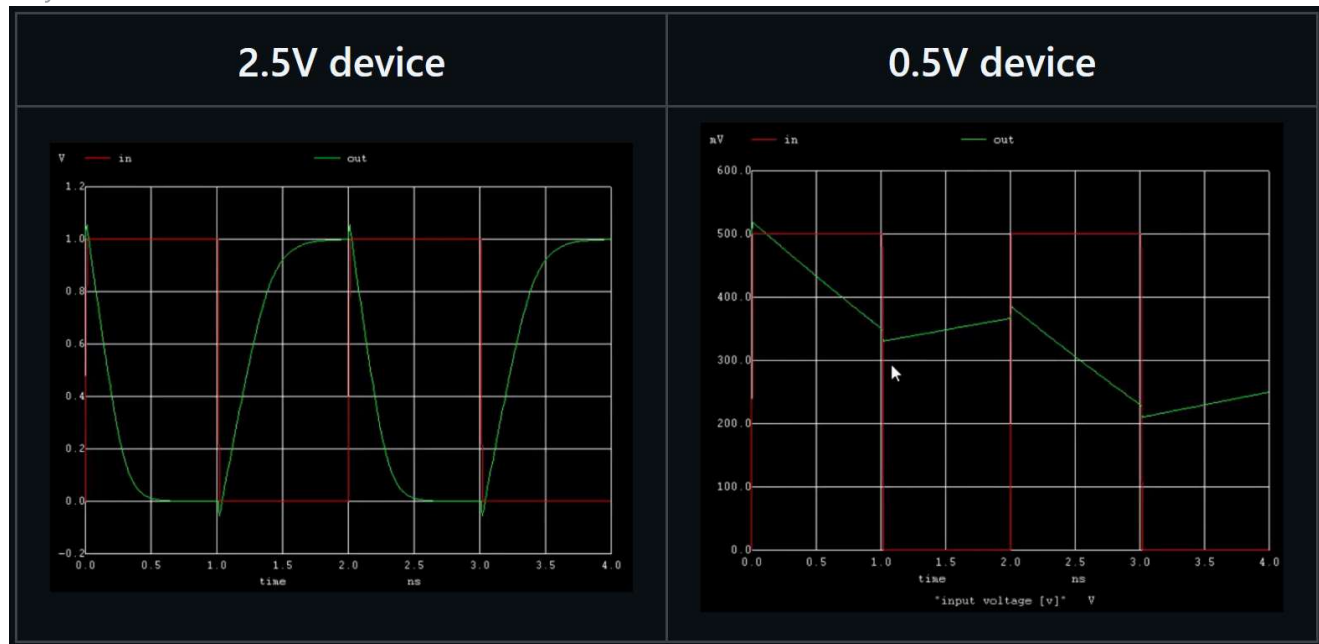
- Define variable: `power_supply = 2.5`
- Use `alter VDD = power_supply`
- Reduce VDD in steps of 0.5 V
- Sweep from 2.5 V $\rightarrow$ 0.5 V

Observations

- VTC shape remains similar as VDD scales down.
- Switching behavior is preserved.
- Even at 0.5 V, inverter still operates.
- Threshold and slope adjust proportionally with VDD.

Lecture 2: Advantages and disadvantages using low supply voltage

As VDD scales down (e.g., 2.5 V $\rightarrow$ 0.5 V), CMOS inverter behavior changes in both beneficial and limiting ways.



Advantages

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